

Nested Sampling for physical scientists

Greg Ashton ^{1,2} Noam Bernstein ³ Johannes Buchner ⁴ Xi Chen ⁵ Gábor Csányi ⁶ Farhan Feroz ⁷ Andrew Fowlie ⁸ Matthew Griffiths ⁹ Michael Habeck ¹⁰ Will Handley ^{11,12} Edward Higson ¹³ Michael Hobson ¹² Anthony Lasenby ^{11,12} David Parkinson ¹⁴ Livia B. Pártay ¹⁵ Matthew Pitkin ¹⁶ Doris Schneider ¹⁷ Leah South ¹⁸ Joshua S. Speagle ^{19,20,21} John Veitch ²² Philipp Wacker ¹⁷ David J Wales ²³ David Yallup ^{11,12}

We review Skilling’s nested sampling (NS) algorithm for Bayesian inference and more broadly multi-dimensional integration. After recapitulating the principles of NS, we survey developments in implementing efficient NS algorithms in practice in high-dimensions, including methods for sampling from the so-called **constrained prior**. We outline the ways in which NS may be applied and describe the application of NS in three scientific fields in which the algorithm has proved to be useful: cosmology, gravitational-wave astronomy, and materials science. We close by making recommendations for best practice when using NS and by summarizing potential limitations and optimizations of NS.

Author addresses
¹ School of Physics and Astronomy, Monash University, VIC 3800, Australia ² Department of Physics, Royal Holloway, University of London, TW20 0EX, United Kingdom ³ Center for Materials Physics and Technology, U. S. Naval Research Laboratory, Washington, DC 20375, USA ⁴ Max Planck Institute for Extraterrestrial Physics, Giessenbachstrasse, 85741 Garching, Germany ⁵ Department of Computer Science, University of Bath, Bath, BA2 7PB, United Kingdom ⁶ Engineering Laboratory, University of Cambridge, Trumpington Street, Cambridge CB2 1PZ, United Kingdom ⁷ Independent researcher ⁸ Department of Physics and Institute of Theoretical Physics, Nanjing Normal University, Nanjing, Jiangsu 210023, China. andrew.j.fowlie@NJNU.edu.cn ⁹ Concr Ltd, Babraham Hall House, Cambridge, CB22 3AT ¹⁰ Microscopic Image Analysis Group, Jena University Hospital, Jena, Germany ¹¹ Astrophysics Group, Cavendish Laboratory, J.J. Thomson Avenue, Cambridge, CB3 0HE, United Kingdom ¹² Kavli Institute for Cosmology, Cambridge, CB3 0HA, United Kingdom ¹³ The D. E. Shaw Group, 1166 Avenue of the Americas, New York, NY 10036, USA ¹⁴ Korea Astronomy and Space Science Institute, Yuseong-gu, Daedeok-daero 776, Daejeon 34055, Korea ¹⁵ Department of Chemistry, University of Warwick, Coventry, CV4 7AL, United Kingdom ¹⁶ Department of Physics, Lancaster University, Lancaster, LA1 4YB, United Kingdom ¹⁷ Department of Mathematics, Friedrich-Alexander-Universität Erlangen-Nürnberg, 91058 Erlangen, Germany ¹⁸ School of Mathematical Sciences, Queensland University of Technology, Brisbane, QLD 4000, Australia ¹⁹ Dunlap Institute for Astronomy and Astrophysics, University of Toronto, 50 St. George Street, Toronto, ON M5S 3H4, Canada ²⁰ David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto, 50 St. George Street, Toronto ON M5S 3H4, Canada ²¹ Department of Statistical Sciences, University of Toronto, 100 St. George St, Toronto, ON M5S 3G3, Canada ²² School of Physics and Astronomy, University of Glasgow, Glasgow, G12 8QQ, United Kingdom ²³ Yusuf Hamied Department of Chemistry, Lensfield Road, Cambridge, CB2 1EW, United Kingdom

1 Introduction

The nested sampling algorithm (NS; [1, 2]) was introduced by Skilling in 2004 in the context of Bayesian inference and compu-

tation (described in box 1). The NS algorithm solves otherwise challenging high-dimensional integrals by evolving a collection of live points through the parameter space. The algorithm was immediately adopted in cosmology, owing to the fact that it partially overcomes three major difficulties in the traditional algorithm for Bayesian computation, **Markov chain Monte Carlo (MCMC; see for example ref. [3, 4])**. First, it simultaneously returns results for model comparison and parameter inference. Second, it is successful in **multi-modal** problems. Third, it is naturally self-tuning, permitting it to be applied immediately to new problems. In the 15 years since, the theoretical properties of the algorithm and connections to other computational methods have been partially clarified, and efficient implementations, variants and cross-checks of NS have been developed. The range of applications now extends beyond cosmology and into many other branches of science.

Here the review of those developments and applications is structured as follows. Below in **Introduction**, we recapitulate the origins and principles of NS. We summarize implementations and variants of the NS algorithm, including the developments since its inception, in **Experimentation** and results from NS in **Results**. We describe scientific applications from cosmology, gravitational-wave astronomy, particle physics and materials science in **Applications**. We outline best practices when using NS, including **Reproducibility and data deposition** and discuss issues with the technique in **Limitations and optimizations**. We close by looking forward to the future of NS and Bayesian computation in **Outlook**. Further details are presented in **Supplementary Information**, including a glossary in Supplementary Information **A** and a simple numerical example in Supplementary Information **G**.

1.1 Multi-dimensional integrals

Since NS is primarily an algorithm for integration, let us write a general multi-dimensional integral of a function L over parameters Θ as

$$Z = \int L(\Theta) \, \mathrm{d}\mu(\Theta). \tag{1}$$

In many scientific problems we need to be able to integrate in high dimensions and for challenging **integrands**. We assume

物理科学家的嵌套采样

GregAshton ^{1,2} Noam Bernstein ³ Johannes Buchner ⁴ Xi Chen ⁵ Gábor Csányi ⁶ Farhan Feroz ⁷ Andrew Fowlie ⁸ Matthew Griffiths ⁹ Michael Habeck ¹⁰ Will Handley ^{11,12} EdwardHigson ¹ ³ Michael Hobson ¹² 安东尼·拉斯本 ^{11,12} 大卫·帕金森 ¹⁴ Livia B. Pártay ¹⁵ Matthew Pitkin ¹⁶ DorisSchneider ¹⁷ Leah South ¹⁸ 约书亚·斯皮格尔 ^{19,20,21} 约翰·维奇 ²² 菲利普瓦克 ¹⁷ 大卫·J·威尔士 ²³ 大卫·亚卢普 ^{11,12}

我们回顾了斯金林（Skilling）的嵌套采样（NS）算法在贝叶斯推理和多维度积分中的应用。在回顾NS原理后，我们调查了在高维度实践中实现高效NS算法的发展，包括从所谓约束先验中进行采样的方法。我们概述了NS的应用方式，并描述了NS在三个科学领域中的应用，该算法已被证明非常有效：宇宙学、引力波天文学和材料科学。最后，我们提出了使用NS的最佳实践建议，并总结了NS的潜在局限性和优化方向。

作者地址
¹ 莫纳什大学物理与天文学院, VIC 3800, 澳大利亚 2 Department of Physics, Royal Holloway, University of London, TW20 0EX, United Kingdom ³ Center for Materials Physics and Technology, U. S. Naval Research Laboratory, Washington, DC 20375, USA ⁴ Max Planck Institute for Extraterrestrial Physics, Giessenbachstrasse, 85741 Garching, Germany ⁵ Department of Computer Science, University of Bath, Bath, BA2 7PB, United Kingdom ⁶ 剑桥大学工程实验室, 特朗普顿路, Cambridge CB2 1PZ, 英国 ⁷ 独立研究员 ⁸ 南京师范大学物理系和理论物理研究所 大学, 南京, 江苏省210023, 中国. andrew.j.fowlie@NJNU.edu.cn ⁹ Concr Ltd, Babraham Hall House, 剑桥, CB22 3AT ¹⁰ 显微镜图像分析小组, 耶拿大学医院, 耶拿, 德国 ¹¹ 天体物理学小组, 卡文迪什实验室, J.J. 汤姆森大道, 剑桥, CB3 0HE, United Kingdom ¹² 卡弗里宇宙学研究所, 剑桥, CB3 0HA, 英国 ¹³ D. E. Shaw集团, 1166美洲大道, 纽约, NY 10036, 美国 ¹⁴ 韩国天体与空间科学研究所, 尤胜区, 大德大路776, 大田34055, 韩国 ¹⁵ 化学系, 华威大学, 考文垂, CV4 7AL, 英国 ¹⁶ 物理系, 兰开斯特大学, 兰开斯特, LA1 4YB, 英国 ¹⁷ 数学系, 弗里德里希-希历山大学埃尔兰根-纽伦堡分校, 91058埃尔兰根, 德国 ¹⁸ 数学科学学院, 昆士兰科技大学, 布里斯班-本, 昆士兰州4000, 澳大利亚 ¹⁹ Dunlap Institute for Astronomy and Astrophysics, University of Toronto, 50 St. George Street, Toronto, ON M5S 3H4, Canada ²⁰ David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto, 50 St. George Street, Toronto ON M5S 3H4, Canada ²¹ Department of Statistical Sciences, University of Toronto, 100 St. George St, Toronto, ON M5S 3G3, Canada ²² School of Physics and Astronomy, University of Glasgow, Glasgow, G12 8QQ, United Kingdom ²³ 尤素福·哈米德化学系, Lensfield路, 剑桥, CB2 1EW, 英国

1 引言

嵌套采样算法（NS； [1, 2]）由Skillingin于2004年在贝叶斯推断的背景下引入。

嵌套采样算法（NS； [3, 4]）由Skillingin于2004年在贝叶斯推断和计算（如框1所述）的背景下提出。NS算法通过在参数空间中演化一组活点来求解原本具有挑战性的高维积分。由于该算法部分克服了贝叶斯计算传统算法（马尔可夫链蒙特卡洛，MCMC；例如参考文献 [3, 4]）的三大主要困难，因此它很快被宇宙学领域所采用。首先，它同时返回模型比较和参数推断的结果。其次，它在多模态问题中表现成功。第三，它具有天然自我调优特性，允许其立即应用于新问题。在过去的15年里，该算法的理论性质及其与其他计算方法的关系得到了部分阐明，并开发了高效的实现、变体和NS的交叉验证。目前，该算法的应用范围已超越宇宙学，扩展到科学领域的许多其他分支。

本文的综述结构如下。在引言部分，我们回顾了NS的起源和原理。在实验部分，我们总结了NS算法的实现和变体，包括其诞生以来的发展，并在结果部分展示了NS的实验结果。在应用部分，我们描述了来自宇宙学、引力波天文学、粒子物理学和材料科学中的科学应用。在使用NS时，我们概述了最佳实践，包括可重复性和数据存档，并在局限性和优化部分讨论了该技术的问题。最后，在展望部分，我们展望了NS和贝叶斯计算的未来。更多细节见补充信息，包括补充信息A中的词汇表和补充信息G中的简单数值示例。

1.1 多重积分

由于 NS 主要是一种积分算法，让我们写一个函数 L 在参数 Θ 上的通用多维积分

$$Z = \int L(\Theta) \, \mathrm{d}\mu(\Theta). \tag{1}$$

在许多科学问题中，我们需要能够在高维度进行积分，并且对于具有挑战性的被积函数。我们假设

Box 1 | Bayesian inference

Although NS is a general purpose algorithm for integration, its major application has been integrals in Bayesian inference, and we describe NS using that language. In Bayesian inference [5–11] our state of knowledge is quantified by probability and we learn from data by updating probabilities using Bayes’ theorem,

$$\Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B)}.$$

To use this to learn from data about a model and its parameters, we write it as

$$P(\Theta) = \frac{L(\Theta)\pi(\Theta)}{Z},$$

where the **prior**, $\pi(\Theta) \equiv \Pr(\Theta)$ represents what was known about a model’s parameters before seeing the data and the **posterior**, $P(\Theta) \equiv \Pr(\Theta|D)$, represents what is known after learning from the data. The observed data, D , was encoded into the **likelihood** function, $L(\Theta) \equiv \Pr(D|\Theta)$.

The denominator, $Z \equiv \Pr(D)$, is the **evidence** value that appears in **Bayesian model comparison** [12]. It may be written,

$$Z = \int L(\Theta)\pi(\Theta)\mathrm{d}\Theta,$$

and so is also known as the marginal **likelihood** and as the normalizing constant, since it normalizes the **posterior** such that $\int P(\Theta)\mathrm{d}\Theta = 1$. The ratio of evidences computed for different models is known as a Bayes factor,

$$B_{10} = \frac{Z_1}{Z_0}.$$

The Bayes factor tells us how we must update the relative plausibility of two models in light of data.

that the **integrand** is positive, $L(\Theta) \geq 0$.

Often, Z may be a physical quantity such as the total mass of an object distributed with density L across volumes $\mathrm{d}\mu(\Theta)$. Whilst NS is a general method for integration, for concreteness, we view all such applications through the lens of Bayesian inference (see box 1), with $\mathrm{d}\mu(\Theta) \equiv \pi(\Theta)\mathrm{d}\Theta$ seen as an element of *prior probability* with π the **prior**, normalized by its nature to $\int \pi(\Theta)\mathrm{d}\Theta = 1$. The **integrand**, L , is the modulating **likelihood** function (hence the symbol) and Z is the **evidence**. In scientific inference problems, the integral could be over tens if not hundreds of parameters, required to model fundamental effects as well as the calibration and systematics of complicated experimental measurements [13].

We may rewrite the **integrand** through the elementary factorization known as Bayes’ theorem,

$$L(\Theta) \times \pi(\Theta) = Z \times P(\Theta), \tag{2}$$

where

$$P(\Theta) = \frac{L(\Theta)\pi(\Theta)}{Z}, \tag{3}$$

is the **posterior**, normalized to $\int P(\Theta)\mathrm{d}\Theta = 1$. Notation apart,

however, all we are doing here is decomposing the **integrand** into a magnitude Z and a shape $P(\Theta)$.

Historically, Bayesian computation focused on only the shape $P(\Theta)$, partly owing to controversies around **Bayesian model comparison** such as its sensitivity to the choices of **prior** [12], and partly due to computational difficulties [14]. However, shapes and magnitudes both matter, especially in the general setting of multi-dimensional integration beyond **Bayesian model comparison**. NS [1, 2] surmounts the challenge by computing shapes and magnitudes simultaneously.

1.2 Simplifying multi-dimensional integrals

Before introducing NS, let us attempt to simplify the general integral in eq. (1). Consider traditional Riemann-style integration. This decomposes the space into volume elements $\Delta\Theta$, typically small cubes, and performs a sum over them. Small cubes, however, rapidly become infeasible in multi-dimensional integration because their cost grows exponentially with dimension — this is the “**curse of dimensionality**.”

We don’t, however, need to decompose our space into little cubes; our cells can be any shape we want. The integrals needed for quantification,

$$Z = \int L(\Theta)\pi(\Theta)\mathrm{d}\Theta = \lim_{|\Delta\Theta| \rightarrow 0} \sum L(\Theta)\pi(\Theta)\Delta\Theta, \tag{4}$$

are defined as limiting sums over volume elements which should be small enough to keep $P(\Theta)$ almost constant *regardless of shape*. We may therefore combine the cells in which the **integrand** is almost constant. Schematically, we may write

$$Z = \sum L(X)\Delta X, \tag{5}$$

where ΔX is the volume of cells that share **likelihood** $L(X)$ weighted by the **prior** $\pi(\Theta)$. This is illustrated schematically in fig. 1a and works whether the **integrand** is **uni-** or **multi-modal**.

We can reach eq. (5) more concretely by noting that the **evidence** is the expectation of a non-negative random variable, such that it may be written as

$$Z = \int X(L)\mathrm{d}L, \tag{6}$$

where the volume variable X ,

$$X(L^\star) = \int_{L>L^\star} \pi(\Theta)\mathrm{d}\Theta, \tag{7}$$

is the volume enclosed by contour $L^\star$. This result can be readily proven by integration by parts (see also sec. 21 in ref. [17]). Applying integration by parts again to eq. (6), we obtain the familiar NS **evidence** identity,

$$Z = \int_0^1 L(X)\mathrm{d}X, \tag{8}$$

providing that $L(X)$, the inverse of $X(\lambda)$, indeed exists and that the **evidence** is finite. This formalizes the schematic eq. (5). We discuss this result more formally in box 2.

Box 1 | 贝叶斯推理

尽管NS是一种通用的积分算法，但其主要应用一直是贝叶斯推理中的积分，我们使用那种语言来描述NS。在贝叶斯推理中 [5–11]，我们的知识状态由概率量化，我们通过使用贝叶斯定理更新概率来从数据中学习，

$$\Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B)}.$$

要使用这种方法从模型及其参数的数据中学习，我们将它写成

$$P(\Theta) = \frac{L(\Theta)\pi(\Theta)}{Z},$$

其中先验分布， $\pi(\Theta) \equiv \Pr(\Theta)$ 表示在看到数据之前对模型参数的了解，后验分布， $P(\Theta) \equiv \Pr(\Theta|D)$ ，表示在从数据中学习后所了解的内容。观测数据， D ，被编码到似然函数， $L(\Theta) \equiv \Pr(D|\Theta)$ 中。

分母， $Z \equiv \Pr(D)$ ，是贝叶斯模型比较中出现的证据值 [12]。它可以写成

$$Z = \int L(\Theta)\pi(\Theta)\mathrm{d}\Theta,$$

因此也被称为边缘似然，以及归一化常数，因为它归一化后验分布，使得 $\int P(\Theta)\mathrm{d}\Theta = 1$ 。对不同模型计算的证据比率被称为贝叶斯因子，

$$B_{10} = \frac{Z_1}{Z_0}.$$

贝叶斯因子告诉我们，在给定数据的情况下，我们必须如何更新两个模型的相对可能性。

该被积函数为正， $L(\Theta) \geq 0$ 。

通常， Z 可能是一个物理量，例如物体在密度 L 分布的体积 $\mathrm{d}\mu(\Theta)$ 上的总质量。虽然 NS 是一种通用的积分方法，但为了具体起见，我们通过贝叶斯推理的视角（参见框 1）来看待所有此类应用，其中 $\mathrm{d}\mu(\Theta) \equiv \pi(\Theta)\mathrm{d}\Theta$ 被视为先验概率的一个元素，而 π 先验概率，由于其性质而被归一化为 $\int \pi(\Theta)\mathrm{d}\Theta = 1$ 。被积函数 L 是调节似然函数（因此使用该符号）， Z 是证据。在科学推理问题中，积分可能涉及数十个甚至数百个参数，这些参数需要用来模拟基本效应以及复杂实验测量的校准和系统误差 [13]。

我们可以通过贝叶斯定理这个基本因子分解重写被积函数，

$$L(\Theta) \times \pi(\Theta) = Z \times P(\Theta), \tag{2}$$

哪里

$$P(\Theta) = \frac{L(\Theta)\pi(\Theta)}{Z}, \tag{3}$$

这是后向的，归一化到 $\int P(\Theta)\mathrm{d}\Theta = 1$ 。除符号外，

然而，我们在这里所做的一切只是将积分项分解为一个幅度 Z 和一个形状 $P(\Theta)$ 。

历史上，贝叶斯计算主要关注 $P(\Theta)$ 的形状，部分原因是围绕贝叶斯模型比较的争议，例如其对先验 [12], 选择敏感，部分原因是计算困难 [14]。然而，形状和幅度都重要，特别是在贝叶斯模型比较之外的多维积分的通用设置中。NS [1, 2] 通过同时计算形状和幅度来克服这一挑战。

1.2 简化多维积分

在介绍 NS 之前，让我们尝试简化等式 (1) 中的通用积分。考虑传统的黎曼风格积分。它将空间分解为体积元素 $\Delta\Theta$ ，通常是小的立方体，并对它们进行求和。然而，在多维积分中，小立方体很快变得不可行，因为它们的成本随维度呈指数增长——这就是“**维度的诅咒**”。

然而，我们不需要将空间分解成小立方体；我们的单元可以是任何我们想要的形状。用于量化的积分，

$$Z = \int L(\Theta)\pi(\Theta)\mathrm{d}\Theta = \lim_{|\Delta\Theta| \rightarrow 0} \sum L(\Theta)\pi(\Theta)\Delta\Theta, \tag{4}$$

它们被定义为对体积元素的限制和，这些体积元素应该足够小，以保持 $P(\Theta)$ 几乎恒定，无论形状如何。因此，我们可以将积分几乎恒定的单元组合起来。示意图中，我们可以写成

$$Z = \sum L(X)\Delta X, \tag{5}$$

其中 ΔX 是具有似然 $L(X)$ 并由先验 $\pi(\Theta)$ 加权的单元体积。这在图 1a 中示意性地说明，并且无论被积函数是单峰或多峰的，它都适用。

通过注意到证据是非负随机变量的期望，我们可以更具体地得到等式 (5)，因此可以写成

$$Z = \int X(L)\mathrm{d}L, \tag{6}$$

其中体积变量 X ，

$$X(L^\star) = \int_{L>L^\star} \pi(\Theta)\mathrm{d}\Theta, \tag{7}$$

是 $L^\star$ 包围的体积。这个结果可以通过分部积分很容易地证明（另见参考文献 [17] 第 21 节）。再次对等式 (6) 应用分部积分，我们得到熟悉的 NS 证据恒等式，

$$Z = \int_0^1 L(X)\mathrm{d}X, \tag{8}$$

只要 $L(X)$ ，即 $X(\lambda)$ 的逆确实存在，并且证据是有限的。这形式化了示意图等式 (5)。我们在框 2 中更正式地讨论这一结果。

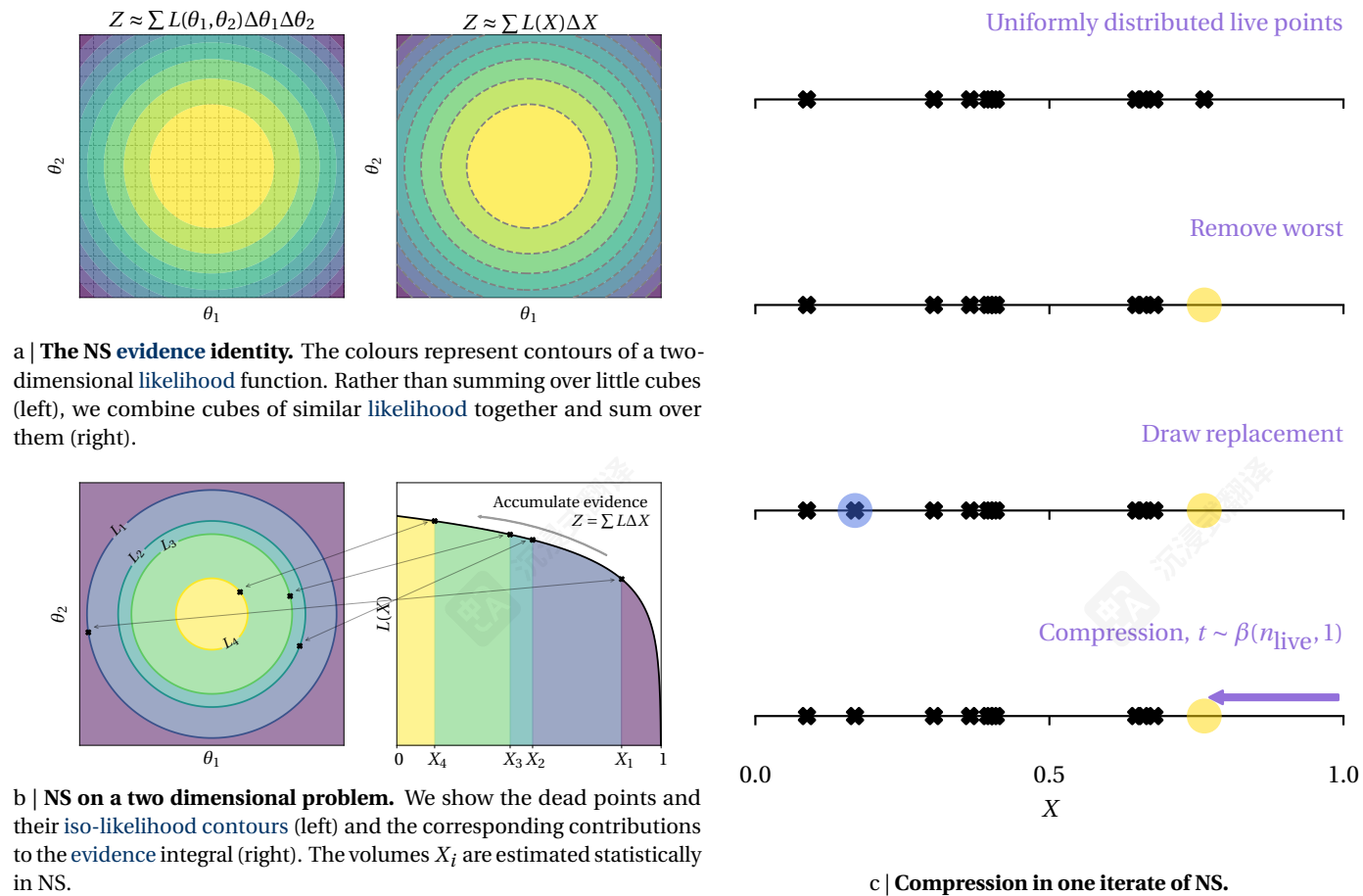


Figure 1 | Illustrations of NS algorithm.

1.3 Nested sampling

As the multi-dimensional integral in eq. (7) is impractical in high dimension, some sort of statistical estimation is inevitable. NS starts with an ensemble of n_{live} random locations Θ drawn from the **prior**, $\pi(\Theta)$, each of which has its **likelihood** $L(\Theta)$ which we can place in ascending order. Crudely, if we discarded the lowest half of the values, the survivors would be random samples taken within the restricted volume $L > \text{Median}[L]$, which would statistically be roughly half the original volume. This allows us to make a statistical estimate of the volume variable in eq. (7). Repeating that n_{iter} times would yield compression by a factor of about $2^{n_{\text{iter}}}$. This is the exponential behaviour required to overcome the **curse of dimensionality**.

Thus NS works by statistical estimates of the compression, which is a general and fundamental operation that can be used in various ways not limited to those in table 1. The **evidence** identity in eq. (6) isn't required in every application; many applications only use the compression in eq. (7). We present scientific applications in [Applications](#).

1.4 Formulation

We now present the NS algorithm in more detail. We assume that there are no regions of constant **likelihood** resulting in like-

lihood plateaus (see [Limitations and optimizations](#) for further discussion). The NS algorithm begins by drawing an ensemble of n_{live} samples from the **prior**. We compute the **likelihood** for each sample. We denote the smallest **likelihood** by L^* and we discard that point. The remaining live points are now distributed over a compressed volume; we denote the factor by which the volume compressed by t . Finally, a replacement point is drawn from the **prior** subject to $L > L^*$, that is, from the **constrained prior**,

$$\pi^*(\Theta) \propto \begin{cases} \pi(\Theta) & \text{if } L(\Theta) > L^* \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

This leaves a new ensemble with n_{live} samples obeying a **likelihood** constraint $L > L^*$.

As they are drawn from the **constrained prior**, the volumes X associated with the live points are uniformly distributed. Thus the compression associated with the discarded outermost sample, t , corresponds to the smallest of n_{live} uniform random variables. This follows a $\text{Beta}(n_{\text{live}}, 1)$ distribution,

$$P(t) = n_{\text{live}} t^{n_{\text{live}}-1}. \quad (10)$$

The first factor accounts for the fact that any live point could be the outermost and the second factor for the fact that $n_{\text{live}} - 1$

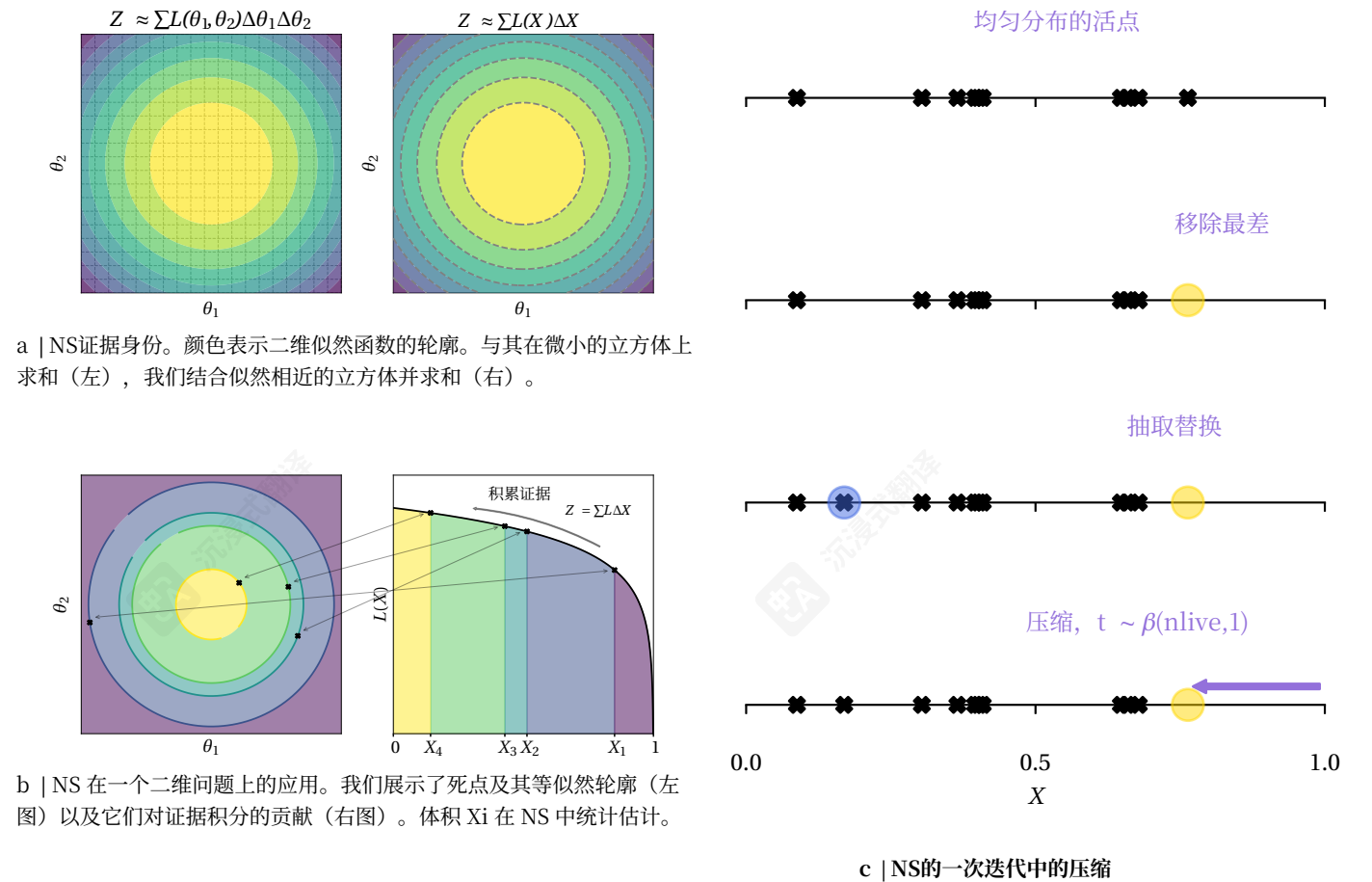


图 1 | NS算法的插图

1.3 嵌套采样

由于等式 (7) 中的多维积分在高维中不切实际，某种形式的统计估计是不可避免的。NS 从先验 $\pi(\Theta)$ 中抽取 n_{live} 个随机位置 Θ 的集合开始，每个位置都有其似然 $L(\Theta)$ ，我们可以将它们按升序排列。粗略来说，如果我们丢弃最低一半的值，那么幸存者将是随机样本，它们在受限体积 $L > \text{Median}[L]$ 内，其统计上大致是原始体积的一半。这使我们能够对等式 (7) 中的体积变量进行统计估计。重复 n_{iter} 次将产生约 $2^{n_{\text{iter}}}$ 的压缩因子。这是克服维度灾难所需的指数行为。

因此，NS 通过对压缩进行统计估计来工作，这是一种通用且基本操作，可用于多种方式，不仅限于表1中所示的那些。等式(6)中的证据身份在每种应用中并非都必需；许多应用仅使用等式(7)中的压缩。我们在“应用”部分介绍了科学应用。

1.4 配方

我们在此更详细地介绍NS算法。我们假设不存在恒定似然率的区域，从而导致似然平台（参见限制和优化以进行进一步讨论）。

NS算法首先从先验分布中抽取 n_{live} 个样本的集合。我们计算每个样本的似然值。我们用 L^* 表示最小的似然值，并丢弃该点。剩余的活点现在分布在一个压缩的体积上；我们用 t 表示体积压缩的因子。最后，根据 $L > L^*$ 从先验分布中抽取一个替代点，即从约束先验分布中，

$$\pi^*(\Theta) \propto \begin{cases} \pi(\Theta) & \text{if } L(\Theta) > L^* \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

这留下了一个新的集成，包含 n_{live} 个样本，这些样本服从似然约束 $L > L^*$ 。

由于它们是从约束先验中抽取的，与活动点相关联的体积 X 是均匀分布的。因此，与丢弃的最外层样本 t 相关的压缩量对应于 n_{live} 个均匀随机变量的最小值。这遵循 $\text{Beta}(n_{\text{live}}, 1)$ 分布，

$$P(t) = n_{\text{live}} t^{n_{\text{live}}-1}. \quad (10)$$

第一个因素解释了任何活点都可能是最外层的这一事实，第二个因素解释了 $n_{\text{live}} - 1$ 这一事实。

NS application	Details
Integration	Perform the general multi-dimensional integral eq. (1) for positive integrands.
Global optimization	Maximize the likelihood , L , by compressing to $\hat{\Theta}$, the maximum of $L(\Theta)$, with no restriction to uni-modal distributions. This may require strict settings and be more computationally expensive than integration; see ref. [146, 240] for further discussion.
Bayesian inference	NS simultaneously computes the posterior and the Bayesian evidence , allowing parameter inference and model comparison.
Approximate Bayesian Computation	Perform efficient Approximate Bayesian Computation by applying NS to the joint space of parameters and data [241]
Statistical thermodynamics	If we use the Boltzmann factor as the likelihood , i.e., $L(\Theta) = e^{-\beta E(\Theta)}$, where E is the energy of the state Θ among N , we may use NS to compute the partition function . By accumulating for several temperatures T in parallel, one can plot thermodynamic functions such as the specific heat $C_V = T \text{d}S/\text{d}T$ as functions of temperature without needing multiple runs.
Rare event sampling	The volume variable X may be interpreted as the probability of a rare event [22, 23] or used to compute a p -value in frequentist statistics [161].

Table 1 | **Applications of NS.**

uniformly distributed samples lie above the outermost sample at t (the remaining sample lies at t).

We started from unconstrained samples drawn from the full original volume $X_0 = 1$. As we repeat this process of replacing the outermost points, the successive compressions by factors $t_1, t_2, t_3, \dots$ lead to exponentially decreasing inferred volumes $X_1, X_2, X_3, \dots$

$$X_0 = 1. \quad X_{i+1} = t_{i+1} X_i. \quad (11)$$

We illustrate a single iteration of NS in fig. 1c. If we are most interested in the magnitude of the **evidence**, $\log Z$, we should consider,

$$\log t \approx \langle \log t \rangle = -\frac{1}{n_{\text{live}}}, \quad (12)$$

as under repeated multiplication $X_k = t_1 t_2 \dots t_k$ it's the *logarithms* that add. For a more complete inference, the compression factors t may be sampled directly from $\text{Beta}(n_{\text{live}}, 1)$. See Supplementary Information B for further discussion. NS therefore estimates volumes through probability not geometry, topology, or even dimension. We do not get a definite compression value, only a distribution of what it might have been.

Having obtained estimates of the volume $X(L^*)$ at each of n_{iter} iterations, we may accumulate the **evidence** via eq. (6) or eq. (8), for example by the trapezium rule,

$$Z = \sum_{i=1}^{n_{\text{iter}}} w_i L_i^*, \quad (13)$$

where the weights are

$$w_i = \frac{1}{2} (X_{i-1} - X_{i+1}). \quad (14)$$

The sum in eq. (13) converges to the desired integral [18–20]. This is the magnitude; we obtain shape by assigning weights to

each sample (see ref. [18] for discussion),

$$P_i = \frac{w_i L_i^*}{Z}, \quad (15)$$

normalized such that $\sum P_i = 1$. These are the **posterior** weights of the dead points; the shape may be recovered by for example a weighted histogram or other density estimation methods. We show the whole algorithm schematically in algorithm 1 and the summation for a two-dimensional problem in fig. 1b.

Compared to the sketch of NS in section 1.3, we replace a single live point per iteration, rather than half of the live points. Whilst the number of replacements can be varied, one replacement is optimal (though see considerations in **Parallelization**). This is because for $r \ll n_{\text{live}}$, replacing r points per iteration would reach the **posterior bulk** in about r times fewer iterations. While the computational expense wouldn't change as r replacements are required per iteration, the error estimates would scale as $\sqrt{r}$ because reducing the number of iterations increases the relative Poisson noise in the number of iterations.

In the last decade or so, analogies between NS and statistical mechanics (see box 3) and other statistical methods, including sequential Monte Carlo (SMC; [21]) and subset simulation in rare event sampling [22–25], among others [26–30], have been recognized. The connections to SMC and an SMC variant of NS are discussed in box 4. There are, of course, other strategies for computing the evidence; see ref. [31–34] for reviews. Notable examples include approximating the integrand by a tractable function [35], Chib's method using density estimation and Gibbs' sampling [36], importance sampling [37], and techniques that re-use **MCMC** draws [38]. Broadly speaking, NS lies in a class of algorithms that form a path of bridging distributions, and evolves samples along that path [39, 40]. NS stands out because the path is *automatic* and *smooth* — we compress in $\log X$ by on average $1/n_{\text{live}}$ at each iteration — and

NS应用程序	详细信息
集成 全局优化	对正的积分项执行一般多维积分公式(1)。通过压缩最大的最大似然度L不受单峰分布的限制。这可能需要严格的设置，并且比积分计算更昂贵；参见参考文献 [146, 240] 进行进一步讨论。
贝叶斯推理	NS同时计算后验分布和贝叶斯证据，允许参数推断和模型比较。
近似贝叶斯计算	通过将NS应用于参数和数据 [241]的联合空间，执行高效的近似贝叶斯计算。
统计热力学	如果我们使用玻尔兹曼因子作为似然函数，即 $L(\Theta) = e^{-\beta E(\Theta)}$ ，其中 E 是状态 Θ 在 N 中的能量，我们可以使用 NS 来计算配分函数。通过并行累积多个温度 T，我们可以将比热容 $C_V = T \text{d}S/\text{d}T$ 等热力学函数绘制为温度的函数，而无需多次运行。
稀有事件采样	体积变量 X 可以解释为稀有事件 [22, 23] 的概率，或在频率统计学中用于计算 p 值 [161]。

表 1 | **NS 的应用。**

在t时刻，均匀分布的样本都位于最外层样本之上（其余样本位于t处）。

我们从完整原始体积 $X_0 = 1$ 中抽取的无约束样本开始。随着我们重复替换最外层点的这个过程，连续的压缩因子 $t_1, t_2, t_3, \dots$ 导致推断体积 $X_1, X_2, X_3, \dots$ 指数级减小。

$$X_0 = 1. \quad X_{i+1} = t_{i+1} X_i. \quad (11)$$

我们在图 1c 中说明了 NS 的单次迭代。如果我们最感兴趣的是证据的幅度， $\log Z$ ，我们应该考虑，

$$\log t \approx \langle \log t \rangle = -\frac{1}{n_{\text{live}}}, \quad (12)$$

如下所示，通过重复乘法 $X_k = t_1 t_2 \dots t_k$ ，是指数数相加。为了更完整的推断，压缩因子 t 可以直接从 $\text{Beta}(n_{\text{live}}, 1)$ 中采样。请参阅补充信息 B 以进行进一步讨论。因此，NS 通过概率而非几何、拓扑甚至维度来估计体积。我们不会得到一个确定的压缩值，而只会得到一个可能值的分布。

在获得每个 n_{iter} 迭代处的体积 $X(L^*)$ 的估计值后，我们可以通过等式 (6) 或等式 (8) 累积证据，例如通过梯形规则，

$$Z = \sum_{i=1}^{n_{\text{iter}}} w_i L_i^*, \quad (13)$$

其中权重为

$$w_i = \frac{1}{2} (X_{i-1} - X_{i+1}). \quad (14)$$

式(13)中的和在收敛到期望的积分 [18–20]。这是幅度；我们通过为每个样本分配权重来获得形状（参见参考文献 [18进行讨论），

（见参考文献 [18] 进行讨论），

$$P_i = \frac{w_i L_i^*}{Z}, \quad (15)$$

已归一化，使得 $\sum P_i = 1$ 。这些是死点的后验权重；可以通过加权直方图或其他密度估计方法来恢复形状。我们在算法1中展示了整个算法的示意图，并在图1b中展示了二维问题的求和。

与1.3节中NS的草图相比，我们每次迭代替换一个活动点，而不是一半的活动点。虽然替换次数可以变化，但一次替换是最优的（尽管请参阅并行化中的考虑因素）。这是因为对于 $r \ll n_{\text{live}}$ ，每次迭代替换r个点，大约需要r倍更少的迭代次数才能达到后验主体。虽然随着每次迭代需要r次替换，计算开销不会改变，但误差估计会按 $\sqrt{r}$ 的比例缩放，因为减少迭代次数会增加迭代次数中相对的泊松噪声。

在过去的十年左右，人们已经认识到NS与统计力学（参见框3）以及其他统计方法（包括序贯蒙特卡洛（SMC；[21]）和稀有事件采样中的子集模拟 [22–25]，等 [26–30]，）之间的类比。SMC和NS的SMC变体的联系在框4中讨论。当然，还有其他计算证据的策略；参见参考文献 [31–34] 的综述。值得注意的例子包括通过可处理函数逼近被积函数 [35]，使用密度估计和吉布斯采样的Chib方法 [36]，重要性采样 [37]，以及重复使用MCMC抽样的技术 [38]。总的来说，NS属于一类形成桥梁分布路径的算法，并沿着该路径演化样本 [39, 40]。NS之所以突出，是因为该路径是自动和平滑的——我们在每次迭代中通过平均 $1/n_{\text{live}}$ 压缩 $\log X$ ——并且

Algorithm 1 | **Schematic of the NS algorithm for the general multi-dimensional integral in eq. (1).** Techniques for drawing replacements and stopping criteria are discussed in [Exploration strategies](#) and [Stopping conditions](#) respectively.

```

1 Choose an estimate of the compression factor, e.g.,  $t = e^{-1/n_{\text{live}}}$ 
2 Initialize volume,  $X = 1$  and integral,  $Z = 0$ 
3 Sample  $n_{\text{live}}$  points from the prior — the live points
4 repeat
5   Let  $L^*$  be the minimum  $L$  of the live points
6   Replace live point corresponding to  $L^*$  by one drawn from the prior subject to  $L > L^*$ 
7   Increment the estimate of the integral,  $Z = Z + L^* \Delta X$ , with e.g.,  $\Delta X = (1 - t)X$ 
8   Contract volume,  $X = tX$ 
9 until stopping criteria satisfied
10 Add estimate of remaining evidence, e.g.,  $Z = Z + \bar{L}X$  where  $\bar{L}$  is the average likelihood among the live points
11 return Estimate of integral, Z

```

because along the path we compress through constrained [priors](#), rather than from the [prior](#) to the [posterior](#). This was in fact a motivation for NS as it avoids phase transitions — abrupt changes in the bridging distributions — that cause problems for other methods including path samplers such as annealing. We further discuss NS's historical background and contrast it with annealing in Supplementary Information C.

1.5 Uncertainties

Our estimates of the magnitude and shape in eq. (1) must be accompanied by a discussion of the bias and statistical uncertainty. The latter originates from our noisy estimates of the compression factors. We may estimate the resulting statistical uncertainty in the [evidence](#) by considering the compression required to reach the [bulk](#) of the [posterior](#). This may be quantified by the information content [41, 42]

$$H = \int P(\Theta) \log \left(\frac{P(\Theta)}{\pi(\Theta)} \right) d\Theta \quad (16)$$

known in statistics as the [Kullback-Leibler \(KL\) divergence](#). We can write it using the volume variable as

$$H = \int P(X) \log P(X) dX \quad (17)$$

$$= - \int P(X) \log X dX + \int P(\log X) \log P(\log X) d\log X \quad (18)$$

where $P(X) \equiv L(X)/Z$ is the posterior density of the volume. In eq. (18) we see that the KL divergence equals minus the posterior expectation of $\log X$ minus the differential entropy associated with the posterior of $\log X$. As the first term typically dominates, the KL divergence provides a measure of compression.

Thus from eq. (12) it's likely to take about

$$n_H \approx n_{\text{live}} H \quad (19)$$

iterations to compress to the [bulk](#) of the [posterior](#) at $\log X = -H$, and this count is likely to be subject to $\sqrt{n_H}$ Poisson variability. Neglecting contributions from outside the bulk, we may write

the evidence sum in eq. (13) as

$$Z \approx e^{-n_H/n_{\text{live}}} \sum_{i=n_H}^{n_{\text{iter}}} (1-t)^{i-n_H} L_i^*. \quad (20)$$

We see that the final estimate of $\log Z$ will be plausibly subject to an approximately Gaussian uncertainty from the first factor,

$$\Delta \log Z \approx \sqrt{\frac{H}{n_{\text{live}}}}. \quad (21)$$

Thus NS statistical uncertainties scale as $1/\sqrt{n_{\text{live}}}$ as usual for statistical uncertainties (see ref. [18] for an alternative proof and discussion).

Although NS was first introduced with this estimate [1], it can be unreliable. Authority rests with repeated simulation through eq. (10) of what the compressions might actually have been. See ref. [43] for further discussion of the statistical uncertainty in the NS estimates and Supplementary Information E for discussion of uncertainties in NS estimates of the posterior. As well as this statistical uncertainty, there are four potential sources of bias: bias originating from failure to faithfully sample from the [constrained prior](#), bias originating from the choice of estimator for the compression factor, the generally negligible quadrature error, and the potentially important truncation error in eq. (13). The latter occurs as we stop after a finite number of iterations and is further discussed in [Stopping conditions](#). Provided that NS is appropriately configured, the statistical uncertainty usually dominates.

We see that difficulty in NS does not in fact lie in dimension but in compression from the [prior](#) to the [posterior](#): the compression and the resolution n_{live} alone determine the uncertainty and the run-time. To maintain a given uncertainty, by eq. (21) we require $n_{\text{live}} \propto H$ live points and by eq. (19) $n_{\text{iter}} \propto H^2$ iterations. If the [prior](#) and [posterior](#) are factorizable into a term for each dimension, by eq. (16) the [KL divergence](#) is additive, and so scales linearly with dimension, D , and so run-time goes like $\mathcal{O}(D^2)$. NS beats the exponential scaling with dimension expected from the [curse of dimensionality](#).

Algorithm 1 | **Schematic of the NS algorithm for the general multi-dimensional integral in eq. (1).** Techniques for drawing replacements and stopping criteria are discussed in [Exploration strategies](#) and [Stopping conditions](#) respectively.

```

1 Choose an estimate of the compression factor, e.g.,  $t = e^{-1/n_{\text{live}}}$ 
2 Initialize volume,  $X = 1$  and integral,  $Z = 0$ 
3 Sample  $n_{\text{live}}$  points from the prior — the live points
4 repeat
5   Let  $L^*$  be the minimum  $L$  of the live points
6   Replace live point corresponding to  $L^*$  by one drawn from the prior subject to  $L > L^*$ 
7   Increment the estimate of the integral,  $Z = Z + L^* \Delta X$ , with e.g.,  $\Delta X = (1 - t)X$ 
8   Contract volume,  $X = tX$ 
9 until stopping criteria satisfied
10 Add estimate of remaining evidence, e.g.,  $Z = Z + \bar{L}X$  where  $\bar{L}$  is the average likelihood among the live points
11 return Estimate of integral, Z

```

因为在路径上我们通过约束先验进行压缩，而不是从先验到后验。这实际上是NS的动机，因为它避免了相变——桥梁分布的急剧变化——这些问题会导致其他方法（包括路径采样器如退火）出现问题。我们在补充信息C中进一步讨论了NS的历史背景，并将其与退火进行对比。

1.5 不确定性

我们对等式 (1) 中幅度和形状的估计必须伴随着对偏差和统计不确定性的讨论。后者源于我们对压缩因子的噪声估计。我们可以通过考虑达到后验体的大部分所需的压缩量来估计由此产生的证据统计不确定性。这可以通过信息量 [41, 42] 来量化

$$H = \int P(\Theta) \log \left(\frac{P(\Theta)}{\pi(\Theta)} \right) d\Theta \quad (16)$$

在统计学中被称为Kullback-Leibler (KL) 散度。我们可以使用体积变量将其表示为

$$H = \int P(X) \log P(X) dX \quad (17)$$

$$= - \int P(X) \log X dX + \int P(\log X) \log P(\log X) d\log X \quad (18)$$

其中 $P(X) \equiv L(X)/Z$ 是体积的后验密度。在公式(18)中，我们看到KL散度等于 $\log X$ 的后验期望的负值减去与 $\log X$ 的后验相关的差分熵。由于第一项通常占主导地位，KL散度提供了压缩度的度量。

因此，根据公式(12)，很可能需要大约

$$n_H \approx n_{\text{live}} H \quad (19)$$

在 $\log X = -H$ 时压缩到后体的大部分所需的迭代次数，并且这个计数很可能受到 $\sqrt{n_H}$ 泊松变异性。忽略来百后体外的贡献，我们可以写成

将等式 (13) 中的证据和写成

$$Z \approx e^{-n_H/n_{\text{live}}} \sum_{i=n_H}^{n_{\text{iter}}} (1-t)^{i-n_H} L_i^*. \quad (20)$$

我们看到， $\log Z$ 的最终估计很可能受到第一个因子的近似高斯不确定性影响，

$$\Delta \log Z \approx \sqrt{\frac{H}{n_{\text{live}}}}. \quad (21)$$

因此 NS 统计不确定性尺度与 $1/\sqrt{n_{\text{live}}}$ 类似，这通常是统计不确定性的情况（参见 ref. [18] 以获得另一种证明和讨论）。

尽管 NS 首次引入时使用了这个估计 [1]，但它可能不可靠。权威性在于通过 eq. (10) 重复模拟实际压缩可能是什么。参见 ref. [43] 以进一步讨论 NS 估计中的统计不确定性，以及补充信息 E 中关于后验 NS 估计不确定性的讨论。除了这种统计不确定性之外，还有四个潜在的偏差来源：源于未能忠实地从约束先验中采样而产生的偏差，源于压缩因子估计器选择而产生的偏差，通常可以忽略的求积误差，以及在 eq. (13) 中可能很重要的截断误差。后者发生在我们停止有限次迭代之后，并在停止条件中进一步讨论。如果 NS 正确配置，统计不确定性通常占主导地位。

我们发现，在NS中，困难实际上不在于维度而在于从先验到后验的压缩：压缩和分辨率 n_{live} 本身决定了不确定性以及运行时间。为了保持给定的不确定性，根据等式(21)我们需要 $n_{\text{live}} \propto$ 个活点，根据等式(19)需要 $n_{\text{iter}} \propto H^2$ 次迭代。如果先验和后验可以分解为每个维度的一项，根据等式(16)KL散度是可加的，因此随维度D线性扩展，因此运行时间与 $\mathcal{O}(D^2)$ 成正比。NS克服了维数灾难预期的那种随维度指数增长的扩展。

Box 2 | Mathematical details

The idea of NS is to transform eq. (1) into eq. (8) which can be approximated more efficiently using the described Monte Carlo approach with active samples:

$$Z = \int_{\Omega} L(\Theta) d\mu(\Theta) = \int_0^1 \tilde{L}(X) dX,$$

where Ω is the **parameter domain** and $\tilde{L}$ is an **overloaded** form of L , as described next. To achieve this transformation, we define the **survival function** $X: \mathbb{R} \rightarrow [0, 1]$, $X(\lambda) = \mu(\{z \in \Omega: L(z) > \lambda\})$, that is the μ -**measure** of the λ -**super-level-sets**. Then, we introduce a mapping $\Phi: \Omega \rightarrow [0, 1]$ with

$$\Phi(\Theta) = X(L(\Theta)) = \mu(\{z \in \Omega: L(z) > L(\Theta)\})$$

This mapping Φ is the transformation which allows us to shift the integration from Ω to $[0, 1]$, by virtue of the **push-forward measure** $\mu \circ \Phi^{-1}$ on $[0, 1]$. Now, we can define

$$\tilde{L}: [0, 1] \rightarrow \mathbb{R}, \quad \tilde{L}(\xi) = \sup\{\lambda \in \text{Im } L: X(\lambda) > \xi\}.$$

Thus the integral transformation above which provides an alternative characterization of Z is true because

$$\begin{aligned} \int_{\Omega} L(\Theta) d\mu(\Theta) &= \int_{\Omega} \tilde{L}(X(L(\Theta))) d\mu(\Theta) \\ &= \int_{\Omega} (\tilde{L} \circ \Phi)(\Theta) d\mu(\Theta) = \int_{[0,1]} \tilde{L}(x) d(\mu \circ \Phi^{-1})(x) \\ &= \int_0^1 \tilde{L}(x) dx \end{aligned}$$

Here, $\tilde{L}$ is a generalized inverse of X under suitable assumptions on L , see ref. [15, 16]. This integral transformation holds at least in the case that L has no plateaus of positive **prior measure**, owing to the fact that in this case $\mu \circ \Phi^{-1}$ is indeed the uniform **measure** on $[0, 1]$, that is $d\mu \circ \Phi^{-1}(x) = dx$.

If L has a plateau of non-negligible mass, i.e., there exists a level λ^* such that $\mu(\{z \in \Omega: L(z) = \lambda^*\}) > 0$, then the derivation is more challenging, see also **Limitations and optimizations**.

2 Experimentation

In this section we will discuss how to implement NS, including considerations such as choosing the number of live points, drawing new live points from the **constrained prior**, parallelization and deciding when to stop.

2.1 Choice of the number of live points

As discussed in **Formulation**, the number of live points controls the *rate* of exponential compression during an NS run; we compress inwards by about $\Delta \log X \approx 1/n_{\text{live}}$ per iteration. This means that run-time scales as $\mathcal{O}(n_{\text{live}})$ (eq. (19)) and that the dominant uncertainties on the **evidence** integral (eq. (21)) and **posterior** scale as $\mathcal{O}(1/\sqrt{n_{\text{live}}})$. The above considerations are

Box 3 | Statistical mechanics analogy

There is a strong analogy between Bayesian inference and statistical mechanics. This suggests that NS might be useful in exploring problems that are typically the subject of statistical mechanical analysis. Consider Θ a **microstate** and $E(\Theta) = -\log L(\Theta)$ its energy. Then the *microcanonical ensemble* includes all states with $E(\Theta) = \epsilon$ where ϵ is a constant energy level. The volume of state space corresponding to a given energy ϵ is given by the density of states,

$$g(\epsilon) = \int \delta(E(\Theta) - \epsilon) \pi(\Theta) d\Theta$$

and taking the **prior** to be uniform corresponds to the *ergodic hypothesis*, namely that each **microstate** that is allowed by applicable conservation laws is equally likely to be observed. The Laplace transform of the density of states is the canonical **partition function** $Z(\beta) = \int e^{-\beta E} g(E) dE$, which corresponds to the generalized **evidence** (eq. (38)). The canonical ensemble is an alternative description of thermodynamic states that is based on the *inverse temperature* β rather than the energy level ϵ . In the canonical ensemble, states follow the Boltzmann distribution $p(\Theta | \beta) = \exp\{-\beta E(\Theta)\} / Z(\beta)$. NS in essence tracks the cumulative density of states via:

$$X(L) = \int_{E \leq -\log L} g(E) dE,$$

The practical approximation corresponding to eqs. (13) and (14) is

$$Z(\beta) = \sum e^{-\beta E_i} (X_{i-1} - X_{i+1}) / 2.$$

During NS, states are generated from the **prior** constrained by an upper energy limit $\epsilon = -\log L^*$, which can be achieved with any number of techniques, such as simple rejection sampling, Galilean Monte Carlo and Hamiltonian dynamics [44] or Creutz' microcanonical demon algorithm [45, 46]. The information entropy of the **constrained prior** is the *volume entropy* $\log X(\epsilon)$ (also known as Gibbs entropy). As NS progresses from one energy limit ϵ to the next $\epsilon' < \epsilon$, the volume entropy changes by $\Delta H = \log[X(\epsilon) / X(\epsilon')]$ at a rate that is constant on average: $\langle \Delta H \rangle = 1/n_{\text{live}}$.

Widely-used tempering methods such as simulated annealing work in the canonical ensemble and use the inverse temperature β as a control parameter to weight each **microstates**. Alternatives include the Wang-Landau method [47] and NS, both of which use energy as a control parameter. In contrast to the ensemble property β , a key advantage of $E(\Theta)$ is that it can be evaluated for a single **microstate**. The Wang-Landau method uses a fixed, predefined set of energy bins, while NS constructs a sequence of energy levels at runtime. The sequence is optimal in that it achieves constant thermodynamic speed because changes in volume entropy are constant on average. Therefore, NS elegantly avoids a major problem of canonical annealing methods: designing a good temperature schedule.

represented graphically in fig. 2a.

There is thus a trade-off between run-time and uncertainty. Despite that, there isn't a straight-forward method for choos-

Box 2 | 数学细节

NS的思想是将等式(1)转换为等式(8)，后者可以使用描述的蒙特卡洛方法与主动样本更有效地近似：

$$Z = \int_{\Omega} L(\Theta) d\mu(\Theta) = \int_0^1 \tilde{L}(X) dX,$$

其中 Ω 是参数域，而 $\tilde{L}$ 是 L 的重载形式，如后所述。为了实现这种变换，我们定义了生存函数 $X: \mathbb{R} \rightarrow [0, 1]$ $X(\lambda) = \mu(\{z \in \Omega: L(z) > \lambda\})$ ，即 μ -测度，它表示 λ -超级集。然后，我们引入一个映射 $\Phi: \Omega \rightarrow [0, 1]$ ，

$$\Phi(\Theta) = X(L(\Theta)) = \mu(\{z \in \Omega: L(z) > L(\Theta)\})$$

这个映射 Φ 是允许我们将积分从 Ω 转移到 $[0, 1]$ ，的变换，这是通过在 $[0, 1]$ 上的推前测度 $\mu \circ \Phi^{-1}$ 实现的。现在，我们可以定义

$$\tilde{L}: [0, 1] \rightarrow \mathbb{R}, \quad \tilde{L}(\xi) = \sup\{\lambda \in \text{Im } L: X(\lambda) > \xi\}.$$

因此，上述积分变换为 Z 提供了一种替代的表征，这是因为它成立，

$$\begin{aligned} \int_{\Omega} L(\Theta) d\mu(\Theta) &= \int_{\Omega} \tilde{L}(X(L(\Theta))) d\mu(\Theta) \\ &= \int_{\Omega} (\tilde{L} \circ \Phi)(\Theta) d\mu(\Theta) = \int_{[0,1]} \tilde{L}(x) d(\mu \circ \Phi^{-1})(x) \\ &= \int_0^1 \tilde{L}(x) dx \end{aligned}$$

在这里， $\tilde{L}$ 是在 L 满足适当假设下的 X 的广义逆，参见参考文献 [15, 16]。这种积分变换至少在 L 没有正先验测度的平台的情况下成立，这是因为在这种情况下 $\mu \circ \Phi^{-1}$ 确实是 $[0, 1]$ ，上的均匀测度，即 $d\mu \circ \Phi^{-1}(x) = dx$ 。

如果 L 存在一个质量不可忽略的平台，即存在一个水平 λ^* 使得 $\mu(\{z \in \Omega: L(z) = \lambda^*\}) > 0$ ，那么推导会更加具有挑战性，另见限制和优化。

2 实验研究

在本节中，我们将讨论如何实现 NS，包括选择活点数量等考虑因素，如从约束先验中抽取新的活点、并行化和决定何时停止。

2.1 活点数量的选择

如《配方》所述，活点数量控制着NS运行期间指数压缩的速率；我们每次迭代向内压缩约 $\Delta \log X \approx 1/n_{\text{live}}$ 。这意味着运行时间按 $\mathcal{O}(n_{\text{live}})$ （公式(19)）缩放，证据积分（公式(21)）的主导不确定性以及后验尺度按 $\mathcal{O}(1/\sqrt{n_{\text{live}}})$ 缩放。上述考虑是

方框 3 | 统计力学类比

贝叶斯推理与统计力学之间存在强烈的类比关系。这表明NS可能有助于探索通常属于统计力学分析范畴的问题。考虑 Θ 一个微态和 $E(\Theta) = -\log L(\Theta)$ 其能量。那么，微正则系综包括所有满足 $E(\Theta) = \epsilon$ 的状态，其中 ϵ 是一个恒定的能量水平。给定能量 ϵ 对应的态空间体积由态密度给出，

$$g(\epsilon) = \int \delta(E(\Theta) - \epsilon) \pi(\Theta) d\Theta$$

将先验视为均匀对应于遍历假设，即每个由适用守恒定律允许的微观状态被观测到的概率相等。态密度的拉普拉斯变换是配分函数 $Z(\beta) = \int e^{-\beta E} g(E) dE$ ，它对应于广义证据（公式 (38)）。配分函数是热力学状态的一种替代描述，它基于逆温度 β 而不是能量级 ϵ 。在配分函数中，状态遵循玻尔兹曼分布 $p(\Theta | \beta) = \exp\{-\beta E(\Theta)\} / Z(\beta)$ 。NS 本质上通过以下方式跟踪态密度的累积：

$$X(L) = \int_{E \leq -\log L} g(E) dE,$$

式(13)和式(14)对应的实用近似是

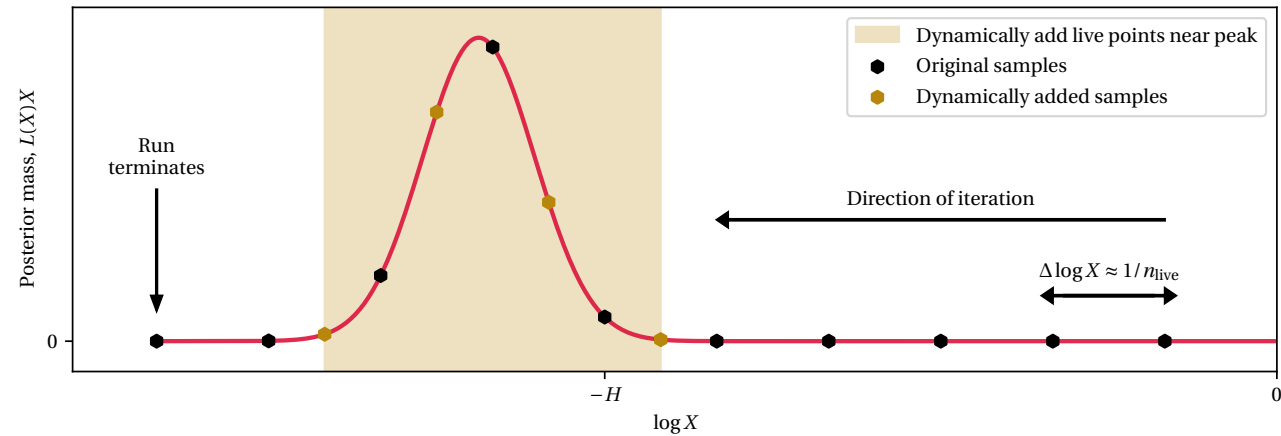
$$Z(\beta) = \sum e^{-\beta E_i} (X_{i-1} - X_{i+1}) / 2.$$

在NS过程中，状态是从受上限能量 $\epsilon = -\log L^*$ 约束的先验分布中生成的，这可以通过任何数量的技术实现，例如简单的拒绝采样、伽利略蒙特卡洛和哈密顿动力学 [44] 或Creutz的微正则 demon 算法 [45, 46]。受约束先验的信息熵是体积熵 $\log X(\epsilon)$ （也称为吉布斯熵）。随着NS从一个能量上限 ϵ 进展到下一个 $\epsilon' < \epsilon$ ，体积熵的变化率 $\Delta H = \log[X(\epsilon) / X(\epsilon')]$ 在平均意义上是恒定的： $\langle \Delta H \rangle = 1/n_{\text{live}}$

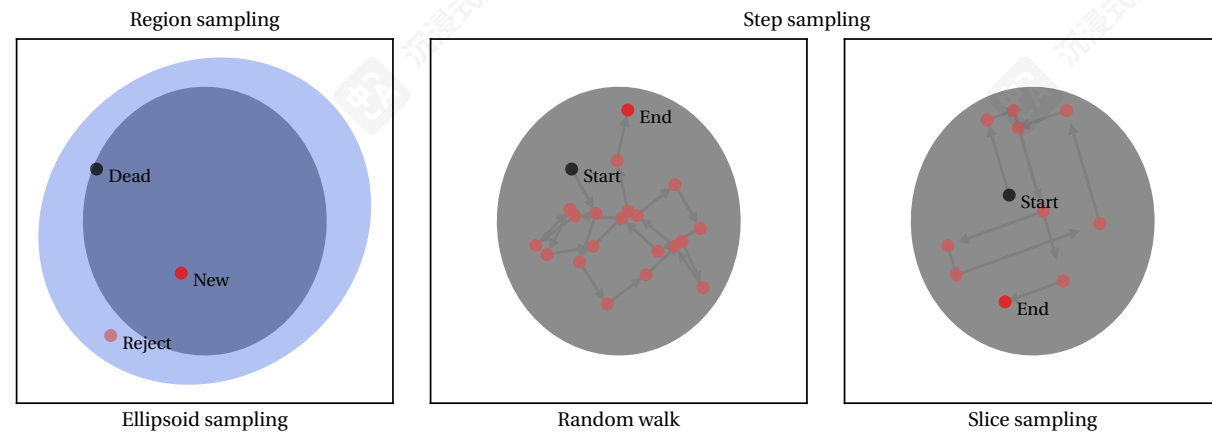
广泛使用的退火方法（如模拟退火）工作在正则系综中，并使用逆温度 β 作为控制参数来加权每个微态。替代方法包括王-兰道方法 [47] 和NS，两者都使用能量作为控制参数。与系综特性 β 相比， $E(\Theta)$ 的一个主要优势在于它可以针对单个微态进行评估。王-兰道方法使用一组固定的、预定义的能量箱，而NS在运行时构建能量级序列。该序列是最佳的，因为它实现了恒定的热力学速度，因为体积熵的变化在平均意义上是恒定的。因此，NS巧妙地避免了正则退火方法的一个主要问题：设计良好的温度调度。

在图2a中用图形表示。

因此，运行时和不确定性之间存在权衡。尽管如此，目前还没有一种直接的方法来选择



a | **Schematic representation of an NS run.** The curve $L(X)X$ shows the relative posterior mass, the bulk of which lies in a tiny fraction e^{-H} of the volume. Most of the original samples lie in regions with negligible posterior mass. In dynamic NS, we add samples near the peak.



b | **Examples of strategies for sampling from the constrained prior.** We must sample from the true iso-likelihood contour (grey ellipse). In region sampling (left) we bound the existing live points (blue ellipse) and draw a new sample from within that bound; some proposals may be rejected. In step sampling (right) we select a live point and perform a sequence of steps inside the contour to obtain an independent draw.

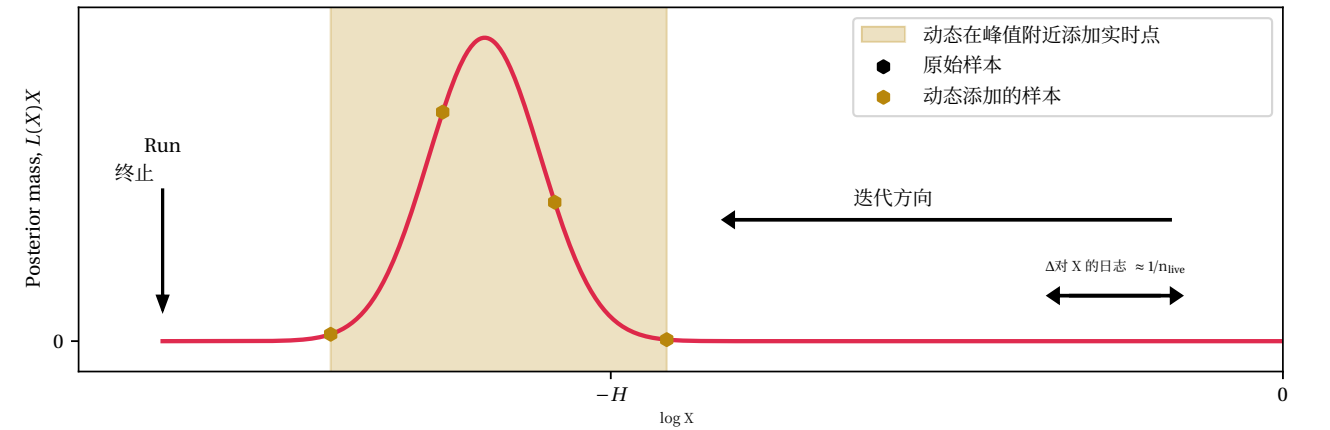
Figure 2 | **Experimentation in NS.** (a) Illustration of an NS run and (b) strategies for sampling from the constrained prior.

ing the number of live points if the required compression isn't known ahead of time. The number should be chosen, furthermore, bearing in mind the alternative role that it plays as a resolution parameter for NS [48, 49], especially in multi-modal problems. In particular, n_{live} should be large enough that at any time the constrained prior splits into disjoint modes, at least one live point lies inside the footprint of each mode. As a rough rule of thumb, if the constrained prior occupies a total volume X , only modes with a footprint greater than about X/n_{live} may be reliably found [8, 49]. This defines a resolution down to which the posterior is reliably sampled. Modes with smaller footprints are typically not located and correctly sampled, and hence will also not contribute to the evidence estimate. Moreover, to sample reliably and efficiently from the constrained prior, it is usually advisable that n_{live} exceeds the dimensionality of the parameter space.

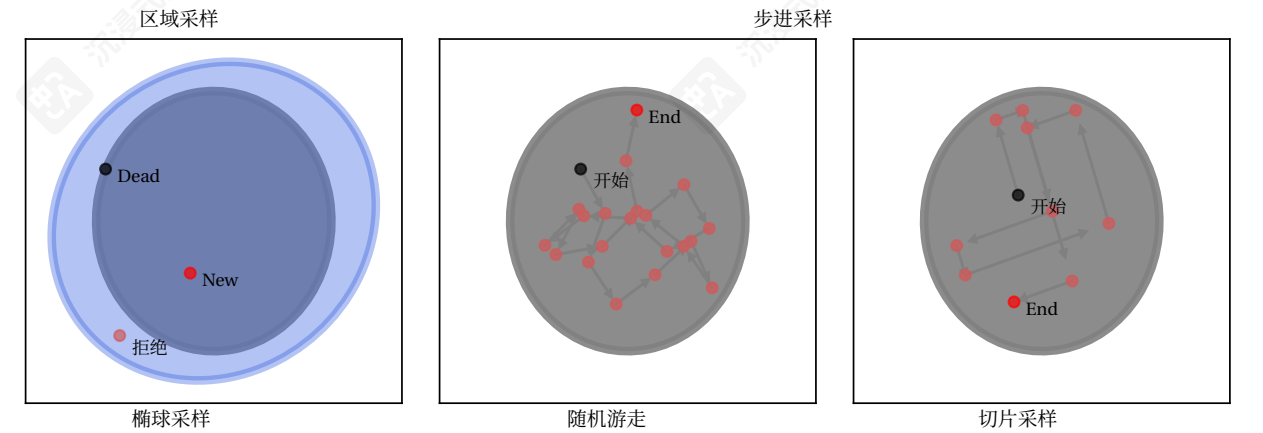
2.2 Dynamic nested sampling

We have so far only considered NS with a fixed number of live points, and noted that the uncertainties in both posterior distributions and evidence estimates are reduced by increasing this number. However, the evidence depends on an accurate estimate of the total compression when we reach the posterior bulk. The posterior, on the other hand, depends only on an accurate estimate of the relative compression once inside the posterior bulk. The former uncertainty cancels in the posterior weights in eq. (15), as they are invariant under rescaling the estimates of the volume variable. This reflects the fact that whereas the evidence may depend strongly on the size of the prior, the posterior usually depends only weakly on its shape.

We are thus motivated to consider a *dynamic* number of live points to efficiently reduce uncertainties in parameter inference. As shown in fig. 2a, we may quickly compress to the posterior bulk using few live points. Upon reaching it, we may increase the number of live points, reducing uncertainty in the bulk of the posterior mass. We denote schemes that vary the number



a | NS运行的概念图示。曲线 $L(X)X$ 显示了相对后验质量，其中大部分位于一个极小体积 e^{-H} 内。大部分原始样本位于后验质量可忽略不计的区域。在动态NS中，我们在峰值附近添加样本。



b | 从约束先验中进行抽样的策略示例。我们必须从真实的等似轮廓（灰色椭圆）中进行抽样。在区域抽样（左图）中，我们限制现有的活点（蓝色椭圆），并在该限制内绘制一个新样本；某些提议可能会被拒绝。在步进抽样（右图）中，我们选择一个活点，并在轮廓内执行一系列步进以获得一个独立的抽样。

图 2 | NS中的实验。(a) NS运行图示和 (b) 从约束先验中进行抽样的策略。

因此，运行时和不确定性之间存在权衡。尽管如此，如果事先不知道所需的压缩量，目前还没有直接的方法来选择活点的数量。此外，选择数量时还应考虑到它作为NS [48, 49]，的分辨率参数的替代作用，尤其是在多模态问题中。具体来说， n_{live} 应该足够大，以确保在任何时候约束先验都会分裂成不同的模式，并且至少有一个活点位于每个模式的影响范围内。作为一个粗略的经验法则，如果约束先验占据的总体积为 X ，那么只有那些影响范围大于约 X/n_{live} 的模式才可能被可靠地找到 [8, 49]。这定义了后验可以被可靠采样的分辨率极限。影响范围较小的模式通常不会被定位和正确采样，因此也不会对证据估计做出贡献。此外，为了可靠且高效地从约束先验中采样，通常建议 n_{live} 超过参数空间的维度。

2.2 动态嵌套抽样

迄今为止，我们仅考虑了具有固定活点数的 NS，并注意到通过增加该数量，后验分布和证据估计中的不确定性都会减少。然而，当达到后验主体时，证据取决于对总压缩的准确估计。另一方面，后验主体本身仅取决于进入后验主体后对相对压缩的准确估计。前者不确定性在等式 (15) 的后验权重中抵消，因为它们在体积变量估计的缩放下保持不变。这反映了这样一个事实：虽然证据可能强烈依赖于先验的大小，但后验通常仅微弱依赖于其形状。

因此，我们被激励去考虑动态数量的活点，以高效地减少参数推断中的不确定性。如图2a所示，我们可以使用少量活点快速压缩到后验主体。到达主体后，我们可以增加活点的数量，从而减少后验质量主体中的不确定性。我们表示那些数量会变化的方案

Box 4 | Sequential Monte Carlo

Connections between NS and a form of rare-event SMC sampler [50] can be exploited to develop an NS-SMC algorithm with desirable theoretical properties [21].

SMC samplers evolve n samples (referred to as “particles”) through a series of distributions π_t for $t = 0, \dots, T$ using reweighting, resampling and mutation steps. Of particular interest is a form of rare-event SMC [50] that involves sampling from the sequentially **constrained prior** distribution. This SMC sampler has the sequence of distributions

$$\eta_t \propto \mathbb{1}(L(\Theta) \geq L_t^*) \pi(\Theta),$$

where $\mathbb{1}(\cdot)$ denotes an **indicator function** and $0 = L_0^* < \dots < L_t^* < \dots < L_{T+1}^* = \infty$. Each iteration t of the sampler involves sampling n particles with replacement from the current set of points satisfying $L(\Theta) \geq L_t^*$ and then diversifying those particles through several iterations of an η_t -invariant **MCMC** kernel.

Noticing the similarities between the two algorithms, ref. [21] built upon these rare-event SMC samplers to create the NS-SMC algorithm for evidence and posterior estimation. Approximation of Z and P in NS-SMC is done by weighting the particles from iteration t to target a shell of the **posterior**,

$$P_t \propto \mathbb{1}(L_t^* < L(\Theta) \leq L_{t+1}^*) L(\Theta) \pi(\Theta).$$

While original NS and NS-SMC both sample from the sequentially **constrained prior** distribution, there are several key differences. Crucially, NS-SMC uses weights based on importance sampling rather than numerical quadrature, it naturally handles the region beyond the largest threshold with no truncation error and it uses a different sampling mechanism. As a result, NS-SMC avoids the issues with bias in NS. Under mild conditions, including the situation where a finite number of **MCMC** steps are used, NS-SMC produces unbiased and consistent estimates of Z and consistent estimates of P as $n \rightarrow \infty$ [21]. Practically, NS can result in noticeably biased estimates of the **evidence** compared to NS-SMC for the same computational cost when the MCMC kernel is inefficient.

of live points as *dynamic* NS [51]. Open-source dynamic NS software packages include dynesty [52] and dyPolyChord [53]; see table 2. The gains are greatest in problems with substantial compression to the **posterior bulk**.

In dynamic NS schemes, the number of live points can be automatically adjusted to maximize a user-specified objective for a fixed computational budget. Usually, the run starts with an exploratory NS run using a constant number of live points. We spend the remaining computational budget by repeatedly increasing the number of live points in the most important regions of volume judged according to the objective (for example the shaded region in fig. 2a). Because dynamic NS re-winds a run and adds extra samples anywhere, running for longer reduces uncertainties and increases the effective sample size, unlike in ordinary NS.

In the original dynamic NS algorithm [51], a user specifies

their objective by assigning a relative importance to reducing uncertainties in the **posterior** and the **evidence**. When focusing on **posterior** inferences, dynamic NS can achieve orders of magnitude reductions in computational cost for a fixed uncertainty. The approach can also improve **evidence** calculation accuracy for a fixed number of samples, and can improve **posterior** inferences and **evidence** calculations simultaneously. This objective was generalised by the reactive NS [54] variant of dynamic NS. This considers the computation as a graph, with the nodes being the live and dead points, and edges indicating the replacement of a point by another. Multiple agents can then add live points (edges) where needed, and optimize towards additional goals, such as the effective sample size, or the number of samples per cluster. Lastly, we note a significant variant of NS, diffusive NS [55], in which the number of live points at a given **likelihood** threshold can change. In this variant, random walks starting from the existing live points are permitted to step down as well as up in **likelihood**, refining the typical **likelihood** in a volume range X .

2.3 Exploration strategies

NS progresses by replacing live points by independent samples drawn from the **constrained prior** in eq. (9), that is the prior restricted to regions in which the **likelihood** exceeds a threshold. This is the major difficulty in efficiently and reliably implementing NS, especially in **multi-modal** problems. Provided one successfully samples from the constrained prior, however, NS works identically in **uni-modal** and **multi-modal** settings.

Whilst we could simply sample from the entire prior until we find a sample for which the **likelihood** exceeds the threshold, this rapidly becomes incredibly inefficient due to the exponential reduction in the volume contained within the **constrained prior** at every iteration. Fortunately, the current set of live points and the estimate of the volume enclosed by the contour may guide our search for new live points. There are two main classes of methods for sampling from the **constrained prior** [56]: region samplers and step samplers. They are illustrated in fig. 2b. Analogous to the choices of **transition kernels** in **MCMC**, the choices here lead to various flavors of NS with different performance characteristics and different behaviour as dimension grows. Although a priori they require just as much tuning as **MCMC**, the live points allow them to build proposal structures and apply clustering algorithms (analogously to ensemble samplers [57]), such that NS is naturally amenable to being robustly self-tuning. As they are guided by them, their reliability and efficiency usually improve when the number of live points is increased; see ref. [54, 58] for numerical investigations. For both region and step samplers, if no current live points lie inside a mode in a **multi-modal** problem, that region of the constrained prior almost certainly won’t be sampled, i.e., they miss that mode (see section 2.1).

Both samplers usually operate in the hypercube, a parameterization in which the prior is uniform over a unit hypercube (though this is not a requirement in NS). This slightly simplifies the problem to that of sampling uniformly from within a contour defined by the threshold. This is usually achieved by the

Box 4 | 序贯蒙特卡洛

NS与一种稀有事件SMC采样器 [50]之间的连接可以被利用来开发具有理想理论性质的NS-SMC算法 [21]。

SMC采样器通过一系列分布 π_t 对 $t = 0, \dots, T$ 的 n 个样本（称为“粒子”）进行演化，使用重加权、重采样和变异步骤。特别值得关注的是一种涉及从顺序约束先验分布中采样的稀有事件SMC [50] 形式。这种SMC采样器具有一系列分布

$$\eta_t \propto \mathbb{1}(L(\Theta) \geq L_t^*) \pi(\Theta),$$

其中 $\mathbb{1}(\cdot)$ 表示指示函数， $0 = L_0^* < \dots < L_t^* < \dots < L_{T+1}^* = \infty$ 。采样器的每次迭代 t 都涉及从满足 $L(\Theta) \geq L_t^*$ 的当前点集中有放回地采样 n 个粒子，然后通过多次迭代 η_t 不变的MCMC核来扩展这些粒子。

注意到两种算法之间的相似性，ref [21] 基于这些稀有事件SMC采样器构建了用于证据和后验估计的NS-SMC算法。NS-SMC中 Z 和 P 的近似通过加权迭代 t 的粒子来瞄准后验的壳层，

$$P_t \propto \mathbb{1}(L_t^* < L(\Theta) \leq L_{t+1}^*) L(\Theta) \pi(\Theta).$$

虽然原始的NS和NS-SMC都从顺序约束先验分布中采样，但存在几个关键差异。关键在于，NS-SMC使用基于重要性采样的权重，而不是数值积分；它能自然地处理最大阈值之外的区域，且没有截断误差；并且它采用不同的采样机制。因此，NS-SMC避免了NS中的偏差问题。在温和条件下，包括使用有限数量的MCMC步骤的情况，NS-SMC产生的 Z 的无偏和一致性估计，以及 P 的一致性估计，当 $n \rightarrow \infty$ [21] 时。实际上，当MCMC核效率低下时，NS在相同的计算成本下，会导致证据估计出现明显的偏差，相比之下NS-SMC则没有。

动态NS [51]的活点。开源的动态NS软件包包括dynesty [52] 和dyPolyChord [53]；见表2。在需要将后验体大幅压缩的问题中，收益最大。

在动态NS方案中，活动点的数量可以自动调整以在固定的计算预算内最大化用户指定的目标。通常，运行开始时会使用恒定的活动点数量进行探索性NS运行。我们通过反复增加在根据目标判断的最重要体积区域（例如图2a中的阴影区域）中的活动点数量来花费剩余的计算预算。由于动态NS会重新启动运行并在任何地方添加额外样本，因此运行时间越长可以减少不确定性并增加有效样本量，这与普通NS不同。

在原始动态NS算法 [51], 中, 用户指定

在原始动态NS算法中, [54] 用户通过为减少后验分布和证据中的不确定性分配相对重要性来指定其目标。当专注于后验推断时, 动态NS可以在固定不确定性下实现计算成本的数个数量级减少。该方法还可以在固定样本数量的情况下提高证据计算精度, 并可以同时改进后验推断和证据计算。这一目标由动态NS的响应式NS [54] 变体推广而来。该变体将计算视为一个图, 其中节点是活点和死点, 边表示用另一个点替换一个点。多个智能体可以在需要的地方添加活点(边), 并优化向附加目标, 例如有效样本量或每个簇的样本数量。最后, 我们注意到NS的一个显著变体——扩散NS [55], 其中在给定似然阈值下的活点数量可以改变。在该变体中, 从现有活点开始的随机游走被允许在似然上下移动, 从而在一个体积范围 X 内细化典型似然。

2.3 探索策略

NS通过用从等式(9)中的约束先验独立抽取的样本替换活动点来推进, 即仅限于似然性超过阈值的区域中的先验。这是高效且可靠地实现NS的主要困难, 尤其是在多模态问题中。然而, 如果成功从约束先验中采样, NS在单模态和多模态设置中工作方式完全相同。

尽管我们可以简单地从整个先验分布中采样, 直到找到一个似然值超过阈值的样本, 但由于在每次迭代中, 受限先验分布中包含的体积呈指数级减少, 这种方法很快变得极其低效。幸运的是, 当前的活点集以及由轮廓包围的体积估计可以指导我们寻找新的活点。从受限先验分布中采样的方法主要分为两类: 区域采样器和步长采样器。它们在图2b中进行了说明。类似于MCMC中转换核的选择, 这些选择导致了具有不同性能特征和不同行为(随着维度增长)的多种NS变体。虽然它们在先验上需要与MCMC一样多的调整, 但活点允许它们构建提议结构并应用聚类算法(类似于集成采样器 [57]), 因此NS自然地易于稳健自调。由于它们受到指导, 当活点数量增加时, 它们的可靠性和效率通常会提高; 参见参考文献 [54, 58] 中的数值研究。对于区域采样器和步长采样器, 如果在一个多模态问题的当前活点不在某个模态内, 那么受限先验分布的那个区域几乎肯定不会被采样, 即它们会遗漏那个模态(参见第2.1节)。

这两种采样器通常在超立方体中运行, 这是一种参数化方法, 其中先验在单位超立方体上是均匀的(尽管在NS中这不是一个要求)。这稍微简化了问题, 使其变为在由阈值定义的轮廓内均匀采样的问题。这通常通过

inverse transformation method. In this case, users specify their **priors** by an inverse-cumulative density function rather than a density function. For example, suppose we desired a Gaussian prior for a parameter, $\theta \sim \mathcal{N}(\mu, \sigma^2)$. We could transform a unit hypercube parameter, $u \sim \mathcal{U}(0, 1)$, using the standard normal distribution's inverse-cumulative density function, Φ^{-1} ,

$$\theta = \mu + \Phi^{-1}(u) \sigma. \quad (22)$$

This in principle allows all manner of **priors**, though see **Limitations and optimizations** for further discussion.

Region sampling

In region sampling, we try to construct a region that bounds the **iso-likelihood contour** defined by the threshold. To find the region, we construct a geometric shape around the current distribution of live points. The shape must contain at least the currently estimated volume. We then draw independent and identically distributed (iid) samples from within that region until we obtain a sample that passes the current **likelihood** threshold. To be confident that the region did not encroach the contour, implementations of region sampling often expand the region by a user-specified factor or by a factor found through cross-validation. For example, dividing the live points into a training and test set and ensuring that the region found from the training set includes points in the test set [56, 59]. The expansion factor improves reliability at the expense of efficiency.

The simplest region sampler would be sampling from the entire unit hypercube, that is the entire prior, which rapidly becomes prohibitively inefficient. Instead, most region samplers attempt to estimate the **constrained prior** by wrapping the live points with one [60, 61] or more overlapping ellipsoids. Using more than one ellipsoid allows complicated and **multi-modal iso-likelihood contours** to be efficiently bounded. An appropriate number of ellipsoids can be found by applying clustering algorithms to the live points to identify distinct **modes**, such as x-means (ref. [62], see **Cosmology**). The shape and location of the ellipsoids may be approximately found from the mean and covariance of the live points they contain, and their volumes may be judiciously expanded by a tuning parameter. This has successfully been implemented in **MultiNest** [48]. Alternatively, MLFriends [56, 59] places an ellipsoid around every live point, and determines the ellipsoid scale by bootstrapping (similar to kernel density estimation with a uniform kernel).

Region samplers suffer from two major limitations when the dimension or complexity of the contour grow. First, the ability to accurately bound a complicated contour strongly depends on the number of live points. For instance, we may fail to identify a substructure with too few points, resulting in overly large regions that bound complicated substructures and in poor efficiency. Alternately, since live points are distributed uniformly within the **constrained prior** rather than near the edge, wrapping a small number of live points can result in overly small estimates of the **constrained prior**. Second, the accuracy and efficiency of region samplers suffer from the **curse of dimensionality**. To see how it strikes, consider region sampling with a single ellipsoid. Suppose that the true contour is a unit hyper-cube.

The smallest ellipsoid that we could construct that enclosed the contour would be a sphere of diameter $\sqrt{D}$. The volume of such a sphere blows up exponentially as dimension increases, leading to $\mathcal{O}(e^{-D})$ efficiency. This follows the general result that an exponentially increasing fraction of volume lies near the boundaries of high-dimensional shapes. As a result, region samplers are efficient and practical only for moderate-to-low dimensionality ($D \lesssim 20$).

Step sampling

Step samplers, by contrast, do not attempt to construct a region that bounds the **iso-likelihood contour** and so avoid some of the issues described above. Instead, they evolve a randomly chosen existing live point through a sequence of steps to a new approximately independent position. The acceptance rule is simply that we accept a transition to Θ' if

$$L(\Theta') > L^* \quad (23)$$

i.e., each step must stay inside the contour (cf. eq. (36)). Such step samplers are akin to running constrained **MCMC** inside NS, and were in fact the originally proposed solution. Strategies for generating new positions vary widely, and currently include:

- random-walk Metropolis [8], where new positions are proposed based on a local target distribution (e.g., a multivariate Gaussian),
- ensemble proposals [63], which use the distribution of all live points to propose new positions using strategies such as differential evolution [64]
- slice sampling variants, where new positions within the **constrained prior** are proposed along a randomly chosen principal axis (slice sampling; [65, 66]), or randomly chosen direction (hit-and-run; [67, 68]), and
- gradient-based trajectories [46, 52, 69–71] that ‘reflect’ off the current **likelihood** constraint.

See ref. [54, 72, 73] for further discussion. In step samplers with a step size parameter, such as random walk Metropolis, the step size is often tuned to ensure a substantial fraction (> 20%) of proposed positions are accepted. This avoids an unacceptable overall sampling efficiency [3, 8]. This tuning may be performed continuously over the course of an NS run [8, 52, 65], although this can introduce biases [71].

It can be challenging to judge the number of steps required to ensure that the live points are independent draws from the constrained prior (see Supplementary Information D for further discussion). While mild violations of this requirement might be inconsequential [52], strong violations lead to unreliable NS **evidence** estimates [74]. The number of iterations required for new samples to approximately de-correlate scales as $\mathcal{O}(D^2)$ for random-walk proposals with tuned step sizes, and $\mathcal{O}(D^1 - D^{5/4})$ for slice sampling or gradient-based trajectories [3]. In practice, the number of steps is often chosen to be aD^b , where a is of order one and b is the anticipated dimensional scaling.

逆变换方法实现。在这种情况下，用户通过逆累积分布函数指定他们的先验，而不是密度函数。例如，假设我们希望为参数 $\theta \sim \mathcal{N}(\mu, \sigma^2)$ 指定高斯先验。我们可以使用标准正态分布的逆累积分布函数 Φ^{-1} 来转换单位超立方体参数 $u \sim \mathcal{U}(0, 1)$,

$$\theta = \mu + \Phi^{-1}(u) \sigma. \quad (22)$$

原则上，这允许所有类型的先验分布， 但请参见限制和优化以进行进一步讨论。

区域采样

在区域采样中，我们尝试构建一个区域，该区域包围由阈值定义的等似然轮廓。为了找到该区域，我们围绕当前活动点的分布构建一个几何形状。该形状必须至少包含当前估计的体积。然后我们从该区域内抽取独立同分布（iid）样本，直到获得一个通过当前似然阈值的样本。为了确保该区域没有侵入轮廓，区域采样的实现通常会通过用户指定的因子或通过交叉验证找到的因子来扩展区域。例如，将活动点分为训练集和测试集，并确保从训练集找到的区域包含测试集中的点 [56, 59]。扩展因子以效率为代价提高了可靠性。

最简单的区域采样器将从整个单位超立方体中采样，即整个先验分布，这会迅速变得低效到不可接受。相反，大多数区域采样器尝试通过用一个或多个重叠的椭球体包裹活动点来估计约束先验分布。使用多个椭球体可以有效地包围复杂和多模态的等似然轮廓。可以通过将聚类算法应用于活动点来识别不同的模式（例如x-means（参考 [62], 参见宇宙学））来找到合适的椭球体数量。椭球体的形状和位置可以通过它们包含的活动点的均值和协方差近似找到，并且它们的体积可以通过调整参数明智地扩展。这已经在MultiNest [48]中成功实现。或者，MLFriends [56, 59] 为每个活动点放置一个椭球体，并通过自举（类似于使用均匀核的核密度估计）确定椭球体尺度）。

当轮廓的维度或复杂度增加时，区域采样器会面临两个主要限制。首先，准确界定复杂轮廓的能力强烈依赖于活动点的数量。例如，如果活动点太少，我们可能无法识别子结构，导致区域过大而包围复杂的子结构，从而效率低下。或者，由于活动点在约束先验中均匀分布而非靠近边缘，少量活动点包裹可能导致约束先验的估计值过小。其次，区域采样器的准确性和效率受维度灾难的影响。要理解其影响，考虑使用单个椭球体的区域采样。假设真实轮廓是一个单位超立方体。

能完全包含轮廓的最小椭球体将是一个直径为 $\sqrt{D}$ 的球体。随着维度增加，这种球体的体积会呈指数级增长，导致 $\mathcal{O}(e^{-D})$ 效率低下。这符合高维形状边界附近体积呈指数增长的普遍规律。因此，区域采样器仅在中等至低维度（ $D \lesssim 20$ ）时高效且实用。

步进采样

相比之下，步进采样器不会尝试构建一个包含等似然轮廓的区域，因此避免了上述一些问题。相反，它们通过一系列步骤将随机选择的现有活动点演化到一个近似独立的新位置。接受规则很简单：如果我们将转换到 Θ' ，则接受该转换

$$L(\Theta') > L^* \quad (23)$$

即，每一步都必须保持在轮廓内（参见公式(36)）。这种步进采样器类似于在NS内部运行约束MCMC，并且实际上是最初提出的解决方案。生成新位置的策略多种多样，目前包括：

- 随机游走Metropolis [8]，其中新位置是基于局部目标分布提出的（例如，多元高斯分布）
（例如，一个多元高斯分布）
多元高斯，
- 使用所有活动点的分布来提出新位置的集成建议 [63]，例如使用差分进化 [64]
- 切片采样变体，其中在约束先验内沿随机选择的主轴提出新位置（切片采样； [65, 66]），或沿随机选择的方向（击中并运行； [67, 68]），以及
- 基于梯度的轨迹 [46, 52, 69–71]，它们会“反射”当前似然约束。

参见参考文献 [54, 72, 73] 以便进一步讨论。在具有步长参数的步长采样器中，例如随机游走Metropolis，步长通常调整以确保相当大比例 (> 20%) 的建议位置被接受。这避免了不可接受的总体采样效率 [3, 8]。这种调整可以在NS运行 [8, 52, 65]，的过程中持续进行，尽管这可能引入偏差 [71]。

要判断确保活点是从约束先验中独立抽取所需的步数可能具有挑战性（有关进一步讨论，请参阅补充信息D）。虽然这种要求的小幅违反可能无足轻重 [52], 但严重违反会导致不可靠的NS证据估计 [74]。对于具有调谐步长的随机游走提议，新样本近似去相关所需的迭代次数按 $\mathcal{O}(D^2)$ 缩放，而对于切片采样或基于梯度的轨迹，则按 $\mathcal{O}(D^1 - D^{5/4})$ 缩放 [3]。在实践中，步数通常选择为 aD^b ，其中a的数量级为1，b是预期的维度缩放。

The computational cost scales linearly with the number of steps. Thus unlike region samplers, step samplers escape the **curse of dimensionality** as their cost shows only polynomial $\mathcal{O}(D^b)$ scaling with dimensionality. Nevertheless, region samplers are often more efficient in low dimensions. As a result, step samplers are more often used when applying NS in high-dimensions ($D \gtrsim 20$).

Table 2 compares the approaches in several publicly available NS implementations that originated in different research fields. For a multitude of programming languages, there are well-documented, free and open source codes. Support for parallelization to computing clusters and checkpointing are also common features. They usually work with the logarithms of the **evidence** and **likelihood**, as the latter may be numerically tiny such that it cannot be represented as a floating point value. In general, sensible defaults for numerous region and step samplers have been found to work across a large variety of problems [52]. We present a simple numerical example in Supplementary Information G.

2.4 Parallelization

To utilize computing resources, we could parallelize the computation of the likelihood function. If that is impossible or impractical, we may wish to design an efficient scheme for parallelizing an NS run itself. As discussed in ref. [51, 52], statistically independent NS runs may be combined into an equivalent NS run with $\sum n_{\text{live}}$ live points. To achieve this, pool the initial live points from the independent runs $a, b, \dots$ together, giving $\sum n_{\text{live}}$ live points drawn from the prior. Remove the worst, supposing it lay at L^* and originated from run a . To draw a replacement from the prior subject to $L > L^*$, simply take the replacement used in run a , since it is already a draw from the prior subject to $L > L^*$. We may continue in this fashion, weaving together independent NS runs to build a new NS run with $\sum n_{\text{live}}$ live points. This allows parallelization of an NS run with n_{live} live points into n_{CPU} independent runs with about $n_{\text{live}}/n_{\text{CPU}}$ live points each. The independent runs themselves proceed linearly and may be ultimately combined, resulting in a speedup of about n_{CPU} . Even simpler, estimates of the **evidence** integrals may be combined by weighted averaging.

However, the reduction in the number of live points per run, $n_{\text{live}}/n_{\text{CPU}}$, impacts the exploration schemes discussed in section 2.3, especially for ellipsoidal rejection sampling where it could lead to inefficient or faulty bounding ellipsoids. It may therefore be desirable to utilize parallelization within individual NS runs. For example, by drawing n_{CPU} candidate replacement points and evaluating their **likelihoods** in parallel. We could subsequently replace the worst n_{CPU} live points at each iteration [75, 76], replace a single live point and consider the other evaluated points at subsequent iterations [65], or replace a single live point and discard as many as $n_{\text{CPU}} - 1$ acceptable candidate points [48]. The latter is wasteful if more than one viable point are likely to be found among the n_{CPU} candidates,

$$\text{Speed-up} \approx \min[n_{\text{CPU}}, 1/\epsilon] \quad (24)$$

If points are considered at subsequent iterations,

$$\text{Speed-up} \approx n_{\text{live}} \log \left(1 + \frac{n_{\text{CPU}}}{n_{\text{live}}} \right) \quad (25)$$

and so a speedup of about n_{CPU} if $n_{\text{CPU}} \ll n_{\text{live}}$. The expression originates from the fact that the threshold increases as the run progresses, meaning that points drawn from the **constrained prior** might not be valid at a subsequent iteration. Lastly, replacing n_{CPU} points in parallel per iteration results in a speedup of about n_{CPU} but increases the variance in the **evidence** integral by about $\sqrt{n_{\text{CPU}}}$. Increasing n_{live} by a factor $\sqrt{n_{\text{CPU}}}$ to maintain the same uncertainty decreases the speedup to about $\sqrt{n_{\text{CPU}}}$.

2.5 Stopping conditions

We must decide when to stop an NS run. The fact that we can only perform a finite number of iterations introduces a truncation error in eq. (13) that we wish to be negligible. Skilling originally proposed to stop NS once we reached the **posterior bulk** at $X \simeq e^{-H}$ at iteration $n_{\text{iter}} \simeq n_{\text{live}}H$, or using an estimate of the remaining **evidence**. Popular NS software later adopted the latter. In **MultiNest**, this was based on the maximum **likelihood** found so far, $\max LX$, whereas **PolyChord** chose the mean **likelihood**, $\bar{LX}$. They stopped once $\Delta Z/Z \leq \text{tol}$, where tol is a user-specified parameter. Upon deciding to stop, the truncation error in the **evidence** was corrected by either adding an estimate of the remaining **evidence** or by killing the live points one by one without replacement and incrementing the **evidence** in the usual manner until no live points remain. The latter is in keeping with the NS approach. For the former the remaining **evidence** may be estimated by $\bar{LX}$. See ref. [43] for further discussion of the statistical properties of this estimate of the remainder. When NS is used to calculate the **partition function** of a material system (box 3), physically motivated stopping conditions can be based on the expected minimum energy (negative $\log$ **likelihood**) or the sampled temperature, which is proportional to the derivative of the limiting energy with respect to NS iteration [77]

None of these approaches guarantees that summation hasn't been terminated too early; there could well be a spike of enormous **likelihood** lurking inward. The computational budget, furthermore, cannot be easily anticipated ahead of time. However, runs that are terminated prematurely may still be used to illustrate what was learned so far about the **posterior**. Proposals to construct termination criteria for a fixed computational budget that result in unbiased estimates of the **evidence** have been suggested [22].

3 Results

NS results in an estimate of the integral in eq. (1) and, in the context of Bayesian statistics, a weighted set of draws from the **posterior** distribution. We discuss some examples in **Applications** and how to report the results in **Reproducibility and data deposition**. The error estimate in eq. (21) depends on the compression and cannot be known ahead of time. If the

计算成本与步数线性缩放。因此与区域采样器不同，步长采样器能够摆脱维度的诅咒，因为它们的成本仅以多项式 $\mathcal{O}(D^b)$ 缩放与维度。尽管如此，区域采样器在低维度中通常更高效。结果，当在高维度中应用NS时，步长采样器更常被使用 ($D \gtrsim 20$)。

表2比较了源自不同研究领域的几种公开可用的NS实现方法。对于多种编程语言，都有文档齐全、免费且开源的代码。支持计算集群并行化和检查点也是常见功能。它们通常使用证据和似然的对数进行计算，因为后者可能非常小，以至于无法表示为浮点数值。一般来说，针对众多区域和步长采样器，已经找到了适用于各种问题的合理默认值 [52]。我们在补充信息G中提供了一个简单的数值示例。

2.4 并行化

为了利用计算资源，我们可以并行化似然函数的计算。如果这不可能或不切实际，我们可能希望设计一个高效的方案来并行化NS运行本身。如参考文献 [51, 52]，中所述，统计独立的NS运行可以组合成一个等效的NS运行，包含 $\{n \sum\}$ 个 $\{n_{\text{live}}\}$ 活点。为此，将独立运行a、b、...的初始活点汇集在一起，得到 $\{n \sum\}$ 个 $\{n_{\text{live}}\}$ 个从先验分布中抽取的活点。移除最差的活点，假设它位于 $L\{n^*\}$ 并源自运行a。根据 $\{n L > L^*\}$ 从先验分布中抽取替代活点，只需使用运行a中使用的替代活点，因为它已经是根据 $\{n L > L^*\}$ 从先验分布中抽取的。我们可以按这种方式继续，将独立的NS运行交织在一起，构建一个新的包含 $\{n \sum\}\{n_{\text{live}}\}$ 个活点的NS运行。这允许将包含 $\{n_{\text{live}}\}$ 个活点的NS运行并行化为 $\{n_{\text{CPU}}\}$ 个独立的运行，每个运行包含约 $\{n_{\text{live}}\}/n_{\text{CPU}}$ 个活点。这些独立的运行本身按线性方式运行，并最终可以组合，从而实现约 $\{n_{\text{CPU}}\}$ 的加速。甚至更简单，证据积分的估计值可以通过加权平均来组合。

然而，每次运行中活点数量的减少，即 $n_{\text{live}}/n_{\text{CPU}}$ ，会影响第 2.3 节中讨论的探索方案，特别是对于椭圆拒绝采样，可能导致低效或错误的边界椭圆。因此，在单个 NS 运行中利用并行化可能是值得考虑的。例如，通过并行绘制 n_{CPU} 个候选替换点并评估它们的似然性。随后，我们可以在每次迭代 [75, 76]，替换最差的 n_{CPU} 个活点，或者替换单个活点并在后续迭代中考虑其他评估点 [65]，或者替换单个活点并丢弃多达 $n_{\text{CPU}} - 1$ 个可接受的候选点 [48]。如果 n_{CPU} 个候选点中很可能存在多个可行点，那么后者会浪费资源，

$$\text{Speed-up} \approx \min[n_{\text{CPU}}, 1/\epsilon] \quad (24)$$

如果点在后续迭代中被考虑，

$$\text{Speed-up} \approx n_{\text{live}} \log \left(1 + \frac{n_{\text{CPU}}}{n_{\text{live}}} \right) \quad (25)$$

因此，如果 $n_{\text{CPU}} \ll n_{\text{live}}$ ，速度提升约为 n_{CPU} 。这个表达式源于阈值随着运行进程的增加而提高的事实，这意味着从约束先验中抽取的点可能在后续迭代中无效。最后，每次迭代并行替换 n_{CPU} 个点会导致速度提升约为 n_{CPU} ，但会增加证据积分的方差约 $\sqrt{n_{\text{CPU}}}$ 。将 n_{live} 增加 $\sqrt{n_{\text{CPU}}}$ 倍以保持相同的 Uncertainty 会降低速度提升至约 $\sqrt{n_{\text{CPU}}}$ 。

2.5 停止条件

我们必须决定何时停止 NS 运行。由于我们只能执行有限次数的迭代，这会在等式 (13) 中引入截断误差，我们希望该误差可以忽略不计。Skilling 最初建议在达到 $X \simeq e^{-H}$ 迭代 $n_{\text{iter}} \simeq n_{\text{live}}H$ 后停止 NS，或使用剩余证据的估计值。流行的 NS 软件后来采用了后者。在 **MultiNest** 中，这是基于迄今为止找到的最大似然值 $\max LX$ ，而 **PolyChord** 则选择了平均似然值 $\bar{LX}$ 。它们在 $\Delta Z/Z \leq \text{tol}$ 时停止，其中 tol 是用户指定的参数。在决定停止后，证据中的截断误差通过添加剩余证据的估计值或逐个无放回地杀死活点并按常规方式递增证据，直到没有活点剩余来纠正。后者符合 NS 方法。对于前者，剩余证据可以通过 LX 估计。参见 ref. [43]，以进一步讨论该剩余估计值的统计特性。当使用 NS 计算材料系统（框 3）的配分函数时，可以基于预期的最小能量（负对数似然）或采样温度（与 NS 迭代 [77] 的极限能量导数成正比）等物理动机的停止条件。

这些方法都无法保证求和没有过早终止；很可能存在一个潜伏在内部的、可能性极高的峰值。此外，计算预算很难提前预估。然而，过早终止的运行仍然可以用来说明目前已知的后验信息。有人提出了构建固定计算预算的终止标准，以获得无偏的似然估计 [22]。

3 结果

NS 方法得到等式 (1) 中积分的估计值，并在贝叶斯统计的背景下，提供后验分布的加权样本集。我们在“应用”部分讨论了一些示例，并在“可重复性与数据沉积”部分说明如何报告结果。等式 (21) 中的误差估计取决于压缩程度，无法提前确定。如果

Code	Methods	Dynamic	Languages	Field	Pub. Year
CosmoNest [60, 61]	ellipsoid	fixed	Fortran	Cosmology	2006
MultiNest [48, 84]	multi-ellipsoid	fixed	Fortran, C/C++, Python	Cosmology	2008
DIAMONDS [249]	multi-ellipsoid	fixed	C++	Astrophysics	2015
nestle [250]	ellipsoid, multi-ellipsoid	fixed	Python	Astrophysics	2015
nessai [90, 91]	normalising flow ellipsoid	fixed	Python	Gravitational waves	2021
(dy)PolyChord [53, 65]	slice	dynamic	Fortran, C/C++, Python	Cosmology	2015
LALInferenceNest [180]	random walk, ensemble, differential evolution	fixed	C	Gravitational waves	2015
Nested_fit [104, 257, 258]	random walk	fixed	Fortran	Atomic physics	2016
cpnest [259]	slice, differential evolution, Gauss, Hamiltonian, ensemble	fixed	Python	Gravitational waves	2017
pymatnest [44]	random walk, Galilean, symplectic Hamiltonian	fixed	Python	Materials	2017
NNest [261]	normalising flow random walk	fixed	Python	Cosmology	2019
DNest5 [55]	user-defined, random walk	diffusive	C++	Astrophysics	2020
BayesianFitting [263]	random walk, slice, Galilean, Gibbs	fixed	Python	Astronomy	2021
dynesty [52]	ellipsoid, multi-ellipsoid, MLFriends & Gauss, slice, Hamiltonian	dynamic	Python	Astrophysics	2020
UltraNest [92]	MLFriends + ellipsoid & Gauss, hit-and-run, slice	reactive	Python, Julia, R, C/C++, Fortran	Astrophysics	2020
jaxns [266]	multi-ellipsoid & slice	fixed	jax	Astronomy	2021

Table 2 | **Comparison of NS codes.** The first two groups are region samplers and step samplers, respectively, whereas the third group offers both. Dynamic implementations allow the number of live points to be changed during a run. We show the language in which the NS code was written followed by any additional languages for which interfaces exist, and the field from which the code originated (though most are general purpose codes).

achieved error is unacceptable, the NS run can be repeated with more live points or combined with a new run. The shape of the [posterior](#) can be found from the [posterior](#) weights by density estimation. There are dedicated software packages for making publication quality figures of marginalized [posterior](#) densities from weighted samples, including [anesthetic](#) [78], [superplot](#) [79], [pippi](#) [80], [dynesty](#) [52], [getdist](#) [81], [corner](#) [82] and [pygtc](#) [83].

There are a few ways to check the results. First, we can check the NS implementation, rather than the particular run. To do so, we can compute the [evidence](#) integral in eq. (1) for problems with known analytic solutions; see ref. [65, 84–86] for examples including a multi-dimensional Gaussian, an eggbox function, the Rosenbrock function [87], Gaussian shells and a mixture of a Gaussian and a log-gamma distribution. Similarly, in some cases, the NS estimates of the volume variable at each iteration, $X(L^*)$, may be checked against analytic results [56]. If discrepancies are found, the implementation is suspect. Alternatively, we can repeat calculations and check whether the distribution of results is consistent with what would be expected if the software was working correctly. Ref. [74] describes procedures for doing this, including tests requiring only two NS runs; these are implemented in [nestcheck](#) [88].

Second, the particular NS run of interest may also be checked

using a test of the [insertion indexes](#) of new live points [89]. If NS draws new live points independently from the [constrained prior](#), as it should, the ranks in [likelihood](#) of each new live point compared to the current live points should be uniformly distributed. This test is implemented in the [anesthetic](#) [78] NS analysis software, which is compatible with [PolyChord](#) and [MultiNest](#), and used on the fly in the [nessai](#) [90, 91] and [UltraNest](#) [92] NS implementations. If this test fails, but other implementation checks pass, the choices of exploration strategy for the problem at hand may be inadequate. Similarly, ref. [93, 94] discuss testing whether live points are uniformly distributed in the unit hypercube in two-dimensional problems. Last, in the context of parameter inference we can compare the posterior samples obtained from different NS implementations or from [MCMC](#) and NS (see for example ref. [95]) or perform [Simulation-Based Calibration](#) to check expected properties of the posterior.

As discussed in [Stopping conditions](#), most NS implementations stop once the estimate of the remaining [evidence](#) appears negligible. This could omit spikes in [likelihood](#) lying inside the remaining unexplored volume. If that is a concern, it may be beneficial to optimize the [likelihood](#) using a local optimization algorithm starting from the live point with the greatest [likelihood](#). The run may be restarted with stricter stopping conditions if local optimization finds a maximum [likelihood](#) that is

Code	方法	动态	语言	领域	出版年份
CosmoNest [60, 61]	椭球体	固定	Fortran	宇宙学	2006
MultiNest [48, 84]	多椭球体	固定	Fortran , C/C++, Python	宇宙学	2008
钻石 [249]	多椭球体	固定	C++	天体物理学	2015
雀巢 [250]	椭球体, 多椭球体	固定	Python	天体物理学	2015
nessai [90, 91]	归一化流椭球体	固定	Python	引力波	2021
(dy)PolyChord [53, 65]	切片	动态	Fortran , C/C++, Python	宇宙学	2015
LALInferenceNest [180]	随机游走 , 集成 , 差分进化	固定	C	引力波	2015
嵌套 _适配 [104, 257, 258]	随机游走	固定	Fortran	原子物理	2016
cpnest [259]	切片, 差分进化 , 高斯, 哈密顿, 集合	固定	Python	引力波	2017
pymatnest [44]	随机游走, 伽利略, 辛哈密顿	固定	Python	材料	2017
NNest [261]	normalising flow random walk	fixed	Python	Cosmology	2019
DNest5 [55]	用户定义, 随机游走	扩散	C++	天体物理学	2020
贝叶斯拟合 [263]	随机游走, 切片, 伽利略, 吉布斯	固定	Python	天文学	2021
dynesty [52]	椭球体, 多椭球体, MLFriends & 高斯, 切片, 哈密顿	动态	Python	天体物理学	2020
UltraNest [92]	MLFriends + 椭球 & 高斯, 逃逸切割	反应式	Python, Julia, R, C/C++, Fortran	天体物理学	2020
jaxns [266]	多椭球体 & 切片	固定	jax	天文学	2021

表 2 | NS码比较。前两组分别是区域采样器和步长采样器，而第三组两者兼具。动态实现允许在运行过程中更改活点数量。我们展示了NS码所使用的语言，随后是存在接口的任何其他语言，以及代码的来源领域（尽管大多数是通用代码）。

达到的误差不可接受，NS 运行可以重复使用更多活点，或与新运行结合。后验分布的形状可以通过密度估计从后验权重中找到。有专门的软件包可用于从加权样本中生成边际后验密度的高质量出版图，包括 [anesthetic](#) [78], [superplot](#) [79], [pippi](#) [80], [dynesty](#) [52], [getdist](#) [81], [corner](#) [82]和 [pygtc](#) [83]。

有几种方法可以检查结果。首先，我们可以检查NS实现，而不是特定的运行。为此，我们可以对具有已知解析解的问题计算等式(1)中的证据积分；参见参考文献 [65, 84–86]，其中包含多维高斯分布、蛋盒函数、Rosenbrock函数 [87], 高斯壳以及高斯和对数伽马分布的混合示例。类似地，在某些情况下，可以检查每次迭代中体积变量NS估计 $X(L^*)$ 与解析结果 [56]是否一致。如果发现差异，则实现可能存在问题。或者，我们可以重复计算并检查结果的分布是否与软件正常工作时预期的结果一致。参考文献 [74] 描述了执行此操作的程序，包括只需要两次NS运行的测试；这些测试在[nestcheck](#) [88]中实现。

其次，也可以检查感兴趣的特定NS运行

使用新活点的插入索引 [89]的测试。如果NS独立地从约束先验中抽取新活点，就像它应该做的那样，每个新活点与当前活点相比在似然度中的排名应该均匀分布。此测试在[麻醉](#) [78] NS分析软件中实现，该软件与PolyChord和MultiNest兼容，并在 [nessai](#) [90, 91] 和[UltraNest](#) [92] NS实现中动态使用。如果此测试失败，但其他实现检查通过，那么针对所处理问题的探索策略选择可能不足。类似地，参考文献 [93, 94] 讨论了在二维问题中测试活点是否在单位超立方体中均匀分布。最后，在参数推断的背景下，我们可以比较从不同的NS实现或从MCMC和NS获得的后验样本（例如参考文献 [95]），或执行基于模拟的校准来检查后验的预期属性。

如《停止条件》所述，大多数NS实现会在剩余证据的估计值看起来可以忽略不计时停止。这可能会忽略剩余未探索体积中存在的似然峰值。如果这是一个问题，优化似然可能是有益的，可以使用局部优化算法从似然最大的活点开始。如果局部优化找到的最大似然值比NS运行期间找到的值大几个数量级，则可以重新启动运行并使用更严格的停止条件。

orders of magnitude greater than that found during the NS run.

Lastly, in many settings we may be concerned about the possibility that **modes** were missed. While the tests described above may indicate whether implementation errors are present, it is in general impossible to know for certain whether all **modes** were identified. We can inspect the **posterior** by eye, or perform a **mode** identification algorithm on the **posterior** samples. If the expected number of **modes** was known and **modes** are evidently missing, the number of live points should be increased.

4 Applications

Here we present the most established NS applications, highlighting its advantages in each case. Besides these established applications, NS is beginning to be utilized in many other areas, including signal processing [96], phylogenetics [97], systems biology [98, 99], acoustics [100, 101], nuclear physics [102, 103], atomic physics [104–108], exoplanet searches [109–113], and geology [114].

4.1 Cosmology

The rapid spread of Bayesian methods in cosmology [9] in the early 2000s was generated by the growth of data, specifically the new cosmic microwave background (CMB) temperature power spectrum measured by the Wilkinson Microwave Anisotropy Probe (WMAP) [115], and the type-Ia supernovae distance measurements [116, 117]. However, these new and powerful cosmological datasets had raised difficult questions regarding the cosmological model. The favoured model of the Universe included a mysterious accelerating force, the dark energy, which accounted for 70% of the energy density today. And the initial spectrum of density fluctuations in the early Universe, which were shown to be Gaussian and adiabatic by WMAP, indicated a period of accelerated expansion at very early times (about 10^{-23} seconds after the Big Bang), known as cosmic inflation [118]. Finally, there was still the question of the missing mass of the Universe, which generates the gravitational fields required for cosmic structure, known as cold dark matter (CDM). All three of these phenomena (cold dark matter, dark energy, and cosmic inflation) had proposed explanations from the field of high-energy theoretical physics, and these model predictions could be combined with the new wealth of cosmological data to be evaluated with respect to each other, model by model. Thus model selection in general, and the computation of the Bayesian **evidence** using NS specifically, became a tool of choice.

The simplest model of cosmic inflation is one driven by a single scalar field, a particle physics object similar (but not identical) to the Higgs boson. The behaviour of the scalar field driving inflation is determined by its potential $V(\phi)$ (ϕ being the value of the scalar field) [118, 119]. The formulation of $V(\phi)$ (as a function of some set of parameters Θ) determines the duration of inflation along with the distribution of anisotropies found in the CMB. Ref. [120] performed a detailed **Bayesian model comparison** between 193 inflationary models using the region

sampler **MultiNest**. They used the cosmological observations (including CMB data from WMAP mission) to discriminate between alternative models for $V(\phi)$ and found slight preference for so-called Small Field Inflation (SFI) models over Large Field Inflation (LFI) models.

Another area where NS has been used extensively in cosmology is the modelling of galaxy clusters [121, 122]. Clusters of galaxies are the most massive gravitationally bound objects in the Universe and therefore, can be used to trace the formation of large scale structure in the Universe. Galaxy clusters can be observed through several methods including X-ray observations, weak gravitational lensing and by exploiting the Sunyaev–Zel’dovich effect (SZ). Weak gravitational lensing involves the distortion of the images of the background galaxies by the presence of a large mass lying along the line-of-sight. Weak lensing allows to probe the total mass distribution, including the dark matter, of the galaxy cluster. Ref. [123] presented a Bayesian approach using NS for the detection of galaxy clusters in SZ data, including estimation of parameters associated with the physical model assumed for the cluster and quantification of the detection using Bayesian model selection, making use of the statistics of the CMB anisotropies in the **likelihood** function. Ref. [124] presented a joint Bayesian analysis of weak lensing and SZ observations of several galaxy clusters, allowing for the estimation of gas fraction of individual clusters. In modern weak lensing surveys [125] such as DES [126] and KiDS [127], NS forms a critical part of their parameter estimation, model comparison and tension quantification pipelines [128] (fig. 3).

The expansion history of the Universe can also be measured through the use of distance measurements, such as observations of type Ia supernovae (SNIa), which can be used as ‘standard candles’. The important results from the Supernova Cosmology Project and the High-Z Supernova Search Team [116, 117] presented **evidence** for the accelerated expansion of the Universe, requiring the existence of a mysterious dark energy acting against gravity. Estimation of cosmological parameters from the observations of SNIa light curves has historically been done using a χ^2 approach (see e.g. [129]) which lacks a rigorous approach for the determination of systematic uncertainties. Ref. [130] introduced a Bayesian hierarchical model for the determination of cosmological parameters. By using NS, they showed that their principled Bayesian approach delivered tighter statistical constraints on the cosmological parameters over 90% of the time, reduced statistical bias by a factor ~ 2 -3 times and that it had better coverage properties than the standard χ^2 approach.

The measurement of the CMB anisotropies by Planck [132] was an increase in the statistical power over the previous experiment (WMAP), but required more sophisticated modeling of galactic foregrounds and instrumental calibration, introducing $\mathcal{O}(20)$ “nuisance parameters”. Metropolis-Hastings techniques for parameter estimation were able to accommodate these by exploiting the fact that these parameters were “fast” in comparison to the cosmological ones; namely, by caching results from previous calculations, these parameters can be changed with negligible computational cost providing the cosmological

大几个数量级。

最后，在许多情况下，我们可能担心遗漏了某些模式。虽然上述测试可以表明是否存在实现错误，但通常无法确定是否所有模式都已识别。我们可以通过肉眼检查后验分布，或对后验样本执行模式识别算法。如果已知预期的模式数量且明显存在缺失模式，则应增加活动点的数量。

4 应用

这里我们介绍了最成熟的NS应用，并突出其在每个案例中的优势。除了这些成熟的应用外，NS正开始被应用于许多其他领域，包括信号处理 [96]，系统发育学 [97]，系统生物学 [98, 99]，声学 [100, 101]，核物理学 [102, 103]，原子物理学 [104–108]，系外行星搜索 [109–113]，以及地质学 [114]。

4.1 宇宙学

贝叶斯方法在宇宙学 [9] 21世纪初的迅速传播是由数据增长驱动的，特别是威尔金森微波各向异性探测器（WMAP） [115]，测量的新宇宙微波背景（CMB）温度功率谱以及Ia型超新星距离测量 [116, 117]。然而，这些新的强大宇宙学数据引发了关于宇宙学模型的难题。宇宙的 **favored** 模型包含一种神秘的加速力量——暗能量，它解释了当今70%的能量密度。此外，早期宇宙密度涨落的初始谱（由WMAP证明为高斯和绝热的）表明在非常早期（大爆炸后约 10^{-23} 秒）存在加速膨胀时期，称为宇宙暴胀 [118]。最后，仍然存在关于宇宙缺失质量的问题，它产生了形成宇宙结构所需的引力场，称为冷暗物质（CDM）。这三种现象（冷暗物质、暗能量和宇宙暴胀）都从高能理论物理学领域提出了解释，并且这些模型预测可以与新的宇宙学数据相结合，逐个模型进行评估。因此，在一般情况下，以及使用NS计算贝叶斯证据时，模型选择成为了一种首选工具。

宇宙暴胀最简单的模型是由单个标量场驱动，这是一种类似（但并不完全相同）希格斯玻色子的粒子物理学对象。驱动暴胀的标量场的行为由其势能 $V(\phi)$ 决定（ ϕ 是标量场的值） [118, 119]。 $V(\phi)$ （作为某些参数集 Θ 的函数）的表述决定了暴胀的持续时间以及CMB中各向异性的分布。Ref. [120] 使用区域sampler MultiNest 对193个暴胀模型进行了详细的贝叶斯模型比较。

他们利用宇宙学观测数据（包括来自WMAP任务的CMB数据）来区分 $V(\phi)$ 的替代模型，并发现对所谓的“小场暴胀”（SFI）模型比对“大场暴胀”（LFI）模型有轻微的偏好。

另一个在宇宙学中广泛使用NS的领域是星系团建模 [121, 122]。星系团是宇宙中最庞大的引力束缚天体，因此可以用来追溯宇宙大尺度结构的形成。通过X射线观测、弱引力透镜和利用太阳耀斑-齐尔多维奇效应（SZ）等多种方法可以观测星系团。弱引力透镜涉及背景星系图像因视线方向上存在大质量而发生的扭曲。弱引力透镜技术可以探测星系团的总质量分布，包括暗物质。参考文献 [123] 提出了一种基于NS的贝叶斯方法，用于在SZ数据中探测星系团，包括估计星系团物理模型相关的参数，并利用贝叶斯模型选择量化探测结果，该方法在似然函数中使用了宇宙微波背景辐射各向异性的统计信息。参考文献 [124] 提出了一种对多个星系团的弱引力透镜和SZ观测数据的联合贝叶斯分析，允许估计单个星系团的气体比例。在现代弱引力透镜调查 [125]，如DES [126] 和KiDS [127]中，NS是它们参数估计、模型比较和张力量化流程 [128]（图3）的关键组成部分。

宇宙的膨胀历史也可以通过距离测量来衡量，例如观测Ia型超新星（SNIa），它们可以作为‘标准烛光’。超新星宇宙学项目和High-Z超新星搜索团队 [116,117] 的重要成果表明了宇宙加速膨胀的证据，这需要存在一种神秘的反重力暗能量。从SNIa光曲线观测中估算宇宙学参数，历史上一直采用 χ^2 方法（例如参考 [129]），该方法在系统不确定性的确定上缺乏严谨性。参考文献 [130] 引入了一种贝叶斯层次模型来确定宇宙学参数。通过使用NS，他们证明了他们的原则性贝叶斯方法在90%以上的时间内提供了更严格的宇宙学参数统计约束，将统计偏差降低了 ~ 2 -3倍，并且其覆盖特性优于标准 χ^2 方法。

Planck [132]对CMB各向异性的测量是相对于前一个实验（WMAP）统计能力的提升，但需要更复杂的银河系前景建模和仪器校准，引入了 $\mathcal{O}(20)$ “干扰参数”。参数估计的Metropolis-Hastings技术能够通过利用这些参数相对于宇宙学参数是“快速”的这一事实来适应这些参数；具体来说，通过缓存先前计算的结果，这些参数可以更改，而不会带来可忽略的计算成本，前提是宇宙学参数保持固定。

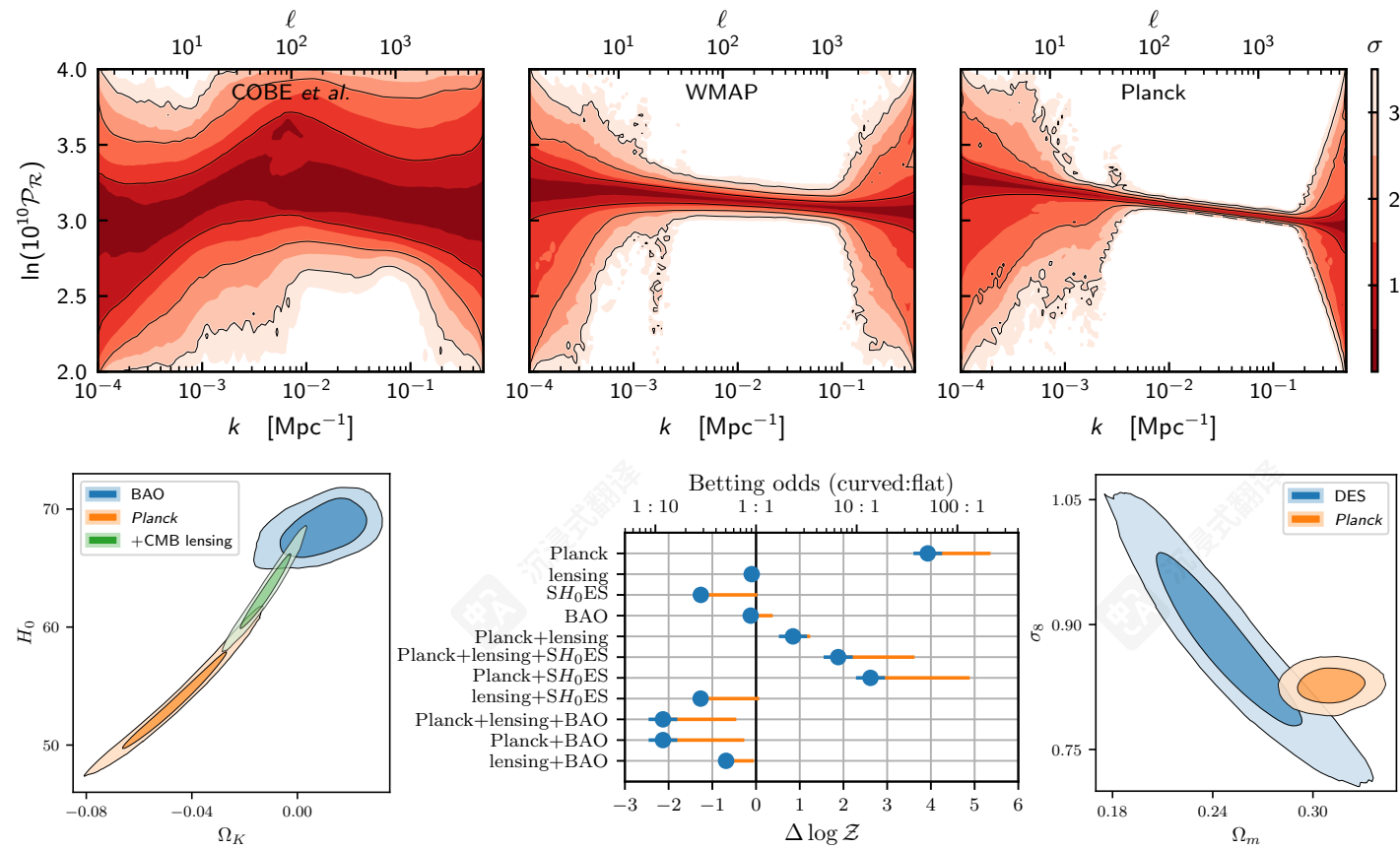


Figure 3 | **Cosmological applications of NS.** *Top row:* Cosmological non-parametric reconstruction of the power spectrum of primordial cosmological fluctuations, as measured by cosmic microwave background satellites across human history. The reconstruction uses a linear spline-based procedure with N movable knots. Each of the locations of the knots along with the cosmological and nuisance parameters are varied in a full NS fit. The **evidence** is then used to marginalize over N to produce the final plots. The standard model of cosmology predicts a featureless tilted power spectrum, so such non-parametric reconstructions are of great interest for astrophysicists searching for **evidence** of beyond standard model physics. *Bottom row:* NS in cosmological tension quantification. Bayesian **evidences** computed by NS can be used to quantify the level of disagreement which may be hidden by marginalization of high-dimensional parameter spaces. *Bottom left:* Planck CMB data is in tension with CMB lensing and BAO in the context of curved cosmologies, so caution should be exercised in combining them [131]. If one includes only CMB data, a **Bayesian model comparison** shows preference for a closed Universe relative to a flat one (*Bottom middle*) in spite of the Occam penalty associated with the additional parameter (orange bar). *Bottom right:* There is also tension between weak lensing data (DES) and the CMB (Planck).

parameters remain fixed.

Region-based samplers such as MultiNest are not able to exploit this hierarchy of parameter speeds, and could not navigate the now high-dimensional cosmological + nuisance parameter space reliably. To address this, cosmologists turned to step sampling based strategies as instantiated in the PolyChord algorithm [65, 133] which uses slice sampling.¹ These were successfully applied throughout the 2015 Planck inflation paper [135] to non-parametric reconstructions [136] (fig. 3) and general inflationary model comparison, in particular to the challenging example of axion monodromy models. The

¹Axial slice sampling had been applied in the then near past to systems biology [134] although unsurprisingly the cosmological authors were not aware of this work. PolyChord improved upon the existing slice-sampling method by implementing covariance-based non-axial steps and mode clustering in addition to the ability to exploit a hierarchy of parameter speeds.

non-parametric reconstruction approach, which only became possible with the ability to reliably (and fully) explore the parameter space of a large number of parameters, was an important model-independent demonstration of the simple power-law behavior of the primordial power spectrum of density perturbations.

Although step sampling was originally introduced in cosmology to exploit a fast-slow hierarchy of parameters, the better dimensional scaling opened up a new range of cosmological analysis possibilities that were inaccessible to region samplers. It has been applied to constraining kinetically dominated inflation models [137], model comparison for the quantum mechanical initial conditions for the Universe [138], additional reconstructions of the dark energy equation of state [139, 140], astronomical sparse reconstruction [141], and played a crit-

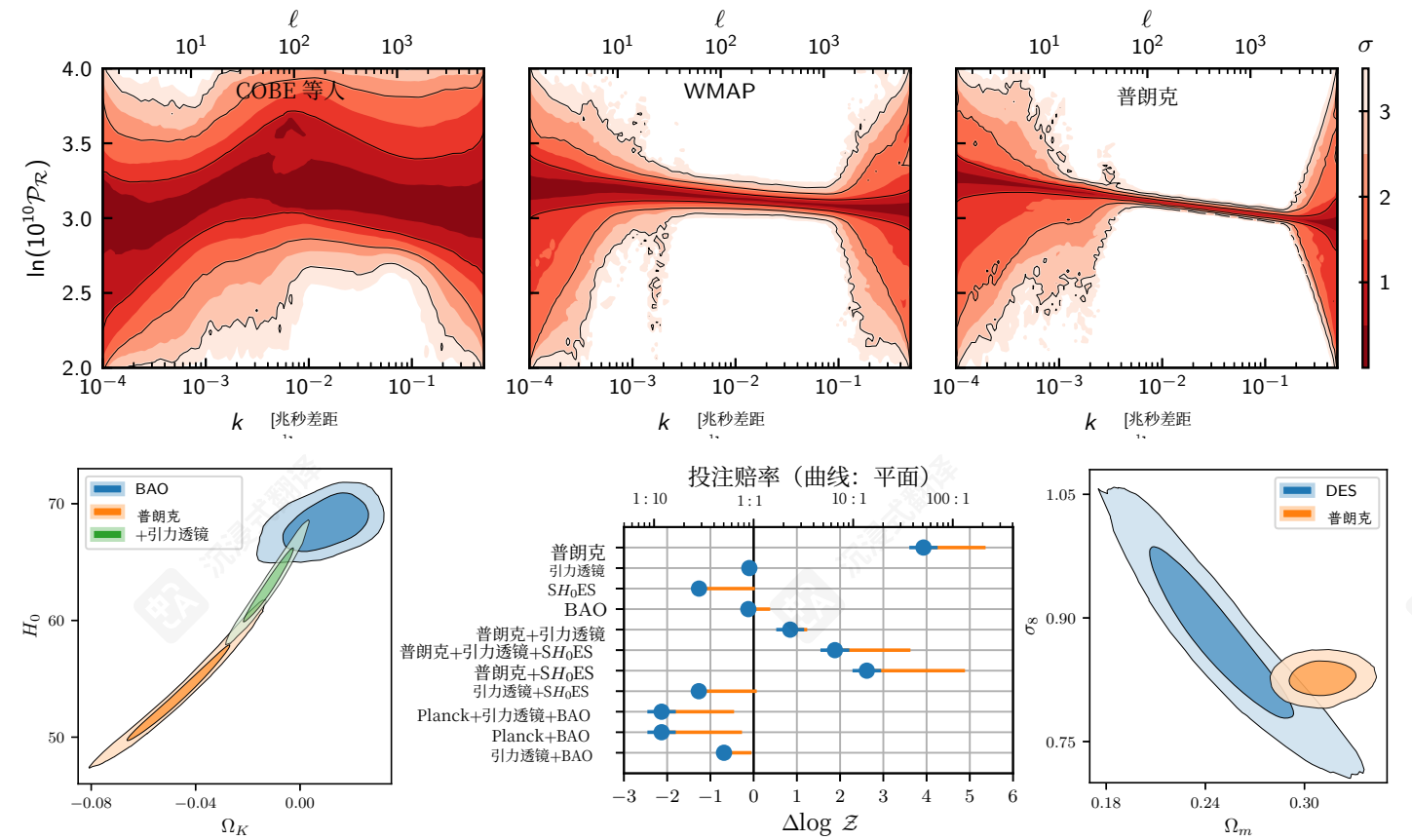


图 3 | NS 在宇宙学中的应用。顶层：宇宙学非参数重建原始宇宙学涨落的功率谱，由跨人类历史的宇宙微波背景卫星测量。重建使用基于线性样条的流程，具有 N 个可移动的节点。节点的每个位置以及宇宙学和干扰参数都在完整的 NS 拟合中变化。然后使用证据对 N 进行边缘化以生成最终图。标准宇宙学模型预测功率谱无特征且倾斜，因此此类非参数重建对寻找超出标准模型物理证据的天体物理学家具有重要意义。底层：宇宙学张力的量化。NS 计算的贝叶斯证据可用于量化可能被高维参数空间边缘化隐藏的分歧程度。左下：在弯曲宇宙学背景下，普朗克 CMB 数据与 CMB 弯曲和 BAO 存在张力，因此在组合它们时应谨慎 [131]。如果仅包含 CMB 数据，贝叶斯模型比较显示，尽管与附加参数相关的奥卡姆惩罚，更倾向于封闭宇宙而非平坦宇宙（中间）。右下：弱透镜数据（DES）与 CMB（普朗克）之间也存在张力。

宇宙学参数保持固定。

基于区域的采样器（如MultiNest）无法利用参数速度的这种层次结构，也无法可靠地导航当前高维的宇宙学 + 噪声参数空间。为此，宇宙学家转向了基于步进的采样策略，例如 PolyChord算法 [65, 133]，该算法使用切片采样¹。这些方法在2015年Planck膨胀论文 [135] 中成功应用于非参数重建 [136]（图3）和一般膨胀模型比较，特别是针对具有挑战性的轴子单流模型。

¹轴向切片采样在不久前已应用于系统生物学 [134]，尽管毫不奇怪，当时的宇宙学家并未意识到这项工作。PolyChord通过实现基于协方差的非轴向步进和模式聚类，在现有的切片采样方法上进行了改进，并增加了利用参数速度层次结构的能力。

非参数重建方法，这只有在能够可靠地（并且完全地）探索大量参数的参数空间时才成为可能，是对原始密度扰动功率谱简单幂律行为的模型无关的重要演示。

尽管步进采样最初是在宇宙学中引入以利用参数的快慢层次结构，但更好的维度缩放开辟了新的宇宙学分析可能性，这些可能性对区域采样器来说是无法触及的。它已被应用于约束动力学主导的暴胀模型 [137]，宇宙量子力学初始条件的模型比较 [138]，暗能量状态方程的额外重建 [139, 140]，天文稀疏重建 [141]，并发挥了关键作用，

ical role in the GAMBIT combined cosmology and particle analyses [142, 143]. Step sampling takes a leading role in the REACH 21cm global cosmology analysis [144], and at the other end of the astrophysical scale it has also been applied to high-dimensional exoplanet analyses [111, 113].

4.2 Particle Physics

Particle physics is a field related to cosmology that has also seen various applications of NS. Around 2010, when NS tools such as **MultiNest** were reaching maturity, the particle physics community was focused on the first results from the Large Hadron Collider (LHC). A particularly favoured theoretical framework was supersymmetry (SUSY; see e.g., ref. [145]). SUSY introduces an array of new particles and unknown parameters that can be fitted to collider data as well as observations from other experiments in a so-called global fit. The physics goals of a global fit are to understand what the model predicts in future experiments and how best to discover the new particles.

These global fits of $\mathcal{O}(10)$ free parameters in SUSY models presented a problem for which NS tools were naturally well-suited, as they were typically **multi-modal**. A package named SuperBayeS [146–148] utilized the **MultiNest** implementation of NS and was used to make a number of early LHC predictions and fits [149–154]. As the LHC results have poured in throughout the 2010s, the theoretical landscape shifted and a wider set of models have been considered using NS (see e.g., ref. [155–158]). More recently, the GAMBIT collaboration has driven many such global fits making use of **MultiNest** and **PolyChord** to sample parameter spaces and compute Bayesian **evidences**, and benchmarked NS against a number of other MC based and gradient-free methods [159, 160]. Related problems such as tuning phenomenological parameters in event generators have seen preliminary work, and could be rich avenues of NS application in the field. Lastly, we note that NS was recently applied to the sampling space, rather than the parameter space, of a statistical model. This enables efficient computation of small p -values that are used in the discovery of new particles at the LHC [161].

4.3 Gravitational waves

Gravitational-wave (GW) astronomy is a field that has blossomed since the first observation in 2015 of two colliding black holes [162] by the LIGO [163] and Virgo [164] interferometers. The signals are produced by non-axisymmetric changes in the gravitational field, typically sourced by the rapid motion of neutron stars and black holes in binary systems [165]. The binary orbits decay through GW emission, increasing the orbital frequency and the rate of energy loss until the objects merge. Such events, known as compact binary coalescences (CBCs), produce signals at frequencies of 10–1000 Hz which are recorded in the detectors as a time-series.

Early development of Bayesian methods for GW analysis was contemporaneous with the first publication of NS, and it was found that NS provided an efficient means to robustly sample the **posterior** distributions of GW signals [166]. This is

important as the **posteriors** provide a rich new astrophysical view: from measuring the masses of neutron stars to the expansion rate of the Universe itself. The small signal-to-noise ratios of observed signals and degeneracies in the model parameter space produce **posterior** distributions that are often **multi-modal** and highly correlated. To date, only NS and **MCMC** approaches have been able to robustly sample the **posteriors** of all observed events. While each has its own advantages, agreement between the NS and **MCMC** approaches has been critical in building confidence in results. However, the NS approach is better-suited to providing robust **evidence** estimates for model comparisons [167]; the efficiency of the NS approach does not depend so much on the use of problem-specific proposals; and massively parallel approaches using the dynesty code have made analyses with more advanced signal models [168] computationally tractable. The success of NS in analysing merging binary signals has inspired many other efforts such as the analysis of continuous signals from individual rapidly-rotating non-axisymmetric neutron stars [169, 170]; detection of unmodelled sources [171]; model selection between different physical mechanisms of core-collapse supernova encoded in the GW signal [172]; and detection of a stochastic superposition of weak merger sources [173].

CBCs consisting of two black holes have been the dominant sources of GW signals seen by the LIGO and Virgo detectors [165]. The signals are modelled through a combination of post-Newtonian approximations to General Relativity and relativistic numerical modelling [174–177]. The signal model, as observed in a detector, is parameterized by 8 parameters that are intrinsic to the binary system (the individual masses, m_1 and m_2 , and their 3D spin vectors) and 7 parameters related to the relative orientation and position of the system with respect to the detector (including the source's luminosity distance D_L and location on the sky). For systems that include at least one neutron star, the signal model's phase evolution requires additional parameters related to the neutron star equation-of-state [178].

Signals from black hole mergers last only a few seconds within the sensitive frequency regime of the detectors. On these timescales the detector data can be approximated as having noise drawn from a stationary Gaussian process described by a known power spectral density, which can either be estimated from data surrounding the signal [180], or inferred directly using a parameterized model of its shape [181]. For inference, a Gaussian **likelihood** function of the form given by Whittle [182] can be used [180]. For multiple detectors, which have independent noise, the **likelihoods** can be coherently combined using the product rule of probability. The prior distributions used are discussed in ref. [180], but are generally set to be uninformative (e.g., uniform over a sphere for sky coordinates), or constrained to be within a physically reasonable range. In addition to the parameters related to the source, the **likelihood** can also contain ~ 60 unknown parameters that relate the frequency-dependent uncertainties in the phase and amplitude calibration of the detectors [183].

For an observed signal, using the **likelihood** and **priors** dis-

in GAMBIT组合宇宙学和粒子分析中发挥了关键作用 [142, 143]. 步进采样在REACH 21cm全球宇宙学分析中扮演着主导角色 [144], 在天体物理学的另一端, 它也被应用于高维系外行星分析 [111, 113].

4.2 粒子物理

粒子物理是一个与宇宙学相关的领域, 也看到了NS的各种应用。大约在2010年, 当像MultiNest这样的NS工具逐渐成熟时, 粒子物理界正关注大型强子对撞机(LHC)的首批结果。一个特别受青睐的理论框架是超对称性(SUSY; 参见例如ref. [145])。SUSY引入了一系列新的粒子以及未知的参数, 这些参数可以拟合对撞机数据以及其他实验的观测结果, 进行所谓的全局拟合。全局拟合的物理目标是要了解模型在未来实验中的预测, 以及如何最好地发现新粒子。

这些SUSY模型中(10)个自由参数的全局拟合提出了一个问题, 而NS工具天然非常适合解决这个问题, 因为它们通常是多模态的。一个名为SuperBayeS [146–148]的软件包利用了NS的MultiNest实现, 被用于进行早期LHC的预测和拟合 [149–154]。随着2010年代LHC结果不断涌现, 理论格局发生了变化, 并使用NS考虑了更广泛的模型(参见例如ref. [155–158])。最近, GAMBIT合作组织利用MultiNest和PolyChord对参数空间进行采样并计算贝叶斯证据, 推动了许多这样的全局拟合, 并将NS与多种基于蒙特卡洛和无需梯度的方法进行了基准测试 [159, 160]。相关问题, 如事件生成器中现象学参数的调谐, 已经进行了初步研究, 这些可能是NS在该领域应用丰富的方向。最后, 我们注意到NS最近被应用于统计模型的采样空间, 而不是参数空间。这能够高效计算用于LHC发现新粒子的小p值 [161]。

4.3 引力波

引力波天文学是一个自2015年首次观测到两个合并的黑洞 [162]以来迅速发展的领域, 观测工作由LIGO [163]和Virgo [164]干涉仪完成。这些信号由引力场中的非轴对称变化产生, 通常由双星系统中中子星和黑洞的快速运动 [165]引发。双星轨道通过引力波辐射衰减, 轨道频率增加, 能量损失速率加快, 直到两个天体合并。这类事件被称为致密双星并合(CBCs), 产生的信号频率在10–1000Hz之间, 被探测器记录为时间序列数据。

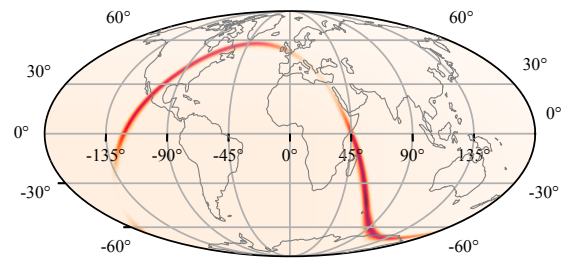
早期贝叶斯方法在GW分析中的发展, 与NS首次发表的时间相同时期, 研究发现NS提供了一种有效的方法, 可以稳健地对GW信号的 [166]后验分布进行采样。这是

早期贝叶斯方法在GW分析中的发展, 与NS首次发表的时间大致相同, 研究发现NS提供了一种高效的方法, 可以稳健地采样GW信号的后验分布 [167]。这一点很重要, 因为后验分布提供了一个丰富的天体物理新视角: 从测量中子星的质量到宇宙本身的膨胀速率。观测信号的小信噪比和模型参数空间中的退相关性, 会导致后验分布通常是多模态且高度相关的。迄今为止, 只有NS和MCMC方法能够稳健地采样所有观测事件的后验分布。虽然每种方法都有其自身的优势, 但NS和MCMC方法之间的一致性对于建立结果的置信度至关重要。然而, NS方法更适合于为模型比较提供稳健的证据估计 [167]; NS方法的效率并不太依赖于使用特定问题的建议; 使用dynesty代码的大规模并行方法, 使得使用更先进的信号模型 [168]的分析在计算上变得可行。NS在分析合并双星信号方面的成功, 激发了其他许多工作, 例如分析单个快速旋转的非轴对称中子星的连续信号 [169, 170]; 未建模源的探测 [171]; 在GW信号中编码的不同物理机制的核心坍缩超新星之间的模型选择 [172]; 以及探测弱合并源的随机叠加 [173]。

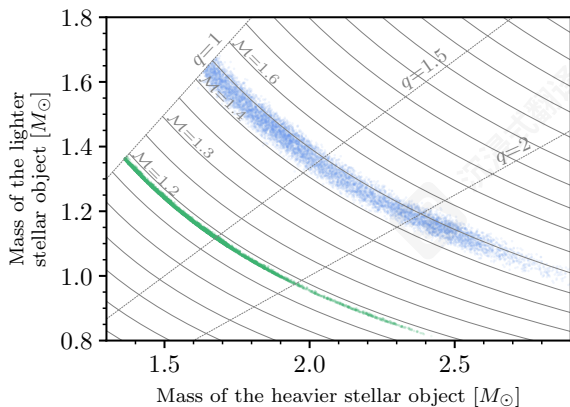
由两个黑洞组成的CBCs一直是LIGO和Virgo探测器观测到的引力波信号的主要来源 [165]。这些信号通过广义相对论的牛顿后近似和相对论数值建模的组合 [174–177]进行建模。在探测器中观察到的信号模型由8个参数参数化, 这些参数是双星系统的固有属性(单个质量 m_1 和 m_2 , 及其三维自旋矢量)以及7个与系统相对于探测器的相对方向和位置相关的参数(包括源的光度距离 D_L 和天空中的位置)。对于包含至少一个中子星的系统, 信号模型的相位演化需要额外的与中子星状态方程相关的参数 [178]。

黑洞合并产生的信号在探测器敏感频段内仅持续几秒钟。在这些时间尺度上, 探测器数据可以近似为具有已知功率谱密度的平稳高斯过程噪声。这可以通过从信号周围的 [180], 数据中估计, 或直接使用其形状的参数化模型 [181]推断。对于推断, 可以使用Whittle [182]给出的高斯似然函数 [180]。对于具有独立噪声的多个探测器, 可以使用概率乘法规则将似然函数相干地组合。使用的先验分布在参考文献 [180], 中进行了讨论, 但通常设置为无信息(例如, 对于天空坐标, 在球面上均匀分布), 或约束在物理上合理的范围内。除了与源相关的参数外, 似然函数还可能包含 ~ 60 与探测器相位和幅度校准的频率相关不确定性相关的未知参数 [183]。

对于观测到的信号, 使用上述似然函数和先验概率,



a | Sky location of the source of the binary black hole merger signal GW151226. The posterior probability distribution from NS is shown in orange [179].



b | Masses of stellar objects involved in two binary neutron star mergers. The posterior samples from NS are shown for the neutron star mergers GW170817 (green) and GW190524 (blue). Solid grey lines mark curves of constant chirp mass $\mathcal{M}$, while dashed lines mark curves of constant mass ratio q . When viewed in $\mathcal{M}$ - q space, the posteriors show only weak correlation.

Figure 4 | Applications of NS in GW astronomy.

cussed above, the joint posterior of all these 15 (or more) parameters can be extracted after application of NS. The posterior distribution of the typical events observed so far are not unimodal or Gaussian. Due to the relatively small signal-to-noise ratio, they exhibit significant degeneracies and correlations. For example, the posterior of the source sky location is largely determined by differences in arrival time of the signal at different detectors. This produces ring-shaped degeneracies as demonstrated in fig. 4a.

The many degeneracies encountered for typical CBC inference have led to significant work to identify optimal parameterization and, where possible, utilized a marginalized likelihood (see ref. [184] for a review). An optimal parameterization involves identifying a mapping between the physical model parameters and combinations of these which reduce the complexity of the target density. Usually, a good reparameterization involves identifying the combinations of physical parameters which are ‘best measured’. As an example, the physical mass of the two component stellar objects are m_1 and m_2 . However, there is a strong banana-like correlation between the two masses (see, e.g. fig. 4b) and an exact degeneracy under ex-

change of m_1 and m_2 . To enable efficient sampling, Veitch *et al* [180] proposed sampling in the chirp-mass, $\mathcal{M}$, and mass ratio q : two algebraic combinations of the component masses. In fig. 4b, we show that the posterior, as viewed in $m_1 - m_2$ space, follows contours of the chirp mass. Physically, this is because the chirp-mass is the most well measured mass parameter (i.e. has the smallest posterior width) followed by the mass ratio.

The chirp-mass and mass ratio parameterization, along with numerous others have greatly improved sampling efficiencies. We note that the choice of sampling parameters, which we choose for computational efficiency, is separate from the choice of prior distributions. If required, a Jacobian transformation [185] can be made to enable sampling in the optimal sampling parameters while setting priors on the physical parameters.

4.4 Materials science

As hinted at in box 3 above, NS can be used to study the thermodynamic properties of molecules and materials, which ultimately derive from the partition function

$$Z(\beta) = \int e^{-\beta H(q,p)} dq dp, \quad (26)$$

where $q \in \mathbb{R}^{3N}$ specifies the spatial coordinates of the N particles in the system, $p \in \mathbb{R}^{3N}$ is the corresponding momentum vector, $H(q,p)$ is the Hamiltonian, and $\beta = 1/k_B T$ is the inverse temperature.

The classical Hamiltonian can be separated into the configuration dependent potential energy, $U(q)$, and momentum dependent kinetic energy, $K(p) = \sum p^2/2m$ (where m is the mass of each particle), giving

$$Z(\beta) = \int e^{-\beta K(p)} dp \int e^{-\beta U(q)} dq. \quad (27)$$

(For the fully quantum treatment using NS, see ref. [186].) The first factor is a Gaussian that may be computed analytically,

$$Z_p \equiv \int e^{-\beta K(p)} dp = \frac{1}{N!} \left(\frac{2\pi m}{\beta \hbar^2} \right)^{3N/2}, \quad (28)$$

and we tackle the second, configuration dependent factor by NS. From the partition function all thermodynamic quantities of relevance can be calculated, for example, the average energy,

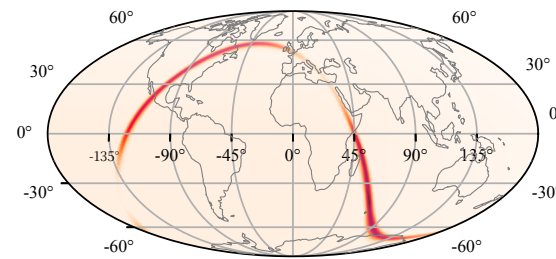
$$\langle H(q,p) \rangle = -\frac{\partial \ln Z(\beta)}{\partial \beta} \quad (29)$$

and the heat capacity

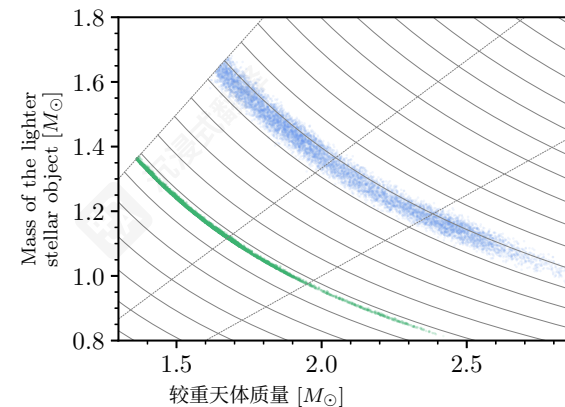
$$C_V = \frac{\partial \langle H(q,p) \rangle}{\partial T}, \quad (30)$$

and these may be interpreted as posterior expectations to be computed after the NS run. In this application,

- the factor $e^{-\beta U(q)}$ plays the role of the likelihood function, parameterized by the inverse temperature, β ;



a | 双黑洞合并信号GW151226的源天体位置。来自NS的后验概率分布以橙色显示 [179]。



b | 参与两个双中子星合并的天体质量。来自NS的后验样本显示于中子星合并GW170817（绿色）和GW190524（蓝色）。实灰线标记恒定chirp质量 $\mathcal{M}$ 曲线，而虚线标记恒定质量比 q 曲线。在 $\mathcal{M}$ - q 空间中，后验分布仅显示弱相关性。

图4 | NS在引力波天文学中的应用。

在应用NS方法后，可以提取出这15个（或更多）参数的联合后验分布。迄今为止观测到的典型事件的后验分布并非单峰或高斯分布。由于信噪比相对较小，它们表现出显著的退化性和相关性。例如，源天球位置的后验分布主要由信号在不同探测器到达时间上的差异决定。这产生了环状退化性，如图4a所示。

典型CBC推断中遇到的多种退化问题，促使研究人员投入大量工作以确定最优参数化方案，并在可能的情况下采用边缘似然（参见文献 [184] 的综述）。最优参数化方案涉及确定物理模型参数与这些参数组合之间的映射关系，从而降低目标密度的复杂性。通常，一个好的重新参数化方案涉及识别那些 ‘最容易测量’ 的物理参数组合。例如，两个组分恒星天体的物理质量分别为 m_1 和 m_2 。然而，这两个质量之间存在强烈的香蕉状相关性（参见例如图4b），在 m [184] 和 m_1 的变化下存在精确的退化。

为了实现高效采样，Veitch等人 [180] 提出了在chirp质量 $\mathcal{M}$ 和质量比 q 空间中进行采样：这两个质量分量的代数组合。在图 4b中，我们展示了后验分布（在 $m_1 - m_2$ 空间中观察）遵循 chirp质量的等高线。从物理上讲，这是因为chirp质量是最容易测量的质量参数（即具有最小的后验宽度），其次是质量比。

chirp质量与质量比参数化，以及其他许多参数，已大大提高了采样效率。我们注意到，为计算效率而选择的采样参数，与选择先验分布是分开的。如果需要，可以进行Jacobian变换 [185]，以在设置物理参数先验的同时，在最优采样参数中进行采样。

4.4 材料科学

如上框3所述，NS可用于研究分子和材料的动力学性质，这些性质最终源自配分函数

$$Z(\beta) = \int e^{-\beta H(q,p)} dq dp, \quad (26)$$

其中 $q \in \mathbb{R}^{3N}$ 指定系统中 N 个粒子的空间坐标， $p \in \mathbb{R}^{3N}$ 是相应的动量矢量， $H(q,p)$ 是哈密顿量， $\beta = 1/k_B T$ 是逆温度。

经典哈密顿量可以分解为与构型相关的势能 $U(q)$ 和与动量相关的动能 $K(p) = \sum p^2/2m$ （其中 m 是每个粒子的质量），得到

$$Z(\beta) = \int e^{-\beta K(p)} dp \int e^{-\beta U(q)} dq. \quad (27)$$

(使用NS进行的进一步量子处理，参见参考文献 [186]。第一项是一个高斯函数，可以解析计算， $Z_p \equiv \int e^{-\beta K(p)} dp = \frac{1}{N!} \left(\frac{2\pi m}{\beta \hbar^2} \right)^{3N/2}$ ，(28)

并且我们通过NS处理第二个、与配置相关的因素。从配分函数中可以计算出所有相关的热力学量，例如平均能量，

$$\langle H(q,p) \rangle = -\frac{\partial \ln Z(\beta)}{\partial \beta} \quad (29)$$

以及热容

$$C_V = \frac{\partial \langle H(q,p) \rangle}{\partial T}, \quad (30)$$

这些可以解释为在NS运行后需要计算的后验期望。在这个应用中，

- 因素 $e^{-\beta U(q)}$ 充当似然函数的角色，由逆温度 β 参数化；

- the prior, according to the ergodic hypothesis, is uniform;
- the dimensionality is commonly on the order of 10^2 to 10^3 or more.

The striking difference from standard Bayesian inference is the inverse temperature parameter β in the **likelihood**. Rather than having a single inference problem for some fixed value of β , almost always we want to know the behaviour of observables *as a function of temperature*, so effectively we have a continuous family of inference problems. And for this, NS (and other density-of-states methods, see box 3) have a remarkable feature: the **likelihood** is a monotonic function of β and hence the entire NS algorithm is invariant to changes in β . In practice this means that a single NS run can be used to calculate observables at all temperatures. For large β values (corresponding to low temperatures), the **partition function** is dominated by the lowest energy minima (highest **likelihood modes**), and for molecules and materials these configurations correspond to the globally stable structures. On the other hand, when β is small (corresponding to high temperatures), the **partition function** is dominated by the large volume associated with high energy states — or in the parlance of materials science, “entropic effects” [187, 188].

Some of the most interesting phenomena in molecular and materials science (and more generally in statistical mechanics) are associated with the above change of regime. Collectively known as *phase transitions*, they are characterized by a dramatic change of where the **bulk** of the **posterior** mass lies, as the temperature (or other system parameters such as pressure) is varied — and this is what makes it very challenging to study them using numerical sampling schemes. Experimentally, phase transitions are often observed indirectly as changes in the expectation value of observables (e.g. discontinuously, in the case of first order phase transitions such as melting and evaporation), or more directly as sharp peaks in response functions, such as the heat capacity or magnetic susceptibility. The promise of NS is to enable the calculation of such response functions in general with high reliability and the minimum of fuss.

There are some aspects of the materials application of NS that turn out to be favourable in comparison with the general inference problem. The first is a rather easy stopping criterion for the NS iterations. For any given model of the potential energy, it is generally not hard to come up with a good global lower bound on the energy, which translates into an upper bound in the **likelihood**. As the **likelihood** values sampled by NS appear to converge, if this is close to the known bound, the iterations can be stopped without the risk of missing the highest **likelihood** mode.

Secondly, convergence of NS with the number of live points and other sampling parameters is desired and observed in terms of convergence of the heat capacity peak locations, and this typically occurs far earlier than the decorrelation of the sampler chains that are used to explore the **constrained prior**.

Thermodynamics of LJ Clusters

The Lennard-Jones (LJ) potential is a simple model for describing the pairwise interactions between atoms and it provides the basis for benchmarking algorithms for modelling materials. In particular, specific sized clusters of LJ atoms exhibit complex thermodynamic properties due to solid-solid transitions (the finite system analogue of first order phase transitions mentioned above) caused by existence of competing low energy minima, in addition to solid-fluid melting [188–190].

We illustrate some features of atomistic energy landscapes in fig. 5a using the example of the cluster of 38 particles (LJ_{38}) along with the corresponding *disconnectivity graph* [191, 192] which help to convey the relationships between the large number of local energy minima (likelihood **modes**). Each leaf of the tree structure corresponds to a local minimum of the energy (or to a closely related set of them), and junctions represent saddle points connecting the minima. As the number of particles in the system is increased, the number of distinct local minima grows exponentially. According to one estimate, LJ_{31} for example has 10^{15} distinct minima, not counting permutation isomers. [188]

Potential energy disconnectivity graphs are good at showing the topology of the energy minima, but to reflect the volumes’ free energy, disconnectivity graphs are required [194, 195]. Using NS, *energy landscape charts* [49] can be generated, and we show one for LJ_{38} in fig. 5a. The vertical axis is the potential energy, and the curve shows a series of (possibly nested) basins, whose width at each potential energy level is proportional to the volume of the corresponding **posterior mode** slice. In order to fit the chart in one diagram, the volumes are also scaled by an exponential factor whose logarithm is shown on the right hand axis. Each basin, encompassing a range of closely related configurations, corresponds to a macroscopic “state” of the system — illustrative structures are shown. The relative volume of the funnel associated with the global minimum (shown in red) and the entropically dominant minimum (in the centre) is 1 : 15 at the energy level where the two separate (at the resolution of this particular NS run), whereas the relative volume reverses to 16 : 1 at the energy level where the known lowest barrier path actually connects the two states [196–198].

It is through these landscape charts that the challenges of thermodynamic sampling can be best understood. If there are not enough live points, even though a few may make it into the global minimum basin where that splits off, there is a danger of that population of walkers “dying out” much before the global minimum is reached. This extinction can happen even while the different basins are still connected in principle, but the volume of states that connect them is so small (or rather, the path connecting them is so narrow) that in practice there is no communication between the basins. This phenomenon is generally referred to as *broken ergodicity* [188].

Note that while small LJ clusters serve as illustration and allow one to develop, benchmark and parameterize various details of sampling algorithms, NS is not the most efficient method for calculating observables in this case. Specialized “bottom-up” algorithms such as Basin-Sampling [188] are significantly more efficient, because they start from a low lying minimum (not

- 根据遍历假说，先前的均匀的；
- 维度通常在 10^2 到 10^3 或更高数量级。

与标准贝叶斯推理的显著区别在于似然中的逆温度参数 β 。与其为某个固定的 β 值有一个单一的推理问题，我们几乎总是想知道可观测量作为温度的函数的行为，因此实际上我们有一个连续的推理问题系列。而对此，NS（以及其他态密度方法，参见框3）有一个卓越的特性：似然是 β 的单调函数，因此整个NS算法对 β 的变化保持不变。在实践中这意味着单个NS运行可用于计算所有温度下的可观测量。对于大的 β 值（对应于低温），配分函数由最低能量极小值（最高似然模式）主导，而对于分子和材料，这些构型对应于全局稳定结构。另一方面，当 β 很小（对应于高温）时，配分函数由高能量状态相关的大体积主导——或者用材料科学的语言来说，“熵效应” [187, 188]。

分子和材料科学（以及更广义的统计力学）中一些最有趣的现象与上述状态变化相关。它们统称为相变，其特征是当温度（或其他系统参数如压力）变化时，后验质量的主要部分发生剧烈变化——这正是使用数值采样方案研究它们时极具挑战性的原因。实验中，相变通常通过可观测量的期望值变化间接观察到（例如，在一级相变如熔化和蒸发的情况下是阶跃式的），或更直接地通过响应函数中的尖锐峰值观察到，如热容或磁化率。NS的承诺是能够以高可靠性和最小麻烦计算这类响应函数。

在NS材料的实际应用中，有一些方面与通用推理问题相比表现出有利之处。首先是NS迭代的一个相当容易的停止标准。对于任何给定的势能模型，通常不难找到一个良好的全局能量下界，这可以转化为似然的上界。当NS采样的似然值似乎收敛时，如果这个值接近已知的界限，就可以停止迭代，而不用担心错过最高似然模式。

其次，期望并观察到NS与活点数量和其他采样参数的收敛性，这体现在热容峰值位置的收敛性上，并且这种情况通常比用于探索约束先验的采样链去相关化发生得早得多。

LJ团簇的热力学

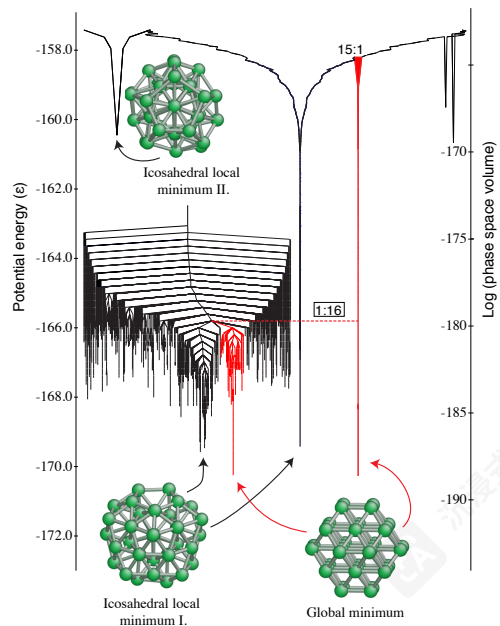
伦纳德-琼斯（LJ）势是一个简单的模型，用于描述原子之间的成对相互作用，并为建模材料的算法基准测试提供了基础。特别是，特定尺寸的LJ原子簇由于存在竞争性低能最小值而表现出复杂的热力学性质，除了固-液熔化 [188–190]之外，还由于固-固转变（上述一阶相变的热力学有限系统类似物）的存在。

我们使用38粒子团簇（ LJ_{38} ）的例子，并结合相应的连通性图 [191, 192]，在图5a中展示了原子尺度能量景观的一些特征，这些图有助于传达大量局部能量最小值（似然模式）之间的关系。树结构的每个叶节点对应于能量局部最小值（或与之密切相关的最小值集），而连接点则代表连接这些最小值的鞍点。随着系统中粒子数的增加，不同局部最小值的数量呈指数增长。据估计，例如 LJ_{31} 就有 10^{15} 个不同的最小值，不计排列异构体。[188]

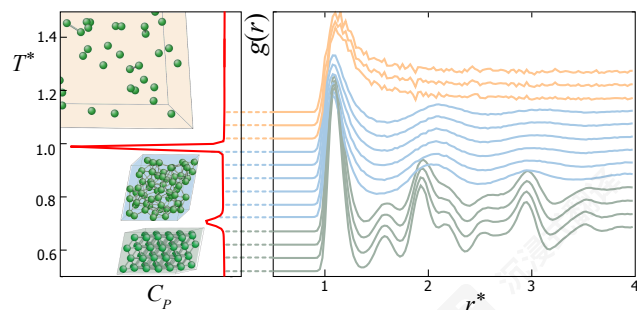
势能不连通图擅长展示能量极小值的拓扑结构，但要反映体积的自由能，就需要不连通图 [194, 195]。使用NS方法，可以生成能量景观图 [49]，我们在图5a中展示了一个 LJ_{38} 的能量景观图。垂直轴是势能，曲线显示了系列（可能是嵌套的）势阱，每个势能水平上的宽度与相应后验模式切片的体积成正比。为了将图适配在一个图中，体积还通过一个指数因子进行缩放，其对数显示在右侧轴上。每个势阱包含一系列密切相关的构型，对应于系统的宏观“状态”——展示了说明性结构。在两个势阱分离的能量水平处（以该NS运行的具体分辨率为准），全局最小值（红色显示）相关的漏斗与熵主导最小值（位于中心）的相对体积为1:15，而在已知最低势垒路径实际连接这两个状态的能量水平处 [196–198]，相对体积则反转至16:1。

正是通过这些景观图，才能最好地理解热力学采样的挑战。如果活点数量不足，即使有少数可能进入那个分支的全局最小值区域，该种群也有可能在全局最小值达到之前就“灭绝”。这种现象甚至可能在不同势阱原则上仍然连通时发生，但连接它们的态空间体积如此之小（或者说，连接它们的路径如此狭窄），以至于在实践中势阱之间没有通信。这种现象通常被称为破 ergodicity [188]。

请注意，虽然小的LJ簇用于说明并允许人们开发、基准测试和参数化采样算法的各个细节，但在此情况下，NS并不是计算可观测量的最高效方法。专门的“自下而上”算法（如盆地采样 [188]）效率要高得多，因为它们从低势阱的极小值开始（不是



a | Potential energy landscape of LJ₃₈, showing the energy landscape chart (see text) obtained by NS, and the disconnectivity graph calculated using discrete path sampling, showing the 200 lowest energy minima [193]. The global minimum basin is shown in red, and the relative phase space volumes of the basins at the separation point and the lowest known connecting path are indicated by the ratios.



b | Constant pressure heat capacity curve (red) peaks are used to locate transitions between the gas (top), liquid (middle) and solid (bottom) phases. Sampled configurations and weights enable calculations of temperature dependent observables such as the radial distribution function (right panel) [77] (note that starred variables correspond to quantities in reduced units).

Figure 5 | **Illustrations of materials science applications.**

necessarily the global minimum) and build a database of all the neighbouring minima by a series of perturbations of the particles and subsequent relaxation. The performance of NS for LJ clusters can be somewhat improved if either (i) the sampler is allowed the use of the pre-generated database of local minima, as in the superposition-enhanced NS (SENS) [199], or (ii)

a large number of independent NS runs each with a single live point are suitably combined, as in Nested Basin-Sampling [71], but neither of these enhancements makes NS competitive for studying small and moderate sized particle clusters.

There is one case for which NS appears to be an effective tool for studying even the smallest clusters, and this is the determination of thermodynamically favourable *transformation paths* between different states at moderately high temperatures where harmonic transition state theory is no longer applicable. [200]

Phase diagrams of materials

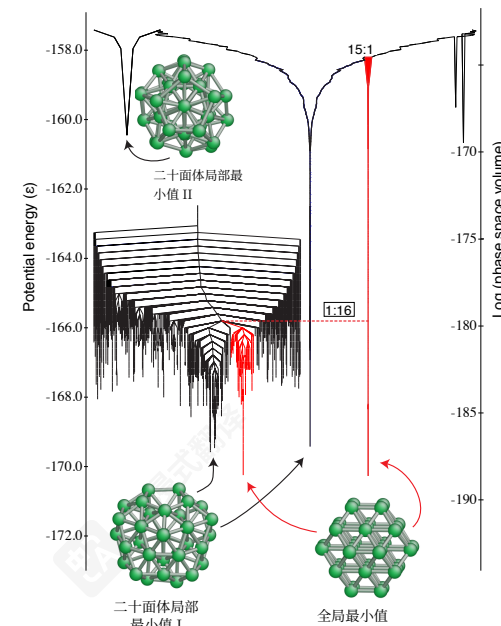
It is in the study of condensed phase systems that NS really comes into its own. There are two reasons for this: one is that it is generally difficult to make efficient sampling moves due to the geometric constraints, the other is that as the system size grows the phase transitions become sharper such that for a hundred or more particles, thermal methods become essentially ineffective.

At high energies and moderate pressures entropy wins and all materials are gases, but as the energy decreases they typically condense into a liquid, then freeze into a solid, and sometimes even undergo solid-phase structural transitions. A specific heat curve is shown as an example in fig. 5b for the periodic Lennard-Jones system. In the thermodynamic limit, the discontinuity of the potential energy across the phase transition would result in a divergence of the heat capacity, but in a numerical simulation they are broadened by finite size effects and appear as sharp peaks. Performing the sampling at a range of pressure values, the loci of the peaks define the boundaries between stability regions of different phases in the pressure-temperature phase diagram. Notice the structural similarity of the solid phase, a closed packed face centred cubic (fcc) crystal, to the narrow global minimum of the LJ₃₈ cluster. As the cluster size grows, the repeated occurrence of these fcc-like clusters hint at the ground state of the infinite crystal. There is no such periodic analogue for the broad icosahedral basin of the clusters because the five-fold symmetry is incompatible with periodicity (periodicity is further discussed in Supplementary Information F).

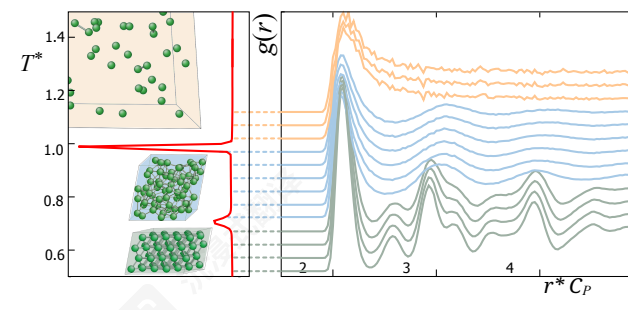
NS has been successful in characterizing the behaviour of a wide range of materials. These include model systems such as the Potts model [201, 202], hard spheres [203], Lennard-Jones [77, 204] and the Jagla potential [205] as well as more chemically realistic potentials for aluminium and the shape memory alloy NiTi [44], as well as lithium [206], and iron [207]. Recently, machine-learning interatomic potentials [208] are now being applied [209] to increase the predictive power of calculated phase diagrams, leveraging the increased accuracy of the potentials. A detailed overview of materials applications can be found in ref. [210].

5 Reproducibility and data deposition

We recommend a set of minimum considerations and reporting standards for NS computations, shown in box 5. You should state clearly the number of live points and stopping conditions.



a | LJ38's potential energy landscape, showing the energy landscape chart (see text) obtained by NS, and the disconnectivity graph calculated using discrete path sampling, showing the 200 lowest energy minima [193]. The global minimum basin is shown in red, and the relative phase space volumes of the basins at the separation point and the lowest known connecting path are indicated by the ratios.



b | Constant pressure heat capacity curve (red) peaks are used to locate transitions between the gas (top), liquid (middle) and solid (bottom) phases. Sampled configurations and weights enable calculations of temperature dependent observables such as the radial distribution function (right panel) [77] (note that starred variables correspond to quantities in reduced units).

图 5 | 材料科学应用的插图。

并非必然是全局最小值)并通过一系列粒子扰动及后续弛豫来构建所有邻近极小值的数据库。对于LJ团簇，若允许采样器使用预生成的局部极小值数据库（如叠加增强的NS (SENS) [199]，)或 (ii)

大量独立的NS运行，每个运行只有一个活点被适当组合，如嵌套盆地采样 [71]，但这两项增强均未使NS在研究小尺寸和中尺寸粒子团簇时具有竞争力。

有一个案例，NS似乎是一种有效的工具，用于研究即使是最小的团簇，这就是在中等高温下确定不同状态之间热力学上有利的转变路径，此时谐振子过渡态理论不再适用。 [200]

材料的相图

在凝聚相系统的研究中，NS真正发挥了它的优势。有两个原因：一是由于几何约束，通常难以进行有效的采样移动；二是随着系统规模的增大，相变变得更加尖锐，以至于对于一百个或更多的粒子，热力学方法基本上变得无效。

在高能量和中等压力下，熵占上风，所有材料都是气体，但随着能量的降低，它们通常会凝结成液体，然后冻结成固体，有时甚至会发生固相结构转变。作为周期性Lennard-Jones系统的示例，在图5b中显示了比热曲线。在热力学极限下，势能在相变处的连续性会导致热容发散，但在数值模拟中，它们由于有限尺寸效应而被展宽，并表现为尖锐的峰值。在一系列压力值下进行采样，峰值的轨迹定义了压力-温度相图不同相稳定区域的边界。请注意固相的结构相似性，一种密堆积面心立方 (fcc) 晶体，与LJ₃₈ 团簇的窄全局最小值。随着团簇规模的增大，这类fcc团簇的重复出现暗示了无限晶体基态。团簇的宽二十面体盆地没有这样的周期性类似物，因为五重对称性与周期性不相容（周期性在补充信息F中进一步讨论）。

NS已成功表征了多种材料的特性。这些材料包括Potts模型 [201, 202]，硬球模型 [203]，Lennard-Jones模型 [77, 204] 以及Jagla势 [205]，还包括更符合化学实际的铝和形状记忆合金 NiTi [44]、锂 [206]，和铁 [207]的势。最近，机器学习原子间势 [208] 正被应用 [209] 以提高计算相图的预测能力，利用势能精度的提高。有关材料应用的详细概述可参见参考文献 [210]。

5 可重复性与数据存档

我们推荐一套用于NS计算的最低考虑事项和报告标准，具体见框5。你应该明确说明活点的数量和停止条件。

In addition, you should report implementation specific settings, for example the number of repeats if using slice sampling, or the enlargement factor if using ellipsoidal sampling. If you are using a public software package, report the version number. Since NS is an MC algorithm, for reproducibility we suggest fixing the random seed so that identical results can be replicated.

Since NS computes an integral, make sure that the **integrand** is explained clearly, including the choices of **likelihood** and prior, and, for example, any overall constant factors that are sometimes omitted from the **likelihood**. To help achieve these goals, consider publishing your computer code alongside your research. Similarly, consider depositing the NS output files publicly. This would permit further scrutiny and re-use of the NS results. The output data should be accompanied with sufficient metadata (e.g., column labels; see ref. [211]).

Although the ultimate result might be for example, a ratio of NS results (a Bayes factor), we recommend reporting the results of all individual NS calculations. To do so, we suggest the triplet of $\log Z$, the estimated uncertainty, and the KL divergence, H . The first two are most relevant for inference, whereas the KL divergence indicates the numerical challenge, as it impacts runtime (eq. (19)) and uncertainty (eq. (21)). You may also wish to report the effective dimensionality or an Occam factor [212].

The NS error estimates are usually reliable, such that if we wish to reproduce NS computations, we should find agreement within uncertainties with the original calculation. As discussed previously, though, it is $\log Z$ rather than Z alone that is distributed with a roughly symmetric Gaussian error.

Box 5 | Check-list for reliability and reproducibility

1. Pick an appropriate technique for sampling from the **constrained prior**; some choices perform more reliably and efficiently in high-dimensions
2. Pick appropriate settings and tailor them to your goals and problem. Relaxed exploration settings (for example fewer steps [65]) may be adequate if you are only concerned about parameter inference.
3. Report software version numbers and settings
4. Describe **priors** and **likelihood** in adequate detail
5. Ideally, publish the code used for the computation, permitting the calculation to be replicated and scrutinized
6. Perform cross-checks on NS run [89], as implemented in e.g., anesthetic. If practical, consider simulation-based calibration to check results
7. Report triplet of $\log Z$, the associated uncertainty, and H for each NS computation
8. Consider publishing the NS output, allowing further re-use and checks of the NS run

6 Limitations and optimizations

6.1 Limitations

Although NS is broadly applicable, there are several potential limitations. The first limitation relates to the prior: we sample from the **constrained prior** and thus require a proper prior. For many NS implementations, this proper prior must be transformed from the unit hypercube. Normalizing flows were recently proposed [213] for cases where this is inconvenient or impossible analytically, including using the **posterior** from an NS run as a prior.

There are, furthermore, limitations related to the **likelihood**. Integration using NS requires a non-negative **integrand** (though the compression itself makes no such restriction). Whilst this condition is always fulfilled in statistical applications, it could be violated when NS is used as a general purpose integrator. We wrote eq. (6) assuming that $L \geq 0$ whereas eq. (8) makes no restriction. NS, however, compresses upwards in **likelihood**, such that positive and negative **likelihood** regions of the integral would be treated differently, with the latter explored at inadequate resolution. NS, furthermore, requires a **tractable likelihood**. For cases in which the **likelihood** is intractable, ref. [99] proposed a **likelihood-free** NS in the context of systems biology, assuming that an unbiased estimator of the **likelihood** was available.

Lastly, plateaus in the **likelihood** function spoil the estimates of the compression [2, 214–216]. That is, sets A with non-negligible prior mass, $\mu(A) > 0$, and constant **likelihood**, which means $L(\Theta) \equiv c$ for all $\Theta \in A$. We may modify the **likelihood** function to remove plateaus by adding a negligible unique iid tie-breaking random draw to the **likelihood** or promoting that draw to a parameter and increasing the dimension of the problem. We may alternatively modify the algorithm itself such that it sums plateaus correctly but reduces to the ordinary NS algorithm in their absence. In the presence of plateaus, the minimum **likelihood** among the live points, $\min L$, may be shared by several live points.

For example, we could modify NS by removing all q points in the plateau and then replacing them all [217]. In this case, the compression factors follow $\text{Beta}(n_{\text{live}} + 1 - q, q)$. This may be applied retrospectively to any NS runs, though suffers from increased uncertainty as the number of live points is temporarily reduced to as few as $n_{\text{live}} - q$. Alternatively, we could first increase the live points by sampling subject to $L > L^*$ until $n_{\text{live}} - 1$ lie at $L > \min L$. Note that $\min L$ may decrease as we add live points at $\min L > L > L^*$. We then remove q points in the outermost contour. The compression factor follows $\text{Beta}(n_{\text{live}}, q)$. This cannot be applied retrospectively but the uncertainties are reduced (at the cost of greater run-time) as the number of live points is temporarily increased beyond n_{live} . This may in fact be seen as a dynamic version of the first modification and we could increase the number of live points according to a different criterion. Lastly, we could split the integral into an integral over plateaus and an integral without plateaus, and perform only the latter with NS [216].

此外，你应该报告特定实现的设置，例如使用切片采样时重复的次数，或使用椭球采样时的放大因子。如果你使用的是公开软件包，请报告版本号。由于NS是一种蒙特卡洛算法，为了可重复性，我们建议固定随机种子，以便能够复制相同的结果。

由于 NS 计算积分，请确保被积函数解释清楚，包括似然函数和先验的选择，以及例如有时从似然函数中省略的任何整体常数因子。为了实现这些目标，请考虑在研究中附带发布您的计算机代码。类似地，请考虑公开存放 NS 输出文件。这将允许对 NS 结果进行进一步审查和再利用。输出数据应附带足够元数据（例如，列标签；参见参考文献 [211]）。

尽管最终结果可能是例如NS结果的比率（贝叶斯因子），我们建议报告所有单个NS计算的结果。为此，我们建议使用对数Z、估计的不确定性和KL散度H这三个元素。前两个与推断最相关，而KL散度则表明数值挑战，因为它影响运行时间（eq. (19)）和不确定性（eq. (21)）。您也可以选择报告有效维度或奥卡姆因子 [212]。

NS误差估计通常是可靠的，因此如果我们希望重现NS计算，我们应该在不确定性范围内与原始计算结果一致。如前所述，不过，分布的实际上是log Z而不是Z本身，其误差大致呈对称高斯分布。

方框 5 | 可靠性和可重复性检查清单

1. 选择合适的从约束先验分布中进行抽样的技术；有些选择在高维空间中表现更可靠和高效
2. 选择合适的设置并根据目标和问题进行定制。如果只关心参数推断，宽松的探索设置（例如步数更少 [65]）可能就足够了
3. 报告软件版本号和设置
4. 详细描述先验分布和似然函数
5. 理想情况下，应发布用于计算的代码，允许重复计算并进行审查计算得以复制和审查
6. 对 NS 运行 [89], 进行交叉检查，例如在模拟环境中实现。如果可行，考虑基于模拟的校准来验证结果
7. 对每个 NS 计算报告 $\log Z$ 的三元组、相关的不确定性和 H
8. 考虑发布 NS 输出，以便进一步重用和检查 NS 运行

6 限制与优化

6.1 限制

尽管 NS 广泛适用，但存在一些潜在限制。第一个限制与先验有关：我们从受限先验中采样，因此需要合适的先验。对于许多 NS 实现，这个合适的先验必须从单位超立方体中转换。最近提出了 [213] 正则化流，用于不方便或无法进行解析的情况，包括使用 NS 运行后的后验作为先验。

此外，还与可能性存在限制相关。使用NS进行积分需要非负的被积函数（尽管压缩本身没有此限制）。虽然此条件在统计应用中始终得到满足，但当NS作为通用积分器使用时可能会被违反。我们假设 $L \geq 0$ 而写出公式(6)，而公式(8)则对此没有限制。然而，NS在可能性上会向上压缩，因此积分的正负可能性区域会被不同对待，后者以不足够的分辨率进行探索。此外，NS还需要一个可处理的似然。对于似然不可处理的情况，ref [99]在系统生物学背景下提出了无似然的NS，假设有一个无偏的似然估计器可用。

最后，似然函数中的平台会破坏压缩 [2, 214–216]的估计。也就是说，具有非忽略先验质量的集合 A ， $\mu(A) > 0$ ，且似然恒定，即 $L(\Theta) \equiv c$ 对所有 $\Theta \in A$ 。我们可以通过向似然添加一个可忽略的唯一独立同分布的破平局随机抽取来修改似然函数，以消除平台，或者将此抽取提升为参数并增加问题的维度。或者，我们可以修改算法本身，使其正确求和平台，但在平台不存在时简化为普通的NS算法。在存在平台的情况下，最小-

在所有活点中，最小似然值 $\min L$ 可能会被多个活点共享同时被几个活点使用。
例如，我们可以通过移除所有高原上的q点，然后用 [217]替换它们来修改NS。在这种情况下，压缩因子遵循 $\text{Beta}(n_{\text{live}} + 1 - q, q)$ 。这可以应用于任何NS运行，但会因临时减少活动点数量至尽可能少的 $n_{\text{live}} - q$ 而增加不确定性。或者，我们可以先通过采样，在 $L > L^*$ 的约束下增加活动点，直到 $n_{\text{live}} - 1$ 位于 $L > \min L$ 。请注意，随着我们在 $\min L > L > L^*$ 处添加活动点， $\min L$ 可能会减少。然后，我们移除最外层轮廓中的q点。压缩因子遵循 $\text{Beta}(n_{\text{live}}, q)$ 。这不能应用于回顾性操作，但不确定性会降低（以运行时间增加为代价），因为活动点数量会临时增加到超过 n_{live} 。这实际上可以看作是第一个修改的动态版本，我们可以根据不同的标准增加活动点数量。最后，我们可以将积分拆分为高原积分和非高原积分，并仅使用NS [216]执行后者。

In addition, there are computational limitations. In ordinary NS running for longer doesn't reduce uncertainties or increase the number of **posterior** samples. If one is only interested in using NS for optimization, this isn't a drawback. In any case, as discussed in **Dynamic nested sampling**, dynamic NS overcomes this problem by rewinding and resuming an NS run. Besides that, NS requires $\mathcal{O}(D^2)$ iterations (eq. (19)), the memory required to store the co-ordinates of every dead point scales as $\mathcal{O}(D^3)$ (with potentially nasty scaling factors for clustering in particular exploration strategies). This means that NS implementations that store every dead point become memory bound at $\mathcal{O}(500)$ dimensions.

We stress, furthermore, that users should be aware of ways in which NS may fail:

1. NS may fail to successfully draw independent samples from the **constrained prior**. Consequently NS results including error estimates may be faulty and anticipated properties of NS, such as convergence, won't hold. In **Results**, we discussed cross-checks on NS results, such as ref. [89], that may identify this issue.
2. Owing to item 1, NS may miss **modes**. **multi-modal** problems pose challenges in Bayesian computation, particularly in **MCMC**, where the chain must make a sequence of unlikely steps between **modes**. Unfortunately, it is in general impossible to know whether all **modes** have been found. Broadly speaking, as NS does not depend on slow transitions between **modes**, it is well-suited to **multi-modal** problems. Once a **mode** is established, furthermore, it won't be abandoned until the **likelihood** threshold exceeds that of the points that were in the mode.
3. NS may sample inefficiently from the **constrained prior**. As discussed in **Exploration strategies**, this may occur in high-dimensional problems with rejection sampling strategies.

Our check-list in box 5 includes checks of NS failures.

6.2 Optimizations

There are several ways NS runs may be optimized to make best use of computing resources (see also **Parallelization**). First, one may utilize fast and slow parameters by breaking the **likelihood** function into fast and slow operations that involve subsets of the D parameters. The parameters associated with fast and slow operations are referred to as fast and slow parameters, respectively; for example, if the **likelihood** function may be written as,

$$L(x, y, z, \dots) = \text{slow}(x) \times \text{fast}(y, z, \dots), \quad (31)$$

then $y, z, \dots$ are fast parameters and x is a slow parameter. When selecting a new point, to minimize runtime we should where possible change the fast parameters and keep the slow parameters constant, allowing caching of the slow operation [218]. This is natural in the slice sampling exploration strategy discussed in **Exploration strategies**, as we may pick slices along

fast directions [65]. If D_s parameters are slow, we require D_s rather than D slow operations per iteration. Similarly, it may be exploited in modified Metropolis algorithms [24, 219] in which the proposals change blocks of parameters at a time.

In addition, one may repartition the **posterior**. NS must compress exponentially through the entire prior volume, which could be slow for a diffuse prior with substantial compression to the **posterior**. This in practice limits NS to $\mathcal{O}(100s - 1000s)$ of dimensions. This prohibits fitting complex hierarchical Bayesian models and deep neural networks, or reaching HMC like dimensionalities of $\mathcal{O}(10^6)$. In some cases, it may be possible to use a narrower prior and correct the **evidence** estimates from NS post-hoc.

This issue may occur in problems in which the prior is unrepresentative [220], i.e., where the observed data lies in the tail of the prior predictive distribution. This may be mitigated by Bayesian automatic prior repartitioning [221] in which one redefines the prior and **likelihood** while leaving their product unchanged. This allows one to keep the **evidence** integral in eq. (1) unchanged, but reduce the prior to **posterior** compression in eq. (16). For example, consider a Gaussian **likelihood**, $\mathcal{N}(x, \sigma^2)$, for a parameter $x \sim \mathcal{U}(-L/2, L/2)$ for $\sigma \ll L$. The compression would be approximately $H \approx \log(\sigma/L)$. Repartitioning the prior and **likelihood** by swapping them, we obtain $H \approx 0$.

The possibility of improving the robustness and efficiency of NS by exploiting the intrinsic degeneracy between the 'effective' likelihood and prior in the formulation of Bayesian inference problems was also discussed in ref. [123], and posterior repartitioning can further be viewed as the vanilla case (when the importance weight function equals to 1) of nested importance sampling proposed in ref. [18].

Lastly, rather than performing new NS runs, one may reuse runs for similar **likelihoods** and **priors**. If the prior or **likelihood** are modified, it may be possible to reweight the **posterior** weights and **evidence** integral [40],

$$P'_i = P_i \times \frac{L'_i \pi'_i}{Z'} \frac{Z}{L_i \pi_i} \quad (32)$$

$$Z' = Z \times \sum P_i \times \frac{L'_i \pi'_i}{L_i \pi_i} \quad (33)$$

where $P_i = w_i L_i$. This re-weighting may be interpreted as a **pseudo-importance sampling** in which the original estimated **posterior** plays the role of the kernel. This is particularly useful for investigating prior sensitivity in the context of Bayesian inference. The effective sample size of draws from the new **posterior** may be used to judged the reliability of this procedure; if the new and original **posterior** distributions differ substantially, a fresh NS run needs to be performed.

There may, furthermore, be cases in which we wish to investigate several similar **likelihood** functions at once; ref. [59] presents a collaborative version of NS that operates on more than one **likelihood** function at once and in which parts of the **likelihood** evaluation may be recycled.

此外, 还有计算限制。在普通NS运行中, 运行时间更长并不会减少不确定性或增加后验样本数量。如果仅对使用NS进行优化感兴趣, 这不是一个缺点。在任何情况下, 正如在动态嵌套抽样中讨论的那样, 动态NS通过回滚和恢复NS运行来克服这个问题。除此之外, NS需要 $\mathcal{O}(D^2)$ 次迭代 (等式(19)), 存储每个死亡点坐标所需的内存按 $\mathcal{O}(D^3)$ 缩放 (对于特定的探索策略中的聚类, 可能存在潜在的糟糕缩放因子)。这意味着存储每个死亡点的NS实现会在 $\mathcal{O}(500)$ 维度时受到内存限制。

此外, 我们强调用户应了解 NS 可能失败的方式:

1. NS 可能无法从约束先验中成功抽取独立样本。因此, 包含误差估计的 NS 结果可能存在错误, 而 NS 的预期属性 (如收敛性) 将不成立。在结果部分, 我们讨论了针对 NS 结果的交叉验证, 例如参考文献 [89], 这些验证可以识别此问题。
2. 由于第1项, NSmaymiss模式。多模态概率问题在贝叶斯计算中构成挑战, 特别是在MCMC中, 链必须在模式之间做出一系列不太可能的步骤。不幸的是, 通常不可能知道是否已经找到了所有模式。总的来说, 由于NS不依赖于模式之间的慢速转换, 因此它非常适合多模态问题。此外, 一旦一个模式建立, 它就不会被放弃, 直到似然阈值超过位于该模式中的点的似然值。
3. NS 可能无法有效地从约束先验分布中采样。如《探索策略》中所述, 这在具有拒绝采样策略的高维问题中可能会发生。

我们的检查清单 (见框5) 包括对NS失败的检查。

6.2 优化

有几种方法可以优化NS运行, 以充分利用计算资源 (另见并行化)。首先, 可以通过将似然函数分解为涉及D参数子集的快速和慢速操作来利用快速和慢速参数。与快速和慢速操作相关的参数分别称为快速参数和慢速参数; 例如, 如果似然函数可以写成,

$$L(x, y, z, \dots) = \text{slow}(x) \times \text{fast}(y, z, \dots), \quad (31)$$

然后 $y, z, \dots$ 是快速参数, 而 x 是一个慢速参数。在选择新点时, 为了最小化运行时间, 我们应该尽可能改变快速参数并保持慢速参数不变, 从而允许缓存慢速操作 [218]。这在《探索策略》中讨论的切片采样探索策略中很自然, 因为我们可以沿着快速方向 [218] 选择切片。

如果 D_s 参数是慢速的, 我们要求 D_s 而不是每迭代 D 次慢速操作。类似地, 这也可以在修改后的 Metropolis 算法 [24, 219] 中利用, 其中提议一次改变参数块的多个参数。

此外, 可以重新划分后验分布。NS必须在整个先验体积中指数级压缩, 这对于具有较大压缩后验的弥散先验来说可能很慢。这在实践中限制了NS的维度为 $\mathcal{O}(100s - 1000s)$ 。这禁止了拟合复杂的分层贝叶斯模型或深度神经网络, 或达到HMC类似的维度 $\mathcal{O}(10^6)$ 。在某些情况下, 可以使用更窄的先验, 并在事后修正NS的证据估计。

这个问题可能出现在先验分布代表性不足的问题中 [220], 即观测数据位于先验预测分布的尾部。这可以通过贝叶斯自动先验重新分配 [221] 来缓解, 其中重新定义先验和似然, 同时保持它们的乘积不变。这使得等式 (1) 中的证据积分保持不变, 但可以减少等式 (16) 中的先验到后验压缩。例如, 考虑一个高斯似然 $\mathcal{N}(x, \sigma^2)$, 对于参数 $x \sim \mathcal{U}(-L/2, L/2)$ 的 $\sigma \ll L$ 。压缩量将大约是 $H \approx \log(\sigma/L)$ 。通过交换先验和似然来重新分配先验和似然, 我们得到 $H \approx 0$ 。

在参考文献 [123], 中还讨论了利用贝叶斯推理问题中“有效”似然和先验之间的内在退化性来提高 NS 的鲁棒性和效率的可能性。后验重新分配还可以进一步被视为参考文献 [18] 中提出的嵌套重要性采样的特例 (当重要性权重函数等于 1 时)。

最后, 与其执行新的NS运行, 不如重用相似似然和先验的运行。如果先验或似然被修改, 则可能可以重新加权后验权重和证据积分 [40],

$$P'_i = P_i \times \frac{L'_i \pi'_i}{Z'} \frac{Z}{L_i \pi_i} \quad (32)$$

$$Z' = Z \times \sum P_i \times \frac{L'_i \pi'_i}{L_i \pi_i} \quad (33)$$

其中 $P_i = w_i L_i$ 。这种重新加权可以被解释为一种伪重要性采样, 其中原始估计的后验分布充当核的作用。这在贝叶斯推理的背景下研究先验敏感性特别有用。新后验分布中抽样的有效样本量可用于判断此程序的可靠性; 如果新旧后验分布差异很大, 则需要执行新的NS运行。

此外, 我们可能还需要同时研究多个相似的似然函数; ref. [59]介绍了一种协作版本的NS算法, 该算法可以同时处理多个似然函数, 并且似然评估的部分结果可以被重复利用。

7 Outlook

The **MCMC** computational revolution of the 1990s solved the problem of computing shapes. Skilling’s remarkable NS algorithm solved the outstanding problem of computing magnitudes at the same time. Although it is naturally expressed in the language of Bayesian inference, NS is a powerful and general-purpose integration algorithm. For that reason, we expect NS to remain relevant long into the future. We are now in the midst of a revolution in data science, and high-dimensional spaces and integration are more important than ever. NS rises to the challenge.

As we reviewed, there have been many theoretical developments in understanding NS, including its convergence, errors, diagnostics and techniques for sampling from the **constrained prior**. We expect them to continue, especially as connections to other statistical methods and machine learning are explored. Indeed, theoretical analysis of NS combined with constrained **MCMC** exploration may be helped by the connections to SMC discussed in box 4, and we anticipate further developments in understanding uncertainties in NS, especially for parameter inference and in the case of a dynamic number of live points. Lastly, NS may be considered a meta-algorithm, as it doesn’t specify an algorithm for sampling from the **constrained prior**. We are already seeing that this opening allows NS to dovetail with developments in machine learning, such as normalising flows that are beginning to be used to sample from the **constrained prior**.

The successes and breadth of applications of NS stem from the fact that it is fundamentally simple. Whilst in the future NS may be understood and improved in the context of more sophisticated computational methods, NS will retain advantage and appeal due to its simplicity.

Acknowledgements

We thank John for his wonderful algorithm. The success of nested sampling may be that simple beats clever; but the beauty of nested sampling is that it is both simple and clever.

We thank Kyle Barbary for discussions. AF was supported by an NSFC Research Fund for International Young Scientists grant 11950410509. LBP acknowledges support from the EPSRC through an Early Career Fellowship (EP/T000163/1). MH acknowledges support from the Carl Zeiss Foundation. NB was funded by the U. S. Naval Research Laboratory’s base 6.1 research program, and CPU time from the U. S. DoD’s HPCMPO at the AFRL and ARL DSRCs. MP acknowledges support from the STFC (ST/V001213/1 and ST/V005707/1). WH was supported by a Royal Society University Research Fellowship.

Author Contributions

Introduction (A. F., D. S., L. S., P. W.); Experimentation (J. B., E. H., J. S. S.); Results (A. F.); Applications (G. A., N. B., X. C., G. C., F. F., M. G., W. H., M. H., A. L., D. P., L. B. P., M. P., J. V., D. W., D. Y.); Reproducibility and data deposition (A. F.); Limitations and optimizations (A. F.); Outlook (A. F.); Overview of the Primer (A. F.).

Competing Interests

There are no competing interests to declare.

Supplementary Information

A Glossary

Bayesian model comparison A method for comparing models based on computing the change in their relative plausibility in light of data using Bayes’ theorem. Pages 2, 12, 13.

bootstrap Techniques that estimate statistical variation by repeatedly drawing samples from the true dataset with replacement. Pages 22.

bulk Region with size of order e^{-H} that contains the overwhelming majority of the posterior mass. Closely related to typical sets. Usually won’t lie near the mode of the posterior, especially in high-dimensions [222, 223]. Pages 4, 5, 7, 8, 10, 16.

constrained prior The **prior** for the parameters restricted to the region in which the **likelihood** exceeds a threshold. Pages 1, 3, 5–11, 16, 18–21.

curse of dimensionality The phenomenon that the difficulty of a problem often increases dramatically with dimension [224]. Pages 2, 3, 5, 9, 10.

evidence A factor in Bayes’ theorem that may be written as an integral and that plays an important role in **Bayesian model comparison**. In NS this is the multi-dimensional integral we wish to compute. Pages 2–14, 19, 21.

indicator function A function $\mathbb{1}(\cdot)$ that takes the value 1 if the condition $\cdot$ holds and 0 otherwise. Pages 8.

insertion indexes The indexes at which the elements of a list must be inserted into an ordered list to maintain ordering. Pages 11.

7 Outlook

20世纪90年代的MCMC计算革命解决了计算形状的问题。Skilling的NS算法同时解决了计算量级这一悬而未决的问题。尽管NS算法自然地用贝叶斯推理的语言来表达，但它是一种强大且通用的积分算法。因此，我们预计NS算法将在未来长期保持其重要性。我们现在正处于数据科学革命的浪潮中，高维空间和积分比以往任何时候都更加重要。NS算法迎难而上。

正如我们所回顾的，在理解NS方面已经有许多理论发展，包括其收敛性、误差、诊断以及从约束先验中采样的技术。我们预计这些发展会继续，特别是随着与其他统计方法和机器学习的联系被探索。事实上，NS的理论分析结合约束MCMC探索可能会受益于框4中讨论的与SMC的联系，我们预计在理解NS的不确定性方面会有进一步发展，特别是在参数推断和动态数量的活点情况下。最后，NS可能被视为一种元算法，因为它没有指定从约束先验采样的算法。我们已经看到，这个开口使NS能够与机器学习的发展相结合，例如那些开始用于从约束先验采样的归一化流。

NS的成功和应用广度源于它本质上很简单。虽然在未来NS可能在更复杂的计算方法的背景下被理解和改进，但由于其简单性，NS将保持其优势和吸引力。

致谢

我们感谢John的精彩算法。嵌套采样的成功或许在于简单胜过花哨；但其魅力在于它既简单又巧妙。

我们感谢 Kyle Barbary 的讨论。AF 由国家自然科学基金会国际青年科学家研究基金（项目编号 11950410509）资助。LBP 感谢英国工程与物理科学研究委员会（EPSRC）通过早期职业奖学金（EP/T000163/1）提供的支持。MH 感谢蔡司基金会提供的支持。NB 由美国海军研究实验室的 6.1 基础研究项目资助，以及美国国防部 HPCMPO 在 AFRL 和 ARL DSRCs 的 CPU 时间支持。MP 感谢英国科学和技术研究委员会（STFC）的支持（ST/V001213/1 和 ST/V005707/1）。WH 感谢英国皇家学会大学研究奖学金的支持。

作者贡献

引言（A. F., D. S., L. S., P. W.）；实验（J. B., E. H., J. S. S.）；结果（A. F.）；应用（G. A., N. B., X. C., G. C., F. F., M. G., W. H., M. H., A. L., D. P., L. B. P., M. P., J. V., D. W., D. Y.）；可重复性与数据沉积（A. F.）；局限性与优化（A. F.）；展望（A. F.）；引言概述（A. F.）。

利益冲突

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补充信息

术语表

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insertion indexes The indexes at which the elements of a list must be inserted into an ordered list to maintain ordering. Pages 11.

integrand A function that is being integrated, for example, $f(x)$ is the integrand in $\int f(x)dx$. Pages 1, 2, 18, 21, 23.

iso-likelihood contour The set of points for which the likelihood is equal to a particular constant; in two-dimensions, this set forms a contour line. Pages 3, 7, 9, 22.

Kullback-Leibler divergence A measure of difference between two distributions that may be interpreted as the information gained by switching from one to the other. See ref. [222, 225] for pedagogical discussions. Pages 5.

likelihood The probability of the observed data as a function of a model's parameters. Pages 2–5, 8–12, 14–16, 18, 19, 21, 23.

Markov chain A sequence of random states for which the probability of a state depends only on the previous state. Pages 20.

Markov chain Monte Carlo (MCMC) A class of algorithms for drawing a correlated sequence of samples from a distribution using the states of a [Markov chain](#). Pages 1, 4, 8, 9, 11, 14, 19, 20.

measure A probability measure is a function assigning probabilities to events. See ref. [225] for a pedagogical discussion. Pages 6.

microcanonical ensemble Assigns equal probability to states Θ with $E(\Theta) = \epsilon$ and zero probability if $E(\Theta) \neq \epsilon$ such that the energy level ϵ rather than the inverse temperature β characterizes a thermodynamic state. Pages 6.

microstate The state of all degrees of freedom in a physical system, for example, the microstate of a multi-particle system includes the positions and momenta of all particles. Pages 6.

mode A peak in a probability distribution. Pages 7, 9, 12, 13, 16, 19, 21.

multi-modal Problems in which the [integrand](#) contains more than one [mode](#). In inference problems, [modes](#) may correspond to distinct ways in which the model could predict the observed data. Pages 1, 2, 7–9, 14, 19, 21, 22.

overloaded A function with a definition that depends on the type of its argument. In NS the likelihood L is an overloaded function since we consider separate functions $L(\theta)$ and $L(X)$. Pages 6.

parameter domain The set of a priori possible parameters, usually the reals $\mathbb{R}^n$ or a subset thereof. Pages 6.

partition function Normalizing constant $Z(\beta) = \int e^{-\beta E(\Theta)} d\Theta$ that fully characterizes a physical system, because many important thermodynamic variables can be derived from it. Pages 4, 6, 10, 15, 16, 21–23.

posterior A probability density conditioned on the observed data found by updating a [prior](#) using Bayes' theorem. In NS it describes the shape of our [integrand](#). Pages 2, 4–8, 10–12, 14–16, 18–22.

prior A probability density that isn't yet conditioned on observed data. In NS it is a measure in our integral. Pages 2, 3, 5, 6, 9, 14, 15, 18, 19, 23.

pseudo-importance sampling Algorithms in which an importance sampling density is defined a posteriori. Pages 19.

push-forward measure The push-forward measure (or image measure) is the distribution of a random variable under a probability measure. Pages 6.

Simulation-Based Calibration Techniques that use simulations from the model to check the correctness of Bayesian computation [226–228]. Pages 11.

super-level-sets A λ -super-level-set of any function contains all points for which the function value exceeds λ . Pages 6.

survival function A function $F(x)$ associated with a distribution that returns the probability of obtaining a sample greater than x . Pages 6.

tractable An adjective used to describe problems that are feasible to solve under computational, monetary or time constraints. Pages 18.

transition kernel A function that describes the likely steps of a [Markov chain](#). Pages 8.

uni-modal Problems in which the [integrand](#) contains only one [mode](#) (cf. [multi-modal](#)). Pages 2, 4, 8, 15, 21, 22.

B Estimators for compression factor

Besides the estimator $\langle \log t \rangle = -n_{\text{iter}}/n_{\text{live}}$ in eq. (12), we may instead consider,

$$\langle t \rangle = \frac{n_{\text{live}}}{n_{\text{live}} + 1}. \quad (34)$$

Similarly, ref. [22] suggests the estimator

$$\hat{t} = 1 - \frac{1}{n_{\text{live}}}. \quad (35)$$

In any case, the relative differences in $\log t$ are of order $\mathcal{O}(1/n_{\text{live}})$.

integrand A function that is being integrated, for example, $f(x)$ is the integrand in $\int f(x)dx$. Pages 1, 2, 18, 21, 23.

iso-likelihood contour The set of points for which the likelihood is equal to a particular constant; in two-dimensions, this set forms a contour line. Pages 3, 7, 9, 22.

Kullback-Leibler divergence A measure of difference between two distributions that may be interpreted as the information gained by switching from one to the other. See ref. [222, 225] for pedagogical discussions. Pages 5.

likelihood The probability of the observed data as a function of a model's parameters. Pages 2–5, 8–12, 14–16, 18, 19, 21, 23.

Markov chain A sequence of random states for which the probability of a state depends only on the previous state. Pages 20.

Markov chain Monte Carlo (MCMC) A class of algorithms for drawing a correlated sequence of samples from a distribution using the states of a [Markov chain](#). Pages 1, 4, 8, 9, 11, 14, 19, 20.

measure A probability measure is a function assigning probabilities to events. See ref. [225] for a pedagogical discussion. Pages 6.

microcanonical ensemble Assigns equal probability to states Θ with $E(\Theta) = \epsilon$ and zero probability if $E(\Theta) \neq \epsilon$ such that the energy level ϵ rather than the inverse temperature β characterizes a thermodynamic state. Pages 6.

microstate The state of all degrees of freedom in a physical system, for example, the microstate of a multi-particle system includes the positions and momenta of all particles. Pages 6.

mode A peak in a probability distribution. Pages 7, 9, 12, 13, 16, 19, 21.

multi-modal Problems in which the [integrand](#) contains more than one [mode](#). In inference problems, [modes](#) may correspond to distinct ways in which the model could predict the observed data. Pages 1, 2, 7–9, 14, 19, 21, 22.

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posterior A probability density conditioned on the observed data found by updating a [prior](#) using Bayes' theorem. In NS it describes the shape of our [integrand](#). Pages 2, 4–8, 10–12, 14–16, 18–22.

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uni-modal Problems in which the [integrand](#) contains only one [mode](#) (cf. [multi-modal](#)). Pages 2, 4, 8, 15, 21, 22.

用于压缩因子的B估计量

除了估计量 $\langle \log t \rangle = -n_{\text{iter}}/n_{\text{live}}$ in eq. (12), 我们也可以考虑,

$$\langle t \rangle = \frac{n_{\text{live}}}{n_{\text{live}} + 1}. \quad (34)$$

类似地, 参考文献 [22] 建议该估计量

$$\hat{t} = 1 - \frac{1}{n_{\text{live}}}. \quad (35)$$

无论如何, $\log t$ 的相对差异在 $\mathcal{O}(1/n_{\text{live}})$ 级别。

C Historical background

In the 1950s, when computational algorithms were beginning to be developed, the Bayesian revolution lay decades in the future and inquiry was focused at best on **posterior** distributions. The first practical computer codes for this were **MCMC** [229]; first, the Metropolis algorithm [230] and later Metropolis-Hastings [231]. They build a **Markov chain** of correlated points drawn from an unnormalized **posterior**. The Metropolis algorithm accepted a transition $\Theta \rightarrow \Theta'$ only if

$$L(\Theta') > uL(\Theta) \quad (36)$$

where $u \sim \mathcal{U}(0, 1)$. This conditional acceptance of random trial transitions satisfies detailed balance and converges towards an equilibrium. This, however, only gives the shape of the distribution, not the normalizing constant, Z .

Later, in the 1970s, a connection between the **likelihood** and the temperature-dependent energy,

$$L(\Theta) \propto e^{-\beta E(\Theta)} \quad (37)$$

where $\beta = 1/T$, suggested generalizing the **likelihood** to L^β . Upon which, the **evidence** generalized to

$$Z(\beta) = \int L^\beta(\Theta) \pi(\Theta) d\Theta, \quad (38)$$

with $Z = Z(1)$ being the desired **evidence** and $Z(0)$ being 1. Whereas the **posterior** generalized to

$$P^\beta(\Theta) = \frac{L^\beta(\Theta) \pi(\Theta)}{Z(\beta)}, \quad (39)$$

with $P^0(\Theta)$ the prior and $P^1(\Theta)$ the desired **posterior**. These values are connected by the computable differential

$$\frac{d \log Z}{d\beta} = \int \log L(\Theta) P^\beta(\Theta) d\Theta = \langle \log L \rangle_\beta \quad (40)$$

which is simply the log-likelihood averaged over the distribution $P^\beta(\Theta)$ at inverse temperature β . Add that up as the distribution is slowly cooled from $\beta = 0$ to 1, i.e., from the prior to the **posterior**, and the **evidence** arrives as [39, 232, 233]

$$\log Z = \int_0^1 \langle \log L \rangle_\beta d\beta. \quad (41)$$

All that seems to be needed to compute the **evidence** integral is to fix the cooling schedule to track β suitably slowly (hence annealing) from 0 to 1 such that the **posterior** distribution $P^\beta(\Theta)$, represented by a collection of samples drawn from it, changes slowly. This annealing schedule may, however, be problematic. Maintaining the thermodynamics analogy, which is further expanded in box 3, the trouble occurs with changes of state as the system cools. The **evidence** plays the role of the thermodynamic **partition function** and at each inverse temperature the samples are expected to coalesce closely around the local

maximum of the **posterior** distribution $P^\beta(\log X)$. This implies that

$$\frac{d \log L(X)}{d \log X} = -\frac{1}{\beta} \quad (42)$$

In smooth cross-over transitions, volume and energy shrink in tandem as the system cools from $\beta = 0$ (vertical slope) to $\beta = 1$ (diagonal slope) in fig. 6-a, which is what happens in stable physical systems.

If there is a second-order phase transition, however, a tiny cooling around the critical β rapidly compresses the ensemble exponentially all the way from disorder to order. We see this in the iso-thermal region in fig. 6-b. The sensitivity of the system to temperature means that it becomes numerically impossible to use β as the control parameter. Lastly, if there's a first-order phase transition, two phases coexist at the same temperature or in a temperature interval, e.g., steam and water in fig. 6c. Starting from the steam phase, to reach the water phase requires an abrupt first-order phase transition. It thus becomes impossible even in principle to use inverse temperature to change smoothly from the steam phase to the water phase.

The first-order phase transition was connected to the fact that there were multiple solutions to eq. (42) for some β . This implies that the **integrand** in the **evidence** integral expressed as

$$Z(\beta) = \int L^\beta(X) X d \log X, \quad (43)$$

was **multi-modal** for at least some temperatures. Physically, this may occur in a problem with distinct solutions at different **likelihood** levels. In our analogy, we may interpret the **modes** as phases. Traditional annealing methods struggle to move particles between them during a phase transition but with NS, the system's evolution can be controlled through compression instead.

D Step sampling

For detailed balance, the transition kernel in a step sampler must satisfy

$$\int P(\Theta_f | \Theta) \pi^\star(\Theta) d\Theta = \pi^\star(\Theta_f). \quad (44)$$

We are thus drawing a new live point Θ_f from

$$P(\Theta_f | \{\Theta_{\text{live}}\}) = \frac{1}{n_{\text{live}}} \sum_{i=1}^{n_{\text{live}}} P(\Theta_f | \Theta_{\text{live}}^i) \quad (45)$$

where $\{\Theta_{\text{live}}\}$ denotes the set of live points and we average over the possible choices of initial live point indexed by i , Θ_{live}^i . For a draw from eq. (45) to be approximately equivalent to an independent draw from the **constrained prior**, we thus require

$$P(\Theta_f | \{\Theta_{\text{live}}\}) \approx \pi^\star(\Theta_f). \quad (46)$$

C 历史背景

在 20 世纪 50 年代, 当计算算法开始发展时, 贝叶斯革命还有几十年之遥, 研究重点最多也只是后验分布。这个领域的第一个实用计算机代码是 MCMC [229]; 首先是 Metropolis 算法 [230], 后来是 Metropolis-Hastings [23

他们构建了一个由未归一化后验分布抽取的相关点组成的马尔可夫链。Metropolis算法仅接受满足 $\Theta \rightarrow \Theta'$ 的转移。

$$L(\Theta') > uL(\Theta) \quad (36)$$

其中 $u \sim \mathcal{U}(0, 1)$ 。这种对随机试验转移的条件接受满足详细平衡并收敛到平衡状态。然而, 这仅给出了分布的形状, 而不是归一化常数 Z 。

后来, 在 20 世纪 70 年代, 似然与温度相关能量之间建立了一种联系,

$$L(\Theta) \propto e^{-\beta E(\Theta)} \quad (37)$$

其中 $\beta = 1/T$, 建议将似然推广到 L^β 。基于此, 证据被推广为

$$Z(\beta) = \int L^\beta(\Theta) \pi(\Theta) d\Theta, \quad (38)$$

其中 $Z = Z(1)$ 是所需证据, $Z(0)$ 为 1。而后验分布被推广为

$$P^\beta(\Theta) = \frac{L^\beta(\Theta) \pi(\Theta)}{Z(\beta)}, \quad (39)$$

通过 $P^0(\Theta)$ 先验和 $P^1(\Theta)$ 的后验。这些值由可计算的微分连接起来

$$\frac{d \log Z}{d\beta} = \int \log L(\Theta) P^\beta(\Theta) d\Theta = \langle \log L \rangle_\beta \quad (40)$$

这简单来说就是逆温度 β 下 $P^\beta(\Theta)$ 分布的对数似然平均值。当分布从 $\beta = 0$ 缓慢冷却到 1 时, 即从先验到后验, 将它们加起来, 证据就出现了 [39, 232, 233]

$$\log Z = \int_0^1 \langle \log L \rangle_\beta d\beta. \quad (41)$$

计算证据积分似乎只需要将冷却调度固定为从0到1以合适缓慢的速度（即退火）跟踪 β , 使得后验分布 $P^\beta(\Theta)$ ——由从中抽取的样本集合表示——缓慢变化。然而, 这种退火调度可能会有问题。根据进一步在框3中扩展的热力学类比, 问题发生在系统冷却时的状态变化。证据扮演着热力学配分函数的角色, 在每个倒数温度下, 样本预期会紧密聚集在后验分布 β 的局部最大值周围。这意味着

这暗示了

$$\frac{d \log L(X)}{d \log X} = -\frac{1}{\beta} \quad (42)$$

在平滑的交叉过渡中, 体积和能量随着系统从 $\beta = 0$ （垂直斜率）冷却到 $\beta = 1$ （对角斜率）在图 6-a 中同步收缩, 这在稳定的物理系统中会发生 1)。

然而, 如果有二级相变, 则在临界 β 附近微小的冷却会指数级地压缩系综, 从无序到有序。我们在图 6-b 的等温区看到这一点。系统对温度的敏感性意味着使用 β 作为控制参数在数值上变得不可能。最后, 如果存在一级相变, 两个相在相同温度或温度区间共存, 例如图 6c 中的蒸汽和水。从蒸汽相开始, 要达到水相需要发生突变的相变。因此, 即使在原则上也无法使用逆温度从蒸汽相平滑地变化到水相

一级相变与以下事实相关联: 对于某些 β , 方程 (42) 存在多个解。这意味着证据积分中的被积函数表示为

$$Z(\beta) = \int L^\beta(X) X d \log X, \quad (43)$$

在至少某些温度下是多模态的。从物理上看, 这在具有不同似然水平下具有不同解的问题中可能发生。在我们的类比中, 我们可以将模态解释为相。传统的退火方法在相变期间难以使粒子在它们之间移动, 但使用 NS 时, 可以通过压缩来控制系统演化。

D 步采样

对于详细平衡, 步进采样器中的转移核必须满足

$$\int P(\Theta_f | \Theta) \pi^\star(\Theta) d\Theta = \pi^\star(\Theta_f). \quad (44)$$

因此, 我们从 中抽取一个新的活点 Θ_f

$$P(\Theta_f | \{\Theta_{\text{live}}\}) = \frac{1}{n_{\text{live}}} \sum_{i=1}^{n_{\text{live}}} P(\Theta_f | \Theta_{\text{live}}^i) \quad (45)$$

$\{\Theta_{\text{live}}\}$ 表示活点集合, 我们对初始活点索引为 i , Θ_{live}^i 的可能选择进行平均。要使从等式(45)中抽取的样本近似等同于从约束先验中独立抽取的样本, 因此我们需要

$$P(\Theta_f | \{\Theta_{\text{live}}\}) \approx \pi^\star(\Theta_f). \quad (46)$$

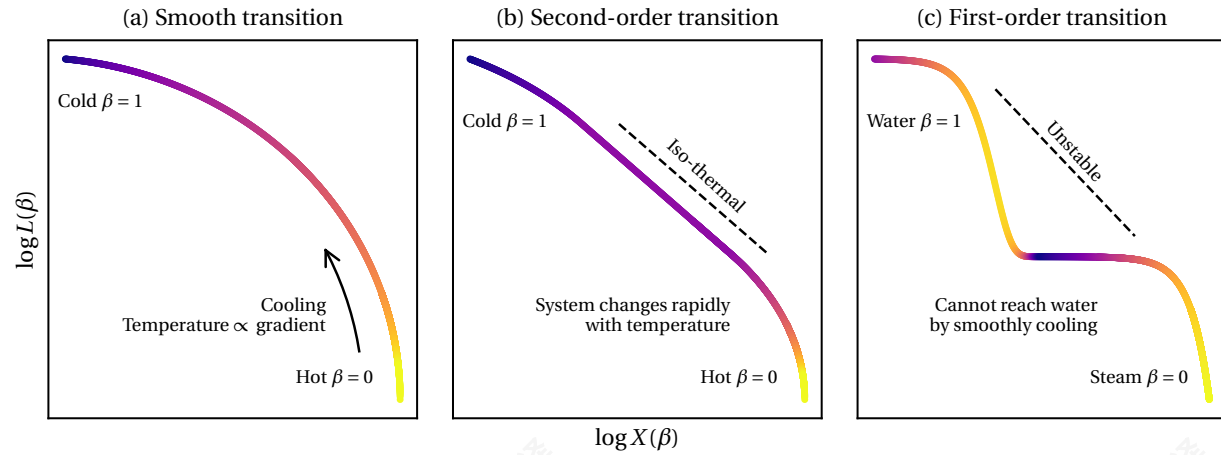


Figure 6 | **Phase transitions in simulated annealing.** We show $\log L(\beta)$ and $\log X(\beta)$ as we cool from $\beta = 0$ to 1 in an ordinary application, a second-order transition and a first-order transition. The colour scale indicates temperature.

We anticipate that this might approximately hold because the live points should be independent draws from the **constrained prior** such that eq. (45) is a Monte Carlo estimate of the **constrained prior** through eq. (44). In any case, to satisfy eq. (46), we may attempt to satisfy a condition of decorrelation between the chosen initial point and the final point,

$$P(\Theta_f | \Theta_{\text{live}}^i) \approx \pi^*(\Theta_f). \quad (47)$$

In **uni-modal** problems, this can be achieved by performing a sufficient number of steps. In **multi-modal** problems, it may be impossible as walks between **modes** are all but impossible. Since we only require eq. (46), however, correlation between the **modes** in which the initial and final points lie may be acceptable. The number of steps required to satisfy eq. (46) may decrease when n_{live} is increased, as Monte Carlo errors in the approximation in eq. (46) shrink as $1/\sqrt{n}$.

Lastly, these considerations show that starting from the last dead point instead of a randomly chosen live point isn't sensible [201], even though it avoids correlation with the existing live points. In **uni-modal** problems, we could require many steps to avoid biasing the draw towards the lowest likelihood levels near the last dead point. In **multi-modal** problems, it would lead to catastrophe, as the fractions of live points in each mode could not change, since a point would always be replaced by a point in the same mode. Consequently, modes would never die and NS would become stuck at the likelihood threshold of the lowest lying mode.

E Uncertainty in the posterior

There is a further source of uncertainty in **posterior** expectations. From the dead points Θ_i , with importance weights P_i , a

posterior mean for some quantity of interest $f: \Omega \rightarrow \mathbb{R}$ may be estimated by [18]

$$\langle f(\Theta) \rangle = \int f(\Theta) P(\Theta) d\Theta = \int f(X) P(X) dX \quad (48)$$

$$\approx \sum_i P_i f(X_i) \approx \sum_i P_i f(\Theta_i), \quad (49)$$

where $f(X)$ is the mean of f over an **iso-likelihood contour**,

$$f(X) = \frac{\int f(\Theta) \delta(L(\Theta) - L(X)) \pi(\Theta) d\Theta}{\int \delta(L(\Theta) - L(X)) \pi(\Theta) d\Theta}. \quad (50)$$

There are thus two sources of uncertainty in eq. (49): the statistical uncertainty in the estimates of the importance weights P_i , as before, and the fact that eq. (49) replaces the mean around the contour by a single draw from around the contour, $f(X_i) \approx f(\Theta_i)$. As before, the first source can be accounted for by simulating the compression factors through eq. (10). Whilst both sources are reduced by increasing the number of live points, the second source usually dominates.

Ref. [234] demonstrated that it can be assessed by decomposing an NS run with n_{live} live points into n_{live} NS runs with one live point each called *threads*. The threads can be recombined in many different combinations using resampling methods such as **bootstrap** to generate simulated runs with the same number of live points as the original run. The variance in **posterior** inferences in these simulated runs captures both sources of uncertainty (see UltraNest [92] for an NS implementation of these simulations).

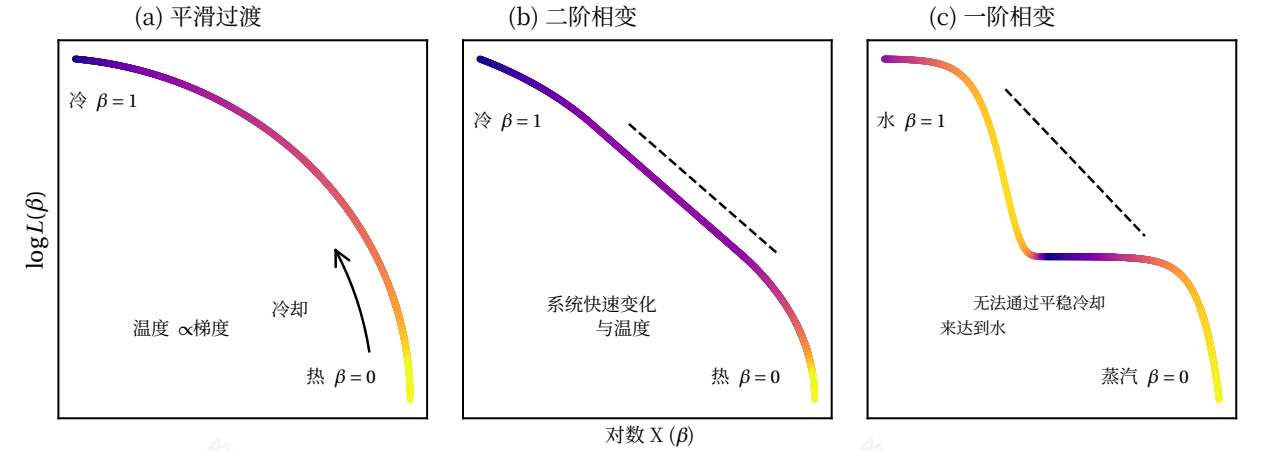


图 6 | 模拟退火中的相变。我们展示了从 $\beta = 0$ 冷却到 1 时 $\log L(\beta)$ 和 $\log X(\beta)$ 的变化，在普通应用中是一个二阶相变和一个一阶相变。颜色标尺表示温度。

我们预计这大致成立，因为活点应该是从约束先验中独立抽取的样本，使得等式(45)通过等式(44)成为约束先验的蒙特卡洛估计。无论如何，为了满足等式(46)，我们可以尝试满足初始点与最终点之间的去相关条件，

$$P(\Theta_f | \Theta_{\text{live}}^i) \approx \pi^*(\Theta_f). \quad (47)$$

在单峰问题中，这可以通过执行足够多的步骤来实现。在多峰问题中，这可能是不可能的，因为模式之间的行走几乎是不可能的。然而，由于我们只需要等式(46)，因此初始点和终点所在的模式之间的相关性可能是可以接受的。当 n_{live} 增加时，满足等式(46)所需的步骤数可能会减少，因为等式(46)中的蒙特卡洛误差随着 $1/\sqrt{n}$ 的缩小而缩小。

最后，这些考虑表明，从最后一个死点而不是随机选择的活点开始并不是明智的 [201]，尽管它避免了与现有活点的相关性。在单峰问题中，我们可能需要许多步骤来避免将抽取偏向到最后死点附近的最低可能性水平。在多峰问题中，这会导致灾难，因为每个模式中的活点分数无法改变，因为一个点总是会由同一模式中的点替换。因此，模式永远不会死亡，NS将会卡在最低峰值的可能性阈值上。

后验不确定性

后验期望中还存在进一步的来源不确定性。从死点 Θ_i ，以重要性权重 P_i ，

某个目标量 f 的后验均值可以通过 [18] 估计： $\Omega \rightarrow \mathbb{R}$

$$\langle f(\Theta) \rangle = \int f(\Theta) P(\Theta) d\Theta = \int f(X) P(X) dX \quad (48)$$

$$\approx \sum_i P_i f(X_i) \approx \sum_i P_i f(\Theta_i), \quad (49)$$

其中 $f(X)$ 是等高似然线上的 f 的均值，

$$f(X) = \frac{\int f(\Theta) \delta(L(\Theta) - L(X)) \pi(\Theta) d\Theta}{\int \delta(L(\Theta) - L(X)) \pi(\Theta) d\Theta}. \quad (50)$$

因此，在公式 (49) 中存在两种不确定性来源：重要性权重 P_i 的统计不确定性，与之前相同，以及公式 (49) 用等高线上的单个样本 $f(X_i) \approx f(\Theta_i)$ 替换了等高线上的均值。与之前一样，第一种来源可以通过通过公式 (10) 模拟压缩因子来解释。虽然增加活点的数量可以减少这两种来源的不确定性，但第二种来源通常占主导地位。

Ref. [234] 证明了可以通过将一个具有 n_{live} 个活动点的 NS 运行分解为 n_{live} 个具有单个活动点的 NS 运行（称为线程）来进行评估。这些线程可以使用重采样方法（如 bootstrap）以多种不同的组合方式重新组合，以生成与原始运行具有相同数量的活动点的模拟运行。在这些模拟运行中后验推断的方差捕获了两种不确定性来源（有关这些模拟的 NS 实现，请参见 UltraNest [92]）。

F Periodic boundary conditions, degrees of freedom and Monte Carlo moves

In any condensed phase, a computationally tractable small finite system would be dominated by surface effects, and eliminating these requires a periodic supercell description. The typical volume per atom changes by orders of magnitude between the gas and condensed phases, and must be allowed to vary for NS to sample the relevant structures. This can be achieved by fixing the pressure P , replacing the potential energy U in the sampling algorithm with the enthalpy $U + PV$, where V is the system volume, and sampling the cell degrees of freedom [44, 77]. The corresponding [partition function](#) is

$$Z(N, P, \beta) = Z_p \beta P \int d\mathbf{h}_0 \delta(\det \mathbf{h}_0 - 1) \times \int_0^\infty dV V^N \int_{(0,1)^{3N}} d\mathbf{s} e^{-\beta(U(V^{1/3}\mathbf{h}_0\mathbf{s}) + PV)},$$

where $\mathbf{h}_0$ is the reduced cell shape matrix (with unit determinant) and $\mathbf{s}$ are the scaled (often called fractional) atomic positions. When generating uniformly distributed configurations V and $\mathbf{h}_0$ must be sampled in addition to the scaled positions $\mathbf{s}$. The expectation value of $V(\beta)$ must be calculated using the same weights as the [partition function](#). To compute an entire pressure-temperature phase diagram the NS run is repeated for each pressure value. After a completed NS run, the expectation value of any observable can be calculated as a function of temperature, e.g. the radial distribution function plotted in fig. 5b, which can be compared to the results of scattering experiments and used to identify each equilibrium phase.

Condensed phase atomic position Monte Carlo moves with a reasonable acceptance probability are also challenging to generate. Single atom moves become inefficient if the resulting energy change cannot be computed in $\mathcal{O}(1)$ time, and the probability of accepting naive collective moves decreases as $1/\sqrt{N}$. A more efficient alternative is Galilean Monte Carlo [44, 69, 70], where a random direction in $3N$ -dimensional configuration space is proposed, the entire system is propagated for a fixed number of steps along straight lines that reflect specularly from the allowed $U(q) < U^*$ boundary. Another is total-energy Hamiltonian Monte Carlo [44, 235], where the kinetic energy is added to U , so the NS iteration constrains *total* energy, and atomic moves are proposed by carrying out short time constant energy molecular dynamics trajectories that are accepted or rejected in their entirety. In both cases, Monte Carlo moves that propose to change the volume and the cell shape are also used.

In condensed phase systems with multiple types of particles the probability for particles to switch places is low, especially in the solid phase, and explicit particle swap proposals are required to ensure mixing. Further, the *composition* of different phases may change discontinuously across phase transitions. The semi-grand-canonical ensemble, with fixed total number of particles but variable composition, describes this situation.

Monte Carlo proposals include changes to particle type, and the energy is augmented by $\sum_i \mu_i N_i$, where i runs over the types, μ_i is a specified chemical potential, and N_i is the number of particles of that type i . Like $V(\beta)$ above, $N_i(\beta)$ is an output of the simulation.

G Example

Let us demonstrate NS using standard NS software on a toy problem using Python. The libraries used in this example may be installed using pip, for example,

```
$ pip install numpy scipy matplotlib pypolychord anesthetic
```

To record our software version numbers,

```
>>> import platform
>>> print(platform.python_version())
3.8.10
>>> from importlib.metadata import version
>>> print(version('pypolychord'))
1.20.1
>>> print(version('anesthetic'))
1.3.6
```

Let us attempt to compute the two-dimensional Gaussian integral,

$$Z = \frac{1}{(2a)^2} \int_{-a}^a \int_{-a}^a e^{-x^2-y^2} dx dy. \quad (51)$$

We may write the [integrand](#) in the form eq. (1) by defining the [likelihood](#)

$$L(x, y) = e^{-x^2-y^2} \quad (52)$$

and the [prior](#),

$$\pi(x, y) = \frac{1}{(2a)^2} \quad (53)$$

for $-a < x < a$ and $-a < y < a$ and vanishing elsewhere. So long as a is greater than about 1, $Z \simeq \pi/(2a^2)$ and $H \simeq -1 - \log \pi - \log 2a^2$. For concreteness, we take $a = 5$ such that $\log Z \simeq -3.46$ and $H \simeq 2.46$

To run NS on this problem we first implement the logarithm of the [likelihood](#) in eq. (52),

```
>>> def loglike(theta):
...     return -(theta**2).sum(), [] # [] is anything else to be saved
```

and a transformation of the unit hypercube representing the prior in eq. (53),

```
>>> def prior(unit_hypercube):
...     a = 5.
...     return 2. * a * unit_hypercube - a
```

Here we map from the unit hypercube $\mathcal{U}(0,1)^2$ to our $\mathcal{U}(-a,a)^2$ prior.

F 周期边界条件、自由度和蒙特卡洛移动

在任何凝聚相中，一个计算上易于处理的有限系统将受表面效应主导，消除这些效应需要采用周期性超胞描述。气相和凝聚相之间，每个原子的典型体积变化达数个数量级，并且必须允许其变化，以便NS能够采样相关结构。这可以通过固定压力P，在采样算法中将势能U替换为焓 $U + PV$ （其中V是系统体积），并采样胞单元自由度 [44, 77]来实现。相应的配分函数是

$$Z(N, P, \beta) = Z_p \beta P \int d\mathbf{h}_0 \delta(\det \mathbf{h}_0 - 1) \times \int_0^\infty dV V^N \int_{(0,1)^{3N}} d\mathbf{s} e^{-\beta(U(V^{1/3}\mathbf{h}_0\mathbf{s}) + PV)},$$

其中 $\mathbf{h}_0$ 是约化单元形状矩阵（具有单位行列式）， $\mathbf{s}$ 是缩放（通常称为分数）的原子位置。在生成均匀分布的构型时，除了缩放位置 $\mathbf{s}$ 之外，还需要对 V 和 $\mathbf{h}_0$ 进行采样。 V 的期望值 $\langle V \rangle$ 必须使用与配分函数相同的权重来计算。为了计算整个压力-温度相图，NS 运行会针对每个压力值重复进行。完成一次 NS 运行后，任何可观测量的期望值都可以作为温度的函数来计算，例如如图 5b 中绘制的径向分布函数，可以将其与散射实验的结果进行比较，并用于识别每个平衡相。

凝聚相原子位置的蒙特卡洛移动在具有合理接受概率的情况下也具有挑战性。如果结果能量变化无法在 $\mathcal{O}(1)$ 时间内计算出来，单个原子移动就会变得低效，并且接受简单集体移动的概率随着 $1/\sqrt{N}$ 的增加而降低。一种更有效的方法是伽利略蒙特卡洛 [44, 69, 70]，其中在 $3N$ 维构型空间中提出一个随机方向，整个系统沿固定数量的直线传播，这些直线会从允许的 $U(q) < U^*$ 边界进行镜面反射。另一种是总能量哈密顿蒙特卡洛 [44, 235]，其中将动能添加到 U 中，因此 NS 迭代约束总能量，原子移动是通过执行短时间的恒定能量分子动力学轨迹来提出的，这些轨迹会整体被接受或拒绝。在这两种情况下，蒙特卡洛移动也会被用于提出改变体积和单元形状的提议。

在含有多种粒子的凝聚相系统中，粒子交换位置的概率较低，特别是在固态中，需要显式粒子交换提议来确保混合。此外，不同相的组成可能在相变过程中发生不连续变化。半巨正则系综，具有固定的粒子总数但可变的组成，描述了这种情况。

蒙特卡洛提议包括对粒子类型的改变，能量增加量为 $\sum_i \mu_i N_i$ ，其中 i 遍历所有类型， μ_i 是指定的化学势， N_i 是该类型 i 的粒子数。与上述 $V(\beta)$ 类似， $N_i(\beta)$ 是模拟的输出。

G 示例

让我们使用 Python 在一个简单的示例上展示标准 NS 软件。本示例中使用的库可以通过 pip 安装，例如，

```
$ pip install numpy scipy matplotlib pypolychord anesthetic
```

To record our software version numbers,

```
>>> import platform
>>> print(platform.python_version())
3.8.10
>>> from importlib.metadata import version
>>> print(version('pypolychord'))
1.20.1
>>> print(version('anesthetic'))
1.3.6
```

让我们尝试计算二维高斯积分，

$$Z = \frac{1}{(2a)^2} \int_{-a}^a \int_{-a}^a e^{-x^2-y^2} dx dy. \quad (51)$$

我们可以通过定义似然 2.2 来将积分函数写成等式 (1) 的形式

$$L(x, y) = e^{-x^2-y^2} \quad (52)$$

以及先验，

$$\pi(x, y) = \frac{1}{(2a)^2} \quad (53)$$

对于 $-a < x < a$ 和 $-a < y < a$ ，并在其他地方为零。只要 a 大于约 1， $Z \simeq \pi/(2a^2)$ ，并且 $H \simeq -1 - \log \pi - \log 2a^2$ 。为了具体起见，我们取 $a = 5$ 使得 $\log Z \simeq -3.46$ 和 $H \simeq 2.46$

要在这个问题上运行NS，我们首先实现等式(52)中的对数似然，

```
>>> def loglike(theta):
...     return -(theta**2).sum(), [] # [] is anything else to be saved
```

并对表示先验的单元超立方体进行转换，如等式(53)所示，

```
>>> def prior(unit_hypercube):
...     a = 5.
...     return 2. * a * unit_hypercube - a
```

在这里，我们将单位超立方体 $\mathcal{U}(0,1)^2$ 映射到我们的 $\mathcal{U}(-a,a)^2$ 先验分布。

We use the PolyChord [65, 133] NS implementation — this uses slice sampling to draw replacement live points from the constrained prior — see table 2 for alternative NS software. We import it by

```
>>> from pypolychord import run_polychord
>>> from pypolychord.settings import PolyChordSettings
```

We specify that our integral is two-dimensional ($\text{ndim} = 2$), that there are no derived quantities to save to disk ($\text{nderived} = 0$), and our NS settings. We wish to use $n_{\text{live}} = 1000$ ($\text{settings.nlive} = 1000$), perform 5 slice sampling steps ($\text{settings.num_repeats} = 5$), and fix our random seed for reproducibility ($\text{settings.seed} = 67$).

```
>>> ndim = 2
>>> nderived = 0
>>> settings = PolyChordSettings(ndim, nderived)
>>> settings.nlive = 1000 # this is  $n_{\text{live}}$ 
>>> settings.num_repeats = 5
>>> settings.seed = 67
```

Finally, we run the NS algorithm,

```
>>> run_polychord(loglike, ndim, nderived, settings, prior)
```

This by default writes our results to files named chains/test* and information to the screen. We may further inspect the results using, for example, anesthetic [78]. First, we load the data from disk and for reproducibility fix the random seed using numpy [236],

```
>>> from anesthetic import NestedSamples
>>> import numpy as np

>>> samples = NestedSamples(root='chains/test')
>>> np.random.seed(71)
```

We may compute and print the triplet $\log Z$, its uncertainty, and the KL divergence,

```
>>> H = samples.D()
>>> logZ = samples.logZ()
>>> uncertainty = (H / settings.nlive)**0.5 # this is eq. (21)
>>> print(logZ, uncertainty)
-3.5109005855438395 0.04994705730822035
>>> print(H)
2.494708533750648
```

Thus in this problem NS estimates $\log Z = -3.51 \pm 0.05$ and $H = 2.5$ in agreement with the analytic results. We may compare these estimates to those found from 1000 simulations ($\text{nsamples}=1000$) of the compression factors,

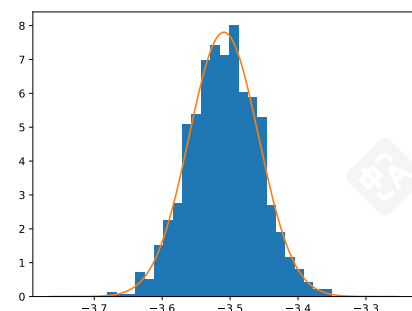
```
>>> draws = samples.logZ(nsamples=1000)
>>> mean = np.mean(draws)
>>> std = np.std(draws)
>>> print(mean, std)
-3.510624723981491 0.05232090274229939
```

showing that the standard NS estimators are reliable in this

case. We may check the assumption that $\log Z$ is approximately Gaussian distributed by histogramming the draws of $\log Z$. Here we use the matplotlib [237] histogramming functionality and the scipy [238] implementation of the normal distribution,

```
>>> import matplotlib.pyplot as plt
>>> from scipy.stats import norm

>>> plt.hist(draws, bins='auto', density=True)
>>> x = np.linspace(mean - 5. * std, mean + 5. * std, 1000)
>>> plt.plot(x, norm(mean, std).pdf(x))
>>> plt.show()
```



We see that $\log Z$ approximately follows a Gaussian distribution.

As a cross-check, we can test whether the insertion indexes of new live points are uniformly distributed using anesthetic,

```
>>> from anesthetic.utils import insertion_p_value
>>> samples._compute_insertion_indexes()
>>> ks = insertion_p_value(samples.insertion, settings.nlive)
>>> print(ks['p-value'])
0.4965573665241407
```

In this case we find p -value of about 0.5, which does not indicate any discrepancy from uniform.

Lastly, we may wish to compute and plot posterior distributions. Here we plot one- and two-dimensional posterior distributions using anesthetic and kernel density estimation ('kde'),

```
>>> samples.plot_2d(samples.columns[:ndim], types={'
lower': 'kde', 'diagonal': 'kde'})
>>> plt.show()
```

我们使用 PolyChord [65, 133] NS 实现——这使用切片采样从约束先验分布中绘制替换的活点——有关替代 NS 软件，请参见表 2。我们通过

```
>>> 从 pypolychord 中导入 run_polychord
>>> from pypolychord.settings import PolyChordSettings
```

我们指定我们的积分为二维 ($\text{ndim} = 2$)，没有要保存到磁盘的导出量 ($\text{nderived} = 0$)，以及我们的 NS 设置。我们希望使用 $n_{\text{live}} = 1000$ ($\text{settings.nlive} = 1000$)，执行 5 次切片采样步骤 ($\text{settings.num_repeats} = 5$)，并固定我们的随机种子以实现可重复性 ($\text{settings.seed} = 67$)。

```
>>> ndim = 2
>>> nderived = 0
>>> settings = PolyChordSettings(ndim, nderived)
>>> settings.nlive = 1000 # this is  $n_{\text{live}}$ 
>>> settings.num_repeats = 5
>>> settings.seed = 67
```

最后，我们运行 NS 算法，

```
>>> 运行_多和弦(loglike, ndim, nderived, settings, prior)
```

这默认将我们的结果写入名为 chains/test* 的文件，并将信息输出到屏幕。我们可以使用例如 anesthetic [78] 进一步检查结果。首先，我们从磁盘加载数据，并使用 numpy [236]，固定随机种子以实现可重复性

```
>>> 从anesthetic导入NestedSamples
>>> import numpy as np

>>> samples = NestedSamples(root='chains/test')
>>> np.random.seed(71)
```

我们可以计算并打印三元组 $\log Z$ 、其不确定性以及 KL 散度，

```
>>> H = samples.D()
>>> logZ = samples.logZ()
>>> uncertainty = (H / settings.nlive)**0.5 # this is eq. (21)
>>> print(logZ, uncertainty)
-3.5109005855438395 0.04994705730822035
>>> print(H)
2.494708533750648
```

因此在这个问题中 NS 估计 $\log Z = -3.51 \pm 0.05$ 和 $H = 2.5$ ，与分析结果一致。我们可以将这些估计值与从 1000 次模拟 ($\text{nsamples}=1000$) 压缩因子中得到的值进行比较，

```
>>> draws = samples.logZ(nsamples=1000)
>>> mean = np.mean(draws)
>>> std = np.std(draws)
>>> print(mean, std)
-3.510624723981491 0.05232090274229939
```

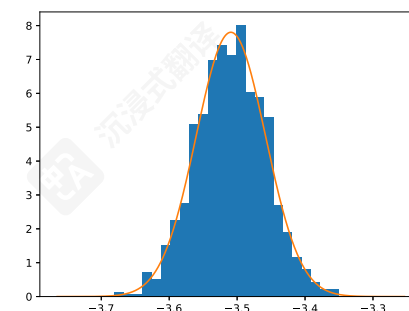
表明标准 NS 估计量在此情况下是可靠的

我们可以通过绘制 $\log Z$ 的样本直方图来检验 $\log Z$ 近似服从高斯分布的假设。这里我们使用 matplotlib 的 [237] 直方图功能

scipy [238] 正态分布的实现，

```
>>> import matplotlib.pyplot as plt
>>> from scipy.stats import norm

>>> plt.hist(draws, bins='auto', density=True)
>>> x = np.linspace(mean - 5. * std, mean + 5. * std, 1000)
>>> plt.plot(x, norm(mean, std).pdf(x))
>>> plt.show()
```



我们看到 $\log Z$ 近似服从高斯分布。

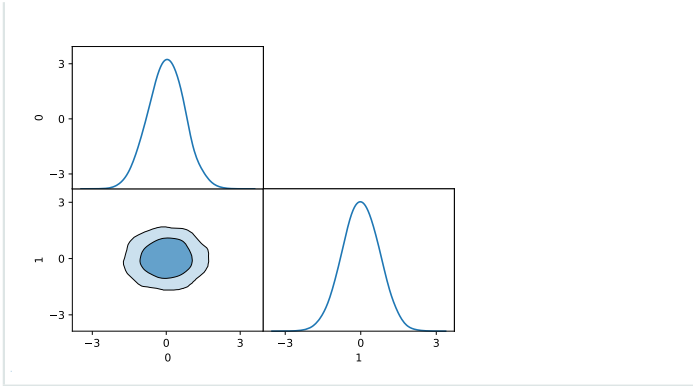
作为交叉验证，我们使用 anesthetic 测试新活点的插入索引是否均匀分布，

```
>>> from anesthetic.utils import insertion_p_value
>>> samples._compute_insertion_indexes()
>>> ks = insertion_p_value(samples.insertion, settings.nlive)
>>> print(ks['p-value'])
0.4965573665241407
```

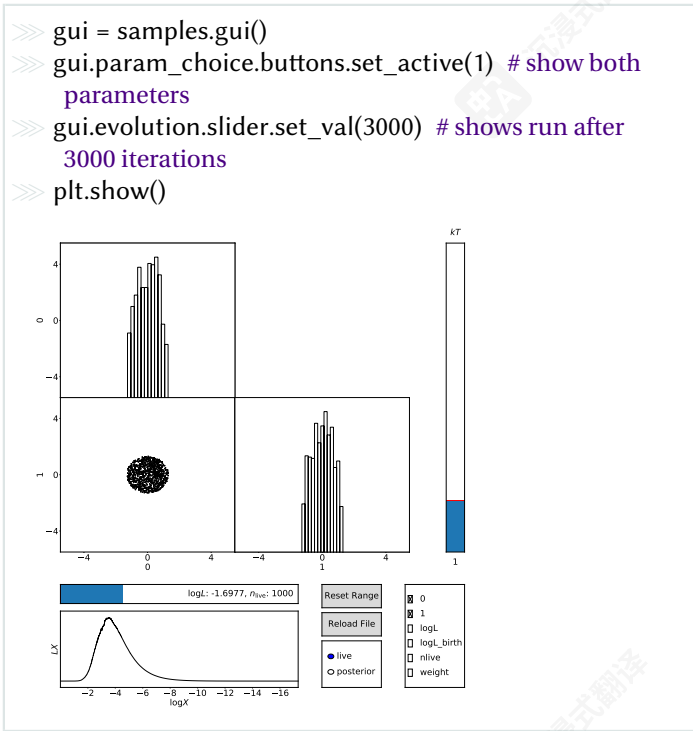
在这种情况下，我们得到 p 值约为 0.5，这并不表明与均匀分布存在任何差异。

最后，我们可能希望计算并绘制后验分布。这里我们使用麻醉剂和核密度估计 ('kde') 绘制了一维和二维后验分布，

```
>>> samples.plot_2d(samples.columns[:ndim], types={'
lower': 'kde', 'diagonal': 'kde'})
>>> plt.show()
```



We may examine the evolution of live points interactively using a graphical user interface (GUI)



This shows, among other things, the distribution of live points at iteration 3000, and we may use the slider to see how it evolved.

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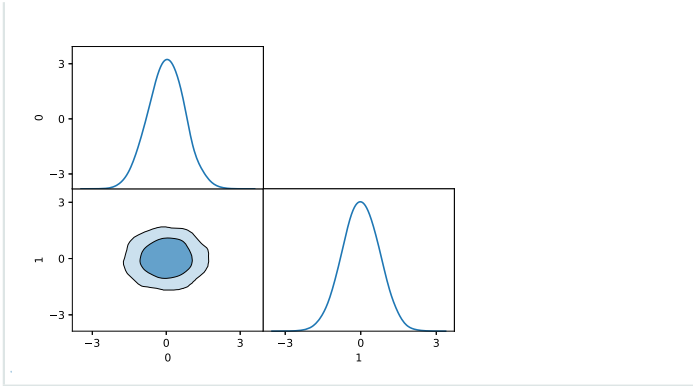
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A classic modern reference for Bayes factors

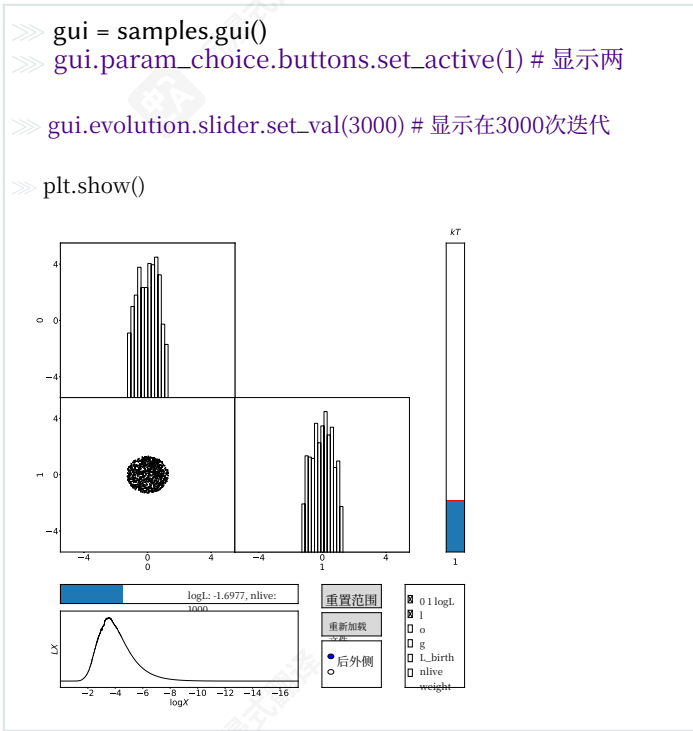
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我们可以使用图形用户界面（GUI）交互式地检查活点的演化。



这显示了在3000次迭代时活点的分布情况，并且我们可以使用滑块查看它是如何演变的。

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